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Abstract

Write here a concise summary of your thesis (150–300 words). The abstract should cover: the physical context and motivation, the methods used, the main results obtained, and their significance. Avoid equations and references if possible.

Keywords: keyword one, keyword two, keyword three, keyword four.

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Chapter 1

Introduction

Snow plays a critical role in Earth energy balance,

1.1 Thesis outline

This thesis is structured as follows:

- *Chapter 2* provides the theoretical background needed to understand the work presented in this thesis. The first part focuses on snow physics. It begins with the nucleation of ice crystals in cold clouds and follows the snowflakes down to the ground, where they accumulate to form a new layer. It then describes snow metamorphism and the energy balance of the snowpack, and finally liquid water percolation and runoff generation. The second part reviews snow measurement techniques, distinguishing between manual observations, automatic monitoring networks and remote sensing, and discussing the strengths and limitations of each approach. The third part examines how climate change is affecting snow cover, outlining the current state of snow dynamics worldwide.

- *Chapter 3* introduces the study area and the three main snow products that cover it. It opens with a description of the Italian hydrological portion of the Greater Alpine Region (GAR), which encompasses the Italian Alps and the Northern Apennines. Each dataset is examined in turn, with particular attention to how it is generated and the estimates it provides over Italy. The chapter closes with a comparison of the three snow products, highlighting the discrepancies between their estimates and thereby motivating the need for a new approach to snow monitoring at the national scale.

- *Chapter 4* describes the model used to generate the snow water equivalent (SWE) dataset. The chapter is structured to closely follow the modelling workflow. First, the precipitation and air temperature datasets used as model inputs are introduced, focusing on the numerical methods employed to generate them and discussing some preliminary results, as they may significantly influence the spatial and temporal distribution of snow cover. The precipitation phase

partitioning scheme and the approach used to estimate snowmelt are then described. The chapter closes with the description of the procedure used to generate the SWE maps.

- *Chapter 5*

Chapter 2

Snow basics

Snow is a fundamental component of the cryosphere, which encompasses all regions of the Earth where water exists in solid form. Alongside sea ice, glaciers, ice caps and ice sheets, snow cover plays an essential role in regulating Earth's climate by reflecting a significant fraction of the incoming solar radiation, consequently reducing the amount of energy absorbed by the ground. In addition, snow also acts as a thermal insulator due to its low thermal conductivity, protecting the soil and the ecosystem from the abrupt temperature fluctuations typical of winter. Snow is also a key factor in the hydrological cycle: water stored in solid form during winter is gradually released into streamflow through spring and summer, ensuring a continuous water supply and mitigating the severity of certain droughts. At the same time, snow has considerable implications for human activities. For instance, it plays a vital role in the economy of the Alpine region by supporting winter tourism through skiing and other winter sports. Its contribution to runoff is equally important, as it supports irrigation and provides a source of drinking water. Meltwater also provides a substantial contribution to hydropower generation. With the ongoing climate change, these dynamics are expected to shift, making the understanding and modelling of snow related processes increasingly important. Earlier snowmelt and reduced solid water storage are among the most direct consequences of a temperature rise, with potential repercussions for multiple aspects of human civilization. The aim of this chapter is therefore to provide a broad overview of snow dynamics, before focusing on snow measurement techniques, which are essential to assess the state of the snowpack.

2.1 Snow physics and dynamics

The formation of snow is the result of a chain of microphysical processes occurring within clouds, from initial ice nucleation to the eventual fall of snowflakes to the ground.

For the atmosphere to release its water content, whether in liquid or solid form, the air must first reach a state of supersaturation with respect to water or ice. This condition is typically achieved through upward motion of air masses which causes

adiabatic cooling of air parcels. The temperature drop strongly reduces air's capacity to hold water vapor, driving the parcel toward, and eventually beyond, saturation (Wallace and Hobbs 2006).

Snowfall can only be produced by clouds that extend above the 0 °C level. Although temperatures within these cold clouds can drop well below freezing point, supercooled water droplets can still coexist with ice particles. The freezing process, known as ice nucleation, occurs through two different mechanisms:

- *homogeneous nucleation*, which occurs in pure water droplets without foreign particles. The first step in the freezing process is the formation of an ice embryo, which must exceed a critical size to produce a decrease in the free energy of the system. If this condition is met, the freezing process continues until the entire water droplet is turned into ice. The average size of the ice embryo increases as temperature decreases, so that below a certain threshold, which is around −40 °C, freezing becomes almost certain. If homogeneous nucleation were the only mechanism responsible for ice particle formation, snowfall would be extremely rare at mid-latitudes. Since this is not the case, an additional mechanism is necessary.

- *heterogeneous nucleation*, which relies on the presence of a foreign particle. Cloud droplets typically contain such impurities acting as freezing nuclei. Their presence promotes the formation of an ice-like structure, thereby enabling freezing at temperatures considerably higher than those required for homogeneous nucleation. It is also possible for foreign particles not to be part of the water droplet itself, but instead to trigger freezing through contact or deposition. Regardless of the specific mechanism involved, the most effective ice nuclei are insoluble in water and have a crystallographic structure similar to ice.

Once nucleated, the ice particle grows through further microphysical processes, whose relative dominance determines its final shape and size. The three main crystal growth mechanisms are:

- *vapor deposition*, which occurs whenever the air is supersaturated with respect to ice. For example, in mixed-phase clouds dominated by supercooled droplets, vapor molecules are deposited onto the ice crystals, causing them to grow significantly faster than the surrounding water droplets. The growth rate reaches a maximum around −14 °C, since at that temperature the difference between the saturated vapor pressure over water and over ice is largest, thus driving vapor molecules more effectively towards ice crystals. Studies on the growth of ice crystals by deposition have shown a strong relationship between

crystal shape and the temperature at which the growth process takes place. This strong temperature dependence produces an alternating pattern between plate-like and column-like ice particles, as shown in Figure 2.1.

- *riming*, which occurs exclusively in mixed-phase clouds. Whenever an ice crystal collides with a supercooled droplet, the latter may freeze onto the former, resulting in an increase in ice mass. The collision frequency depends on cloud density and composition, but typically numerous riming events take place, sometimes producing a graupel, namely an ice particle in which the original ice crystal can no longer be identified. Riming can also lead to the formation of hailstones, which typically range from a few millimeters to several centimetres in diameter.

- *aggregation*, caused by collisions between ice particles. The probability of adhesion is strongly linked to crystal shapes and increases with increasing temperature, reaching a maximum at around $-5\,^{\circ}\mathrm{C}$, where ice surfaces become particularly "sticky".

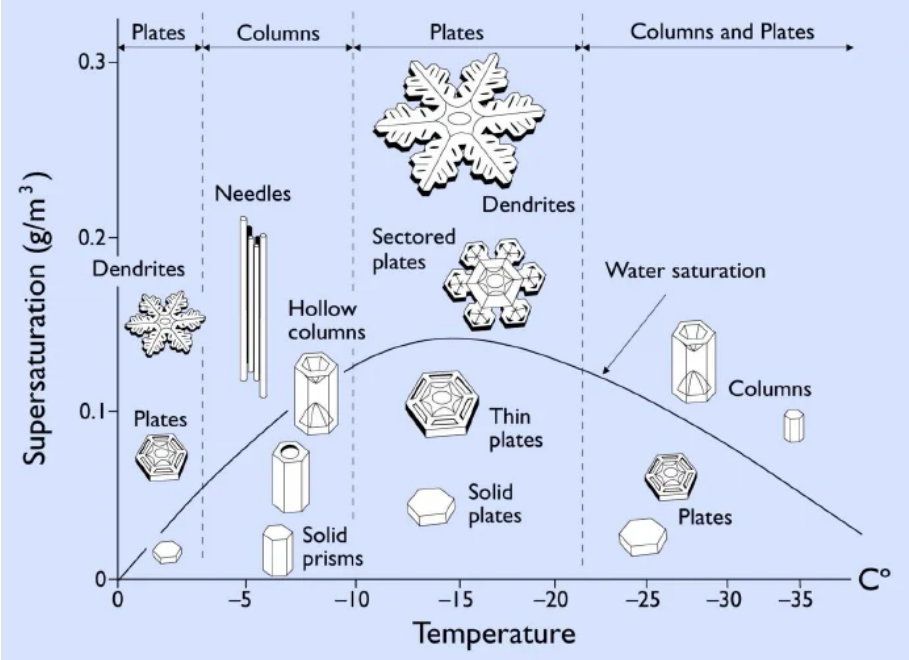

Figure 2.1: The Nakaya diagram illustrates the morphology of ice crystals growing via vapor deposition as a function of cloud's thermodynamic state. Crystal shape is primarily governed by temperature, which determines whether it develops in a planar or columnar structure, while the degree of supersaturation with respect to ice modulates the complexity of the resulting form, with higher supersaturations generally linked to more complex structures. Image taken from Libbrecht 2012.

Once an ice particle grows large enough to overcome updraft forces and turbulent motion within the cloud, it begins to fall toward the ground. The temperature profile encountered by the descending particle ultimately determines the type of precipitation that reaches the surface. For snowfall to occur, the snowflake, composed of up to hundreds of ice crystals, must remain predominantly below freezing and avoid melting completely before reaching the ground. Typically, snowflakes have fall rates of approximately $1.5\,\mathrm{m\,s^{-1}}$. Snowfall amounts are measured either by depth or by snow water equivalent (SWE), defined as the height of the water column resulting from a complete melting of the snow. Snow depth typically increases of approximately $1\,\mathrm{cm\,h^{-1}}$ during snowfall events. Newly fallen snow density ranges from $20\,\mathrm{kg\,m^{-3}}$ to $300\,\mathrm{kg\,m^{-3}}$ depending on wind patterns and weather conditions, with higher temperatures generally leading to higher density values (Armstrong and Brun 2008).

2.1.1 Snow metamorphism and compaction

Accumulation of snowflakes on bare ground or atop older snow results in the formation of a new snow layer. Snow crystals form a solid matrix delimiting pores filled with humid air or liquid water or both, depending on layer temperature. This structure can be clearly appreciated in Figure 2.2, where the coexistence of the solid, liquid and gaseous phases of water is visible. Snow is thus a porous medium, whose properties are closely linked to its texture.

The arrangement of snow crystals evolves over time as a result of thermodynamic interactions among the constituents of the snow layer. This process, known as *snow metamorphism*, is strongly influenced by thermal and meteorological conditions. Since each layer within the snowpack is subject to different burial depths, temperature gradients and moisture presence, it follows a distinct metamorphic pathway. Consequently, the snowpack develops a stratified structure, with individual layers exhibiting different crystal characteristics.

A key factor governing the evolution of ice particles is the presence or absence of liquid water, which motivates the distinction between dry and wet snow metamorphism. The former occurs when the pores are filled only by air, typically saturated with water vapor. The main mechanisms underlying dry snow metamorphism are:

- *equilibrium growth*, driven by crystal shape. Local variations in surface curvature generate corresponding variations in equilibrium vapor pressure, inducing local pressure gradients that drive vapor from convex toward concave (or less convex) regions. Over time this transfer, sustained by sublimation on convex areas and

deposition on concave ones, rounds ice grains, driving snow crystals toward a spherical shape.

- *kinetic growth*, related to the presence of a temperature gradient. Because saturation vapor pressure depends strongly on temperature, such a gradient causes vapor to diffuse from the warmest crystals toward the coldest ones. Here too, sublimation on warm crystals and deposition on cold ones sustain the process, which, when the temperature gradient is strong enough, leads to the formation on facets.

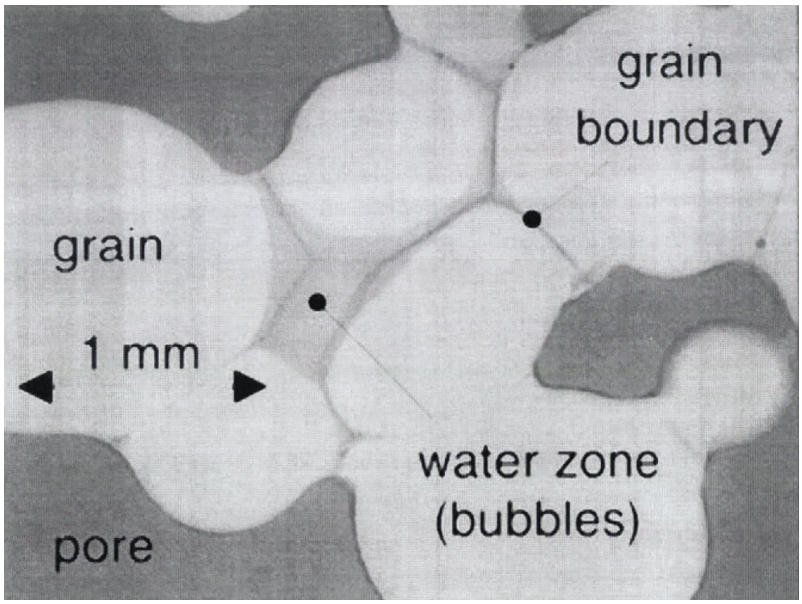

Figure 2.2: Thin section of wet snow under transmitted light, where ice grains delimit pore spaces containing both liquid water and air. The presence of air-filled pores suggest that the sample can potentially retain additional water. Image from Brzoska et al. 1998

These two mechanisms are rarely mutually exclusive: they act simultaneously within the snowpack, with their relative strengths determined mainly by the magnitude of the local temperature gradient compared to the grain's curvature effect. In particular, the kinetic growth of snow particles is a matter of considerable concern in avalanche forecasting, since faceted crystals, if the temperature gradient persists over time, can evolve in depth hoar. This type of layer is notoriously weak and prone to failure, often resulting in large-scale avalanches.

When liquid water is present within the snowpack, the dynamics change substantially: mass exchanges between the three water phases become essential to understand wet snow metamorphism. The main driver of ice particles evolution is, once again, crystal surface curvature, with convex areas associated with lower melting point

temperatures. This topological property leads to the melting of the most convex surfaces and to refreezing on the more concave ones, driving ice particles towards the equilibrium spherical shape. The efficiency of this metamorphism channel is linked to the freedom water has to move within the snowpack: at very low water content mass exchanges between the solid and gaseous phases may still dominate.

Regardless of the specific metamorphic process involved, the evolution of ice crystals ultimately results in the compaction of the snow layer. Wind also plays a crucial role in increasing snow density, primarily through the fragmentation of snowflakes upon collision, which allows for a tighter packing of ice particles. Finally, gravity contributes to snow compaction as well, since buried snow layers must withstand an increasing load exerted by the weight of the overlying layers, resulting in higher deformation. The combined effects of metamorphism and deformation strain are typically described through a Newtonian viscosity approach. Snow viscosity depends on several parameters, including temperature, liquid water content and porosity and its values are generally retrieved by means of empirical relations. Overall, the density of older snow layers can easily exceed values of $400\,\mathrm{kg\,m^{-3}}$, sometimes even reaching $500\,\mathrm{kg\,m^{-3}}$ in specific locations, depending on water content and time of the year (Jonas et al. 2009).

2.1.2 Snowpack energy balance

The evolution of a snowpack throughout the season is governed not only by the microphysical processes described above, but also by the exchange of energy between the snow cover, the atmosphere and the underlying ground. Neglecting horizontal energy transfers, the internal energy of the snowpack evolves in time as:

$$\frac{d\mathcal{H}}{dt} = R_N + H_S + H_L + H_P + G, \tag{2.1}$$

where $\mathcal{H}$ is the internal energy per unit area and R_N is the net radiative flux at the surface. H_S and H_L denote, respectively, the turbulent fluxes of sensible and latent heat, governed by the action of turbulent eddies within the boundary layer. H_P accounts for the energy exchange associated with precipitation, since rainfall carries additional heat that must be distributed within the snowpack. Finally, G represents the energy exchange with the ground, which is especially important during the deposition of the first snow layer. Each term in Equation (2.1) is expressed in $\mathrm{W\,m^{-2}}$ and, by convention, a positive sign indicates an energy flux directed into the snow layer. A schematic representation of the energy balance is provided in Figure 2.3: radiative contributions are discussed in more detail below.

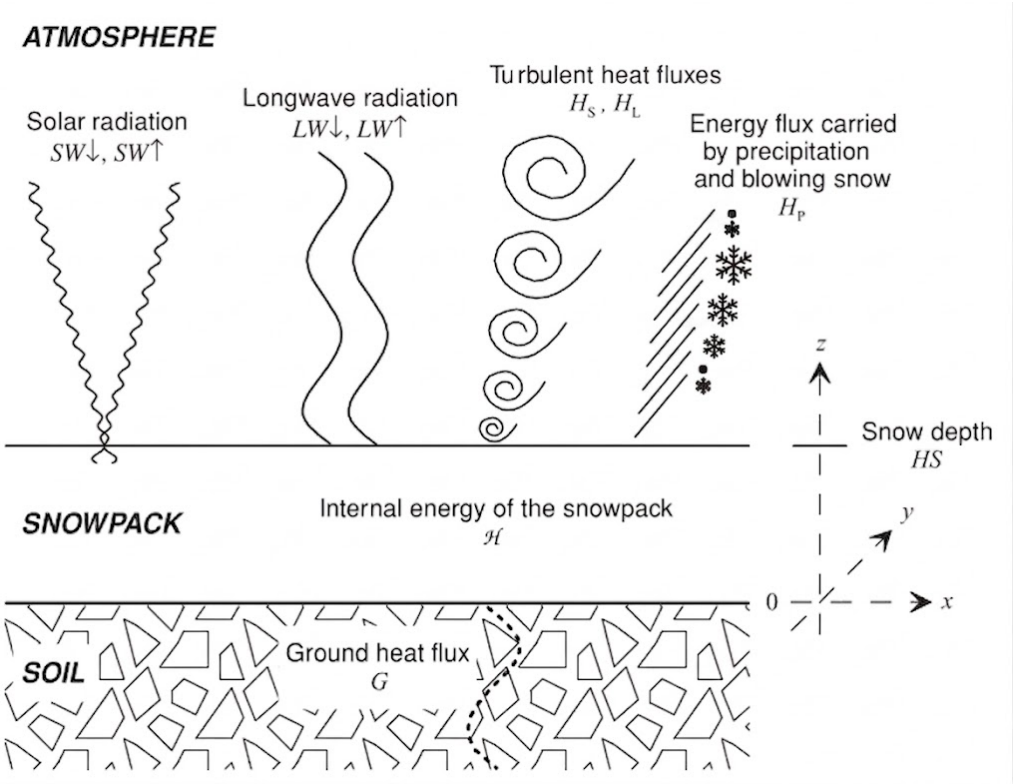

Figure 2.3: Schematic representation of the energy balance of a snowpack. Shortwave (SW) and longwave (LW) radiation, originating from the Sun and from thermal emission respectively, make up the net radiative budget R_N. The internal energy of the snowpack $\mathcal{H}$ can also be altered by the turbulent heat fluxes H_S and H_L, the precipitation heat flux H_P and the ground heat flux G. Image adapted from Armstrong and Brun 2008.

To fully understand the term R_N, it is essential to consider the dual nature of radiation in the atmosphere. Two distinct sources of radiation can in fact be identified: the Sun, with a surface temperature of around 5800 K, and the Earth-atmosphere system, whose temperature ranges from 200 K up to 350 K. According to the Wien's law, the two blackbody spectra peak respectively at roughly 0.5 µm and 10 µm with negligible overlap, thereby justifying a decoupled treatment of the two spectral domains. Solar radiation is commonly referred to as *shortwave* (SW), while the latter is known as *longwave* (LW). Therefore, the term R_N can be explicitly written as

$$R_N = SW\downarrow - SW\uparrow + LW\downarrow - LW\uparrow, \tag{2.2}$$

where a distinction is made between incoming and outgoing fluxes, as denoted by the direction of the arrows.

Incoming shortwave radiation is subject to absorption and scattering by atmospheric constituents. Consequently, the total SW radiation that reaches the Earth's surface is the sum of a direct beam transmitted through the atmosphere and a

diffuse component, resulting from multiple reflection and refraction events. The relative strengths of these two components is strongly dependent on meteorological and seasonal conditions. For example, at the lower solar altitudes typical of winter, the diffuse fraction is larger, since the solar beam must travel through a thicker atmospheric layer, increasing scattering probability. The same occurs at dawn or sunset, when the Sun is close to the horizon. Cloud cover also plays a significant role: under overcast conditions, up to 80% of $SW\downarrow$ can occur as diffuse radiation (DeWalle and Rango 2008). In the shortwave spectral range, snow features an exceptionally high reflectance, thanks to the combination of its microstructure with the optical properties of ice. The ratio between $SW\uparrow$ and $SW\downarrow$, commonly referred to as albedo α, typically exceeds 0.9 for freshly fallen snow, and decreases as impurities are deposited on the snow surface. This parameter is rather important, as it controls the amount of shortwave radiation absorbed, which generally represents the main source of energy for snowmelt. By albedo definition, this quantity can be calculated as

$$SW_{net} = SW\downarrow(1 - \alpha)\,. \tag{2.3}$$

Even at advanced stages of the melt season, snow albedo remains considerably higher than that of the snow-free surroundings. As the snowpack retreats, this bright surface is replaced by darker ground, causing a significant α drop. This results in increased SW absorption, which causes further warming and, in turn, accelerates snowmelt itself in a self-reinforcing loop known as snow-albedo feedback.

Snow behaviour in the thermal infrared range, however, differs substantially from that in the shortwave domain. The snowpack absorbs nearly all incoming longwave radiation, emitted by atmospheric gases, clouds and the surrounding terrain, while radiating according to the Stefan-Boltzmann law and thus behaving as an almost perfect blackbody, with emissivities ranging from 0.97 to 0.99. Longwave radiation exchanges occur both during the day and at night, and are strongly influenced by meteorological conditions. In fact, water vapor significantly absorbs infrared light and, by Kirchhoff's law, is therefore also a strong emitter. As a result, clouds typically increase the downward longwave flux reaching the snow surface, thereby reducing the snowpack's radiative cooling capacity. This effect is especially pronounced at night, when the absence of the shortwave component would otherwise favor significant longwave losses.

Thanks to its relatively low thermal conductivity, the snowpack also shields the ground from rapid atmospheric temperature changes. This property is closely tied to snow microstructure and therefore evolves with metamorphism, as higher snow densities are associated with increased conductivity.

2.1.3 Snow melt and water runoff

The variations in internal energy $\mathcal{H}$, driven by the heat fluxes listed above, ultimately modify the thermal state of the snowpack and determine whether mass is lost through melting or sublimation. The snowpack temperature profile varies on both seasonal and daily scales, typically showing large gradients near the snow surface as a consequence of the low thermal conductivity of snow. The temperature oscillations of the first layers are damped rather quickly with depth, as the thermal state of the deeper bulk remains relatively stable. During winter, snowpacks typically exhibit subfreezing conditions, which poses a significant limitation to runoff generation. In fact, even when the surface temperature rises to 0 °C and the energy balance is positive, the water produced by surface melting refreezes in deeper layers after some percolation. Although this process does not involve any mass loss, it effectively transports latent heat, warming the snowpack and reducing its cold content, which is the energy required to bring the whole snowpack to 0 °C. This isothermal profile is eventually reached and, if a positive energy balance still persists, further melting contributes to filling the pores between ice crystals. This liquid water holding capacity is commonly reported ranging between 3% and 5% of the snowpack mass, although the exact value depends on the snow's porosity and microstructure. Once the water holding capacity is exceeded, the excess water percolates through the snowpack in a process that is mainly gravity driven. Because a typical snowpack is stratified into layers of contrasting grain size and permeability, water flow is rarely uniform as it tends to concentrate along preferential pathways known as flow finger, which are usually triggered by inhomogeneities in microstructure. Typical percolation velocities span more than an order of magnitude, ranging from $1\,\mathrm{cm\,h^{-1}}$ up to $20\,\mathrm{cm\,h^{-1}}$. Ground conditions represent another critical factor in runoff evaluations, since meltwater can in principle infiltrate the soil. If the underlying surface is impermeable, such as frozen ground or bedrock, the excess water instead accumulates and flows downward along the base of the snow cover. If the snowpack is not isothermal throughout, part of this water may still refreeze, further warming the surroundings. The interplay between thermal conditioning, liquid water retention, and preferential flow pathways ultimately governs the timing, volume and rate of snowmelt runoff entering local stream networks.

In high elevation alpine catchments, or at high latitudes, some snow may survive melting through summer. If accumulation persists over multiple years, continued densification ultimately leads to the formation of firn, an intermediate stage between snow and ice characterized by densities ranging from $400\,\mathrm{kg\,m^{-3}}$ to $830\,\mathrm{kg\,m^{-3}}$ and by a porous structure that still permits water percolation. As compaction proceeds

further, most pores become sealed, marking the transition to glacier ice proper. The rate of this transformation depends on both accumulation amounts and temperature, since colder climates results in slower crystal evolution. This process directly links seasonal snow cover to the longer lived components of the cryosphere, such as glaciers and ice sheets (Cuffey and Paterson 2010).

2.2 Snow observation and measurement

A detailed knowledge of the snow cover and of its physical properties is a prerequisite for studying snowpack dynamics and for issuing reliable avalanche forecasts. Among the quantities routinely collected during snow measurement campaigns, snow depth (HS) plays a central role: it is defined as the vertical distance between the top of the snowpack and a reference surface, which for seasonal snowpacks is normally the ground. HS should not be confused with snow thickness (DS), which is conventionally the perpendicular distance to the same reference surface. Provided that the slope angle ϕ of the measurement site is known, the two quantities are related by

$$\mathrm{DS} \;=\; \mathrm{HS}\cos\left(\phi\right), \tag{2.4}$$

so that the two definitions coincide on flat terrain ($\phi = 0$) and progressively diverge as the slope steepens. The geometrical meaning of Equation (2.4) is illustrated in Figure 2.4. Both quantities are expressed in cm and are usually rounded to the nearest integer. While these metrics describe only the geometrical extent of the snowpack, other quantities provide deeper insight into its physical state. The most hydrologically relevant of these is snow water equivalent (SWE), namely the amount of water stored within the snowpack. Being closely related to snow density, it can be expressed as

$$\mathrm{SWE} \;=\; \int_0^{\mathrm{HS}} \rho_s\left(z\right) dz, \tag{2.5}$$

where $\rho_s\left(z\right)$ is the vertical density profile of the snowpack. The result is given in $\mathrm{kg\,m^{-2}}$, and can be converted in m (or mm) by dividing by the water density ρ_w. Although formally correct, Equation (2.5) is of limited practical use, since continuous knowledge of the density profile is generally unattainable. What can realistically be measured is the density of the individual layers, so that SWE is more commonly computed as

$$\mathrm{SWE} \;=\; \sum_{i=1}^{N} \rho_i\left(z\right) \Delta z_i, \tag{2.6}$$

where N is the number of layers and Δz_i is the vertical extent of the i-th layer.

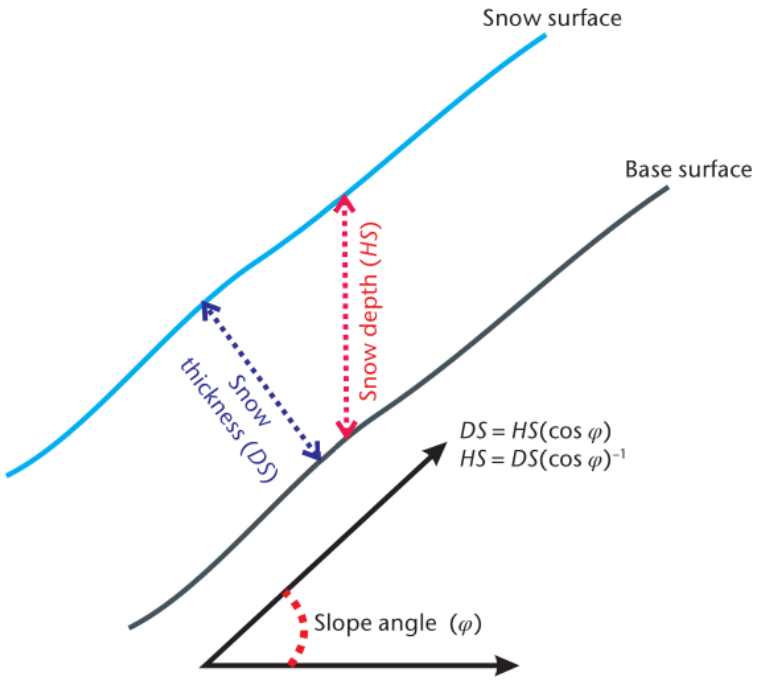

Figure 2.4: Relationship between snow depth (HS) and snow thickness (DS) on a slope. Image adapted from WMO 2024.

Historically, snow metrics were recorded manually, requiring an operator to carry out manual measurements on a daily basis whenever feasible. To ensure continuous data collection even in remote areas, automated weather stations (AWS) have been introduced and are progressively replacing manual observations as well. An automated monitoring network provides a comprehensive overview of snow conditions, but achieving high spatial resolution remains challenging due to high installation costs and maintenance issues. Consequently, certain data can only be obtained through remote sensing, by means of Earth-observing satellites.

While remote sensing provides spatially distributed data across entire regions, manual and automated measurements are taken at specific locations, which must be selected according to strict criteria. To ensure the reliability of these in situ observations, the monitoring site must match the features of the surrounding terrain. In alpine regions, monitoring sites must be free of large boulders and protected from excessive sun exposure, which can cause premature snowmelt and thus yield unrepresentative data. It is equally important to avoid locations prone to preferential snow accumulation driven by local wind patterns (WMO 2024).

2.2.1 Manual snow measurements

Manual snow depth (HS) records are an effective tool for assessing long-term trends, as they can span several decades. Their longevity is due to a straightforward measuring procedure, requiring only a stake or an extendible graduated rod. The snow stake should be vertical, painted white to minimize melt and fixed to the ground before accumulation begins. Snow depth observations should be made from a sufficient distance to avoid any deformation of the snowpack. Because the stake acts as an obstacle to wind and snow redistribution, localized mounds can form around it. Under such circumstances, operator must visually estimate the undisturbed snow surface to recover the actual HS value. For a non-fixed observation carried out with a graduated rod, the snow depth should be evaluated at the same location each time. The observer needs to ensure that the device, manually inserted into the snowpack, reaches the base surface without penetrating the soil, which would result in an overestimate of the actual HS. The rod must be vertical, and an estimate of the slope angle should be paired with every measurement.

Closely linked to snow depth is snowfall depth (HN), which can only be obtained manually. To avoid wind-induced bias this measurement should be carried out in a sheltered location, so that snow accumulates undisturbed on an artificial surface for the prescribed interval (usually 24 hours). Just before clearing the surface, thereby starting the next observation period, a ruler is inserted vertically into the freshly formed layer to obtain a depth measurement.

Although HS and HN define the vertical extension of the snowpack, hydrological applications rely mainly on the snow water equivalent (SWE): the two most common methods for manual measurement are snow pits and snow tubes. A snow pit consists of a hole manually dug into the snowpack down to a reference surface, usually the ground. Initially, the observer performs a stratigraphic analysis to gain deeper insight into the internal structure of the snowpack. The actual SWE measurement is conducted by inserting a sampler of known volume into the exposed snow face; the sample is then weighed to assess snow density. This procedure is usually repeated at multiple depths, either for each identified layer or at fixed spatial intervals, such as every 20 centimetres. This method enables the observer to reconstruct the snow density profile which, when combined with depth information, yields both the layer-by-layer and total snow water equivalent at that specific location. Clearly, the snow pit technique is disruptive and needs to be performed at a different site each time.

If the measurement campaign is focused on a spatial survey rather than a detailed, locally defined inspection, the logical choice is to use a snow tube, as it allows for rapid sampling of the entire snow column to directly retrieve the total SWE, sacrificing

layer-specific information. The snow tube must be inserted vertically into the snowpack until it reaches the ground. An estimate of the extracted core volume is made, and the average density is then computed by weighing the snow column. The conversion to SWE is then straightforward. This device should be used where the snow water equivalent is less spatially homogeneous, such at locations with heterogeneous exposure to both wind patterns and solar radiation.

2.2.2 Automatic snow monitoring

Networks of automatic weather station (AWS) require no manual intervention beyond routine maintenance and instrument calibration, and can provide continuous observations even in remote locations where human operation would be impractical. An AWS is typically equipped with a wide range of sensors, such as:

- *anemometer*, which measures wind speed. The blowing wind causes a small windmill, or a structure consisting of three cups mounted on a spindle, to rotate at a rate proportional to wind intensity. This quantity is particularly important, as empirical relations link it to turbulent heat fluxes, thus playing a significant role in energy balance considerations.

- *wind wane*, typically coupled with the anemometer to assess wind direction. It consists of a horizontal arm bearing a flat vertical vane at one end. Incoming wind gusts push against the vain, causing the arm to rotate freely until it aligns with the wind direction.

- *pyranometer*, which measures shortwave solar radiation. An AWS is typically equipped with both and upward and a downward facing pyranometer, in order to quantity incoming and outgoing SW radiation, thus enabling albedo estimates.

- *pyrgeometer*, which measures the longwave (infrared) radiation. Pyrgeometers likewise come in oppositely facing pairs, so that, combined with a pair of pyranometers, an evaluation of the net radiation balance of the surface is possible.

- *barometer*, used to evaluate the atmospheric pressure, a fundamental quantity for characterizing the dynamical state of the atmosphere.

- *thermo-hygrometer*, designed to simultaneously measure both air temperature and relative humidity at a given height above the ground (typically two meters).

- *rain gauge*, consisting of a circular collector and a funnel that channels the collected precipitation into a measuring device. In regions where solid precipitation is possible, the rain gauge may be heated in order to also evaluate the water equivalent of snowfall.

In alpine catchments and at high latitudes, these sensors are typically complemented by additional instruments specifically dedicated to snow monitoring. Mounted together on the same tripod or mounting structure, as shown in Figure 2.5, they constitute what is referred to as an alpine automatic weather station, a key tool for the study of snow related dynamics.

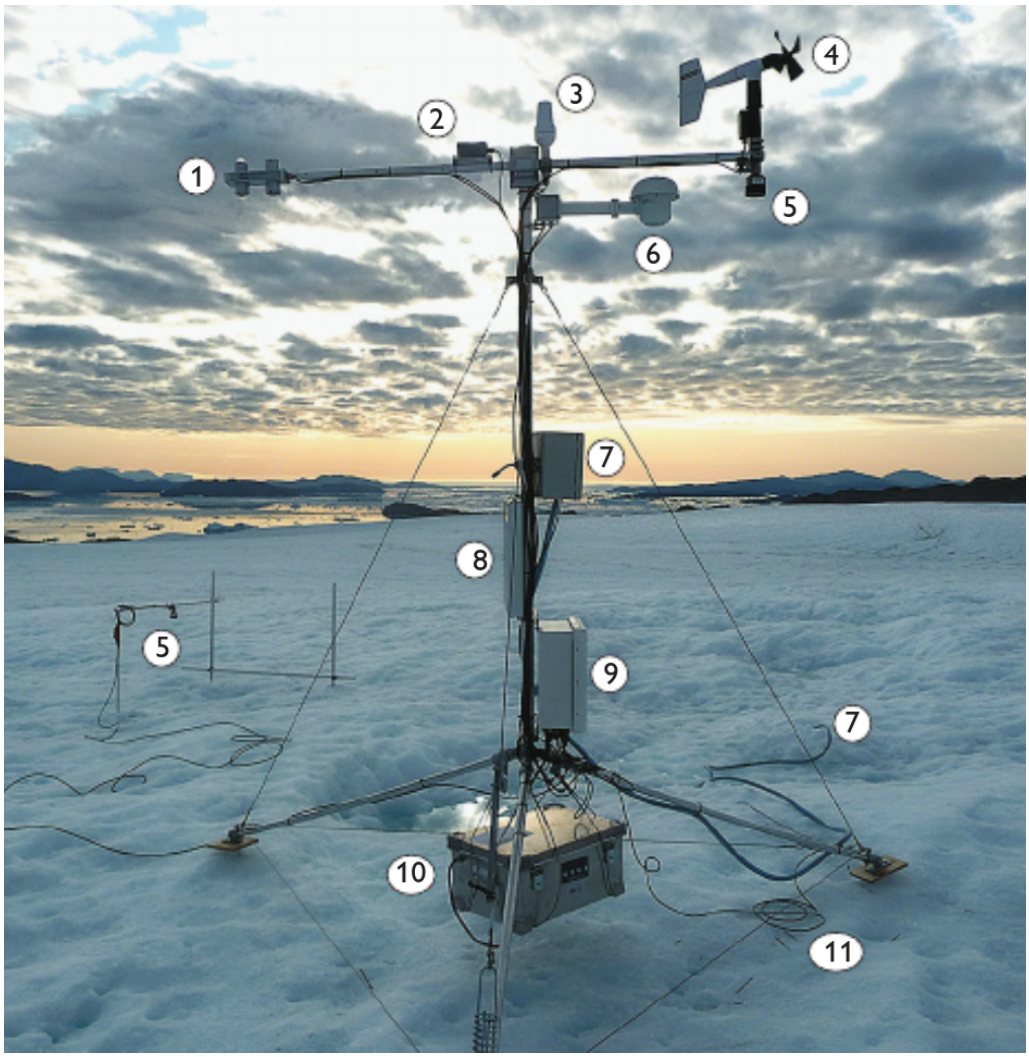

Figure 2.5: The AWS UPE_L which is part of an automated network of the Programme for Monitoring of the Greenland Ice Sheet (PROMICE). The constituents are respectively: (1) radiometer LW + SW, (2) inclinometer, (3) satellite antenna, (4) anemometer and wind vane, (5) snow depth sensor, (6) thermometer and hygrometer, (7) pressure transducer, (8) solar panel, (9) data logger and GPS, (10) battery box and (11) ground temperature sensor. Image adapted from Fausto and As 2012.

Automated HS measurements can be performed using instruments based on either sonic or optical (laser) technology. Sonic sensors transmit an ultrasonic pulse towards the snow surface and detect the returning echo. Since the speed of sound depends on meteorological conditions, a concurrent temperature reading is required to ensure accuracy. Laser instruments, by contrast, determine the distance to the target surface through interferometric analysis of the returned signal. This approach yields significantly higher precision, but at a grater energy cost.

In both cases, the sensor measures the distance to the upper surface of the snowpack, from which the snow depth is then derived as

$$\text{HS} = \text{Zero depth distance} - \text{Distance to target.} \tag{2.7}$$

The zero depth distance must be determined before snow accumulates and constitutes a key step in instrument calibration. To ensure reliable measurements, the sensor should be positioned well above the maximum expected snow depth, preventing burial and the consequent loss of data during winter. Minimizing interference from the support structure is equally important, since the mast itself acts as a wind obstacle and may induce measurement artefacts. For this reason, HS sensors are typically mounted on a horizontal arm stretching outward from the pole, keeping the target area sufficiently far from the mounting structure.

SWE monitoring can likewise be automated, with the most common technique relying on snow pillows. This device consists of a rubber bladder filled with an antifreeze liquid mixture and connected to a pressure transducer, which yields the SWE estimate. The pillow is installed flush with the ground surface, so that it supports the full weight of the overlying snowpack. As snow accumulates, the resulting increase in hydrostatic pressure is transduced into snow water equivalent values. Snow pillows were first introduced in the late 1960s, and have proved to be a valuable tool for SWE measurements (Beaumont 1965). However, they are now being progressively replaced by snow scales due to the toxicity of the antifreeze fluid, which poses an environmental hazard to the installation site.

2.2.3 Remote sensing

Satellite remote sensing (SRS) serves as a useful tool to fill critical gaps in current knowledge regarding climatic variables, enabling researchers to monitor, usually on a daily basis, even remote locations that would otherwise be inaccessible to a human operator. Furthermore, SRS data substantially improve meteorological reanalysis products, which constitute a cornerstone of contemporary climate change research.

Earth observing satellites can be classified into two distinct categories based on their orbital characteristics:

- Polar orbit satellites, which travel roughly from north to south over the poles, targeting a specific longitudinal strip per pass. The entire globe is thus mapped over consecutive orbits. These satellites typically operate at an altitude of approximately 700 km, completing a full revolution in under two hours.
- Geostationary satellites, which occupy a geosynchronous orbit, thereby allowing for real-time meteorological monitoring on a specific target area. Their altitude of approximately 36000 km above Earth's equator results in a significantly lower spatial resolution than that of polar orbit satellites.

The choice of orbit is strictly driven by observational priorities. Consequently, satellites dedicated to cryospheric research favor polar orbit, as maximizing spatial resolution is essential to capture the highly variable distribution of snow coverage across complex terrains.

Satellite-based observations of snow cover extent (SCE) began in 1967, through the efforts of National Oceanic and Atmospheric Administration (NOAA). This monitoring program focused on the Northern Hemisphere, where most of the world's terrestrial snow cover is found, with observations conducted on a weekly basis. Despite the preliminary nature of this work, it highlighted the strong latitudinal dependence of snow cover duration and paved the way for more comprehensive analyses. One of the main limitations of these early observations was their coarse spatial resolution of $\sim 200\,\text{km}$, which was insufficient to adequately characterize snow cover in many mountainous regions (Bormann et al. 2018). A significant advance in sensor technology came with the introduction of the Advanced Very High Resolution Radiometer (AVHRR), which measures Earth radiance in five spectral bands to derive information on temperature, albedo and cloud coverage distribution. The first tests of a spaceborne sensor took place in 1978, but reliable snow cover data have only been available since 1981. This product has been used to estimate snow cover duration and trends since the mid-eighties in various mountainous regions such as the Alps, where SRS data reveal significantly negative trends throughout the 1990s, followed by a partial recovery at the beginning of the new millennium (Hüsler et al. 2014). Although AVHRR offers a substantial improvement in spatial resolution, with pixels of $\sim 1\,\text{km}$, its data reliability is limited by the presence of only one thermal channel for snow-cloud discrimination (Fugazza et al. 2021). To overcome these limitations, the Moderate Resolution Imaging Spectroradiometer (MODIS) was launched in 1999 aboard the Terra satellite, followed by a second unit aboard

Aqua in 2002. Both instruments operate in polar orbit, jointly covering the entire globe every one to two days. MODIS is a 36-band spectral sensor spanning from the visible (0.41 µm) to the long-wave infrared (14.4 µm), with a spatial resolution of 500 m. To detect the presence of snow on the ground, the Normalized Difference Snow Index (NDSI) is used, defined as

$$\mathrm{NDSI} = \frac{\mathrm{G} - \mathrm{SWIR}}{\mathrm{G} + \mathrm{SWIR}}, \tag{2.8}$$

where G and SWIR denote the green and shortwave infrared spectral bands respectively. This normalized difference approaches one for a snow-covered pixel, since snow exhibits very high reflectance in the visible portion of the electromagnetic spectrum, while strongly absorbing radiation in the SWIR. An NDSI value greater than zero indicates the presence of snow in the pixel; in practice, a threshold of 0.4, corresponding to roughly 50 % snow coverage, is commonly adopted to classify a pixel as snow covered (Salomonson and Appel 2006).

Although MODIS has proven to be a robust and well-validated tool for characterizing snow cover, the information it provides remains partial, as it is limited to detecting the presence or absence of snow on the ground. In order to assess how much water is stored in solid form, three main spaceborne techniques have been developed:

- *passive microwave radiometry (PMW)*, which is by far the most common method to remotely evaluate SWE. Since the snowpack attenuates the microwaves emitted by the Earth, comparing signal intensities between two frequencies (~ 18 GHz, and ~ 37 GHz) makes it possible to estimate snow water equivalent. This is the only spaceborne technique that allows for long term studies, as global daily maps have been available since the launch, in 1978, of NASA's Scanning Multichannel Microwave Radiometer (SMMR) (Schilling et al. 2024). Altough the temporal coverage of these series is considerably better than that of other available sources, several factors affect data reliability, such as signal saturation caused by a deep snowpack.

- *gravimetric methods*, which offer a different approach focusing on mass changes. The accumulation of snow on the Earth's surface causes variations in its gravitational field, which can be converted into SWE estimates. This approach is mainly employed by the Gravity Recovery and Climate Experiment (GRACE) mission. Owing to its global coverage, this SWE dataset features a coarse spatial resolution of roughly 300 km, which nonetheless remains useful for continental scale hydrological studies (Schilling et al. 2024).

- *Synthetic Aperture Radar (SAR)*, which consists in the transmission of electromagnetic waves towards Earth and the analyses of the backscattered echoes from the Earth's surface. SAR can penetrate cloud cover and acquire data under all weather conditions. Snowpack properties, such as SWE and snow grain size can be estimated remotely, as longer SAR wavelengths are even capable to penetrate into the snowpack. Furthermore, the phase information recorded by SAR can be exploited to monitor ground deformation and assess the stability of terrain features. Although SAR provides a spatial resolution on the order of tens of metres, its temporal resolution is significantly lower than that of PMW and gravimetric methods, as the revisit time of a specific location is typically five days (Tsai et al. 2019).

2.3 Climate change

Climate change is conventionally defined as an alteration of the climate that can be attributed, directly or indirectly, to human activities that modify the composition of the global atmosphere. This phenomenon causes shifts in the mean and/or in the variability of temperatures and weather patterns, which can be identified through statistical tests and persist over an extended period of time, generally of at least thirty years. Since the anthropogenic signal is superimposed on the natural variability of the climate system, it must be disentangled from a background of spontaneous fluctuations. Over the past few million years, such natural fluctuations have produced a cyclic alternation between glacial periods, lasting roughly 100 000 years, and warmer interglacial conditions. These climatic oscillations were primarily driven by periodic variations in the orbital parameters governing Earth's revolution around the Sun, which caused slight changes in the amount of solar radiation reaching the surface. The effect of this weak radiative forcing on the planetary energy budget was amplified by snow albedo feedbacks, producing temperature changes large enough to modify the extent of the cryosphere on a global scale (Hays et al. 1976).

The ongoing global warming, however, cannot be attributed to natural variability, as it is primarily driven by the large amounts of greenhouse gases released into the atmosphere by human activities. These gaseous compounds, of which carbon dioxide (CO_2) is the best known, partially absorb the longwave radiation emitted by Earth's surface and re-emit it in all directions. A fraction of this energy is thus returned to the surface instead of escaping to space, resulting in an increase in near-surface air temperature. Since the Industrial Revolution the atmospheric CO_2 concentration has risen steadily, from about 278 part per million (ppm) in pre-industrial times to

422.8 ppm in 2024 (NOAA 2025). During the last 60 years, atmospheric CO_2 has risen nearly 100 times faster than during previous natural increases, and present day concentrations exceed those preserved in the ice core records of the last 800000 years, as shown in Figure 2.6.

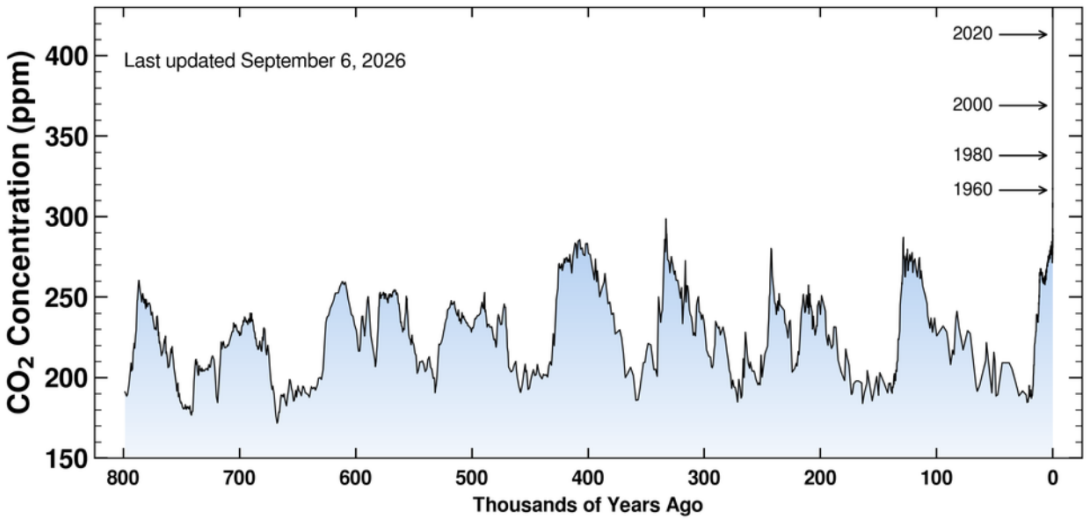

Figure 2.6: Evolution of CO_2 concentration in the atmosphere over the last 800000 years. Image adapted from Oceanography 2026.

Relative to the 1850–1900 climatology, the global mean surface air temperature observed over the 2010–2019 period was about 1.1 °C higher, and anthropogenic forcing alone is estimated to account for 0.9–1.3 °C of this increase, with the best estimate being 1.07 °C (Gillett et al. 2021). The warming has also accelerated significantly in recent years, as the mean increase rate of roughly 0.18 °C per decade typical of the 1970–2010 period has nearly doubled between 2010 and 2023 (Hansen et al. 2025). These trends are illustrated in Figure 2.7, which shows the global surface temperature anomaly relative to the 1880–1920 average.

2.3.1 Impacts on climate system

The increase in temperatures has significant consequences for every physical component of the climate system. A warmer atmosphere holds more water vapour, which provides additional moisture to storms and thus results in more intense precipitation events. At the same time, warming causes a greater share of precipitation to occur as rainfall instead of snowfall, increasing the risk of flooding in early spring and of droughts in summer, as less water is stored in seasonal snowpacks (Trenberth 2011). Global warming also contributes to sea level rise: global mean sea level increased by 20 cm between 1901 and 2018 as a consequence of the thermal expansion of seawater

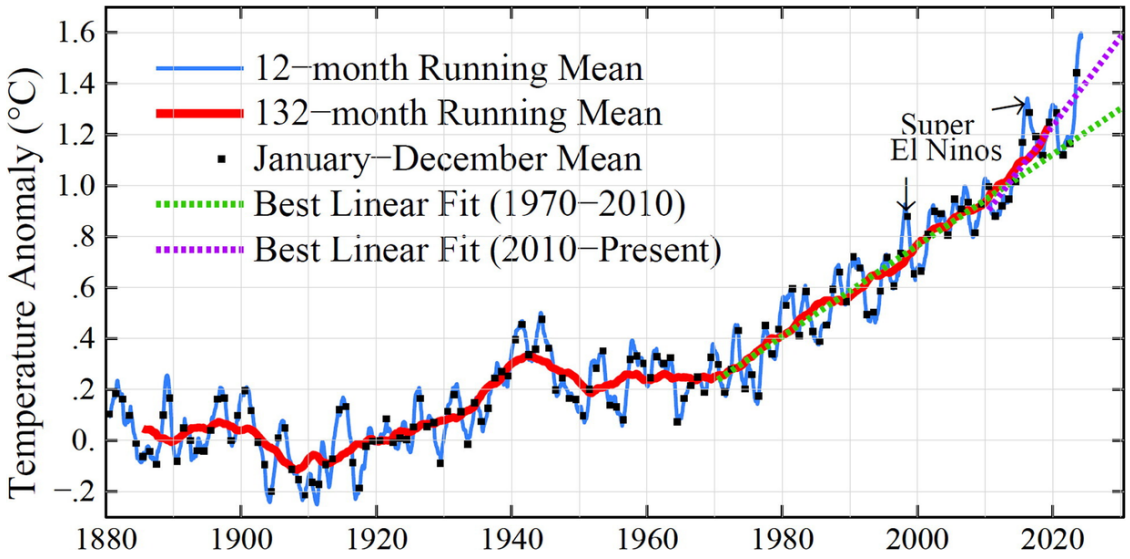

Figure 2.7: Global surface temperature anomaly relative to the 1880–1920 average. Adapted from Hansen et al. 2025.

and of the substantial melting of land based ice masses. The average rise rate nearly tripled over the last century, climbing from 1.3 mm year^{-1} over the 1901–1971 period to roughly 3.7 mm year^{-1} between 2006 and 2018. Furthermore, the continuous warming of the upper ocean layers results in more violent tropical storms and poses an existential threat to maritime ecosystems (IPCC 2023).

Among the many components of the climate system, the cryosphere is arguably the most sensitive to a warming planet, as it can thrive only in cold climates. Over the past four decades, the mean Arctic sea ice extent has been shrinking at an alarming rate of 54 300 km^2 year^{-1}, decreasing from approximately 12.5 million km^2 in 1979 to roughly 10.2 million km^2 in 2021 (Parkinson 2022). This decline is accompanied by a thinning of the ice layer and by a notable shortening of the sea ice season in the southernmost regions of the Arctic, as shown in Figure 2.8. Nowadays, the shores of Siberia remain ice-free for up to three months during summer, and comparable conditions are observed along the coasts of northern Canada. This marked reduction in the extent and persistence of sea ice is further worsening the climate crisis through a vigorous ice albedo feedback: as the disappearing sea ice is replaced by open water, which has a significantly lower albedo, more solar radiation is absorbed, leading to additional warming. Ice loss has also picked up in Greenland and Antarctic ice sheets, as the rate of mass loss has more than tripled with respect to the 1990s, surging from 105 Gt year^{-1} between 1992 and 1996 to 372 Gt year^{-1} between 2016 and 2020 (Mudryk et al. 2020). A similar figure emerges for mass loss of glaciers worldwide, which between 2000 and 2023 lost on average 273 ± 16 Gt annually (Zemp

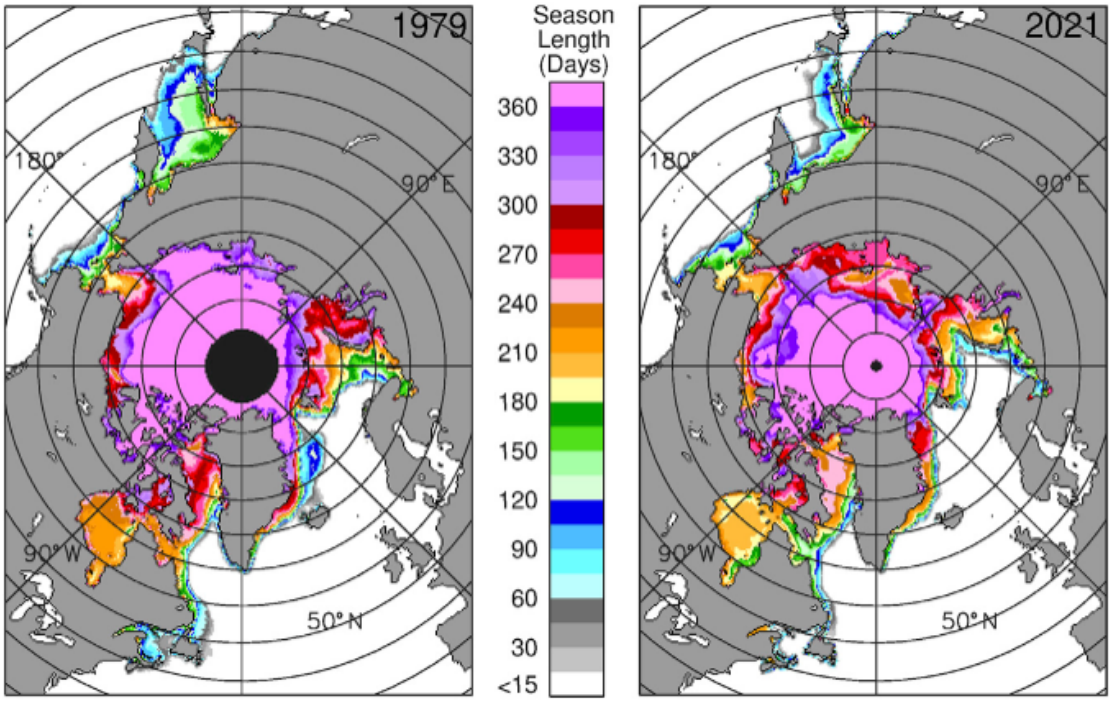

Figure 2.8: Comparison between the sea ice seasons in 1979 and 2021. Image adapted from Parkinson 2022.

et al. 2025). Mountain glaciers are particularly vulnerable to global warning, as rates of environmental change are typically enhanced by elevation. On a global scale, the differences between mountain and lowland trends are 0.21 °C per century for temperature and −11.5 mm per century for total precipitation (Pepin et al. 2025), implying that seasonal snowpacks and glacier ice in these regions will be further affected by a warming climate.

2.3.2 Impacts on the seasonal snow cover

The seasonal snow cover is clearly affected by the consequences of climate change, as air temperature determines whether precipitation falls in solid or liquid form and regulates the amount of energy available for melt. At the hemispheric scale, snow cover duration (SCD) shows a marked negative trend of −0.44 days year^{-1} (Roessler and Dietz 2023). This decline can be attributed to an earlier onset of spring melt: over the 1967–2026 period, the mean snow covered area in April has been shrinking by 0.43 million km^2 per decade, that is by 1.44 % per decade once normalized to the 1991–2020 mean. This loss rate is significantly larger in June, where the decadal reduction accounts for 14.51% of the corresponding monthly average, or to 1.18 million km^2 (NCEI 2026). Total snow mass also exhibits a negative trend at the hemispheric scale, regardless of the period taken into account. This temporal trends,

however, show a significant spatial variability, as peak snow mass over North America is currently declining by $-46\,\mathrm{Gt}$ per decade, whereas no significant trend is detected over Eurasia, where losses on the European portion are compensated by enhanced snowfall in Siberia. This increase in snow mass over Northern Russia is also driven by climate change, as warmer air has a higher water holding capacity causing a shift in precipitation patters (Pulliainen et al. 2020).

A similar picture emerges when focusing on mountainous regions, as aggregate trends over global mountain areas are slightly negative, with an average reduction in snow cover extent of 3.6 ± 2.7 % and a decrease in snow cover duration of 15.6 ± 11.6 days over the 1982–2020 period. Despite the overall reduction at the global scale, the results show substantial spatial variability across mountain ranges and some positive trends driven by climatic forcing may still occur, as visible in Figure 2.9. For instance,

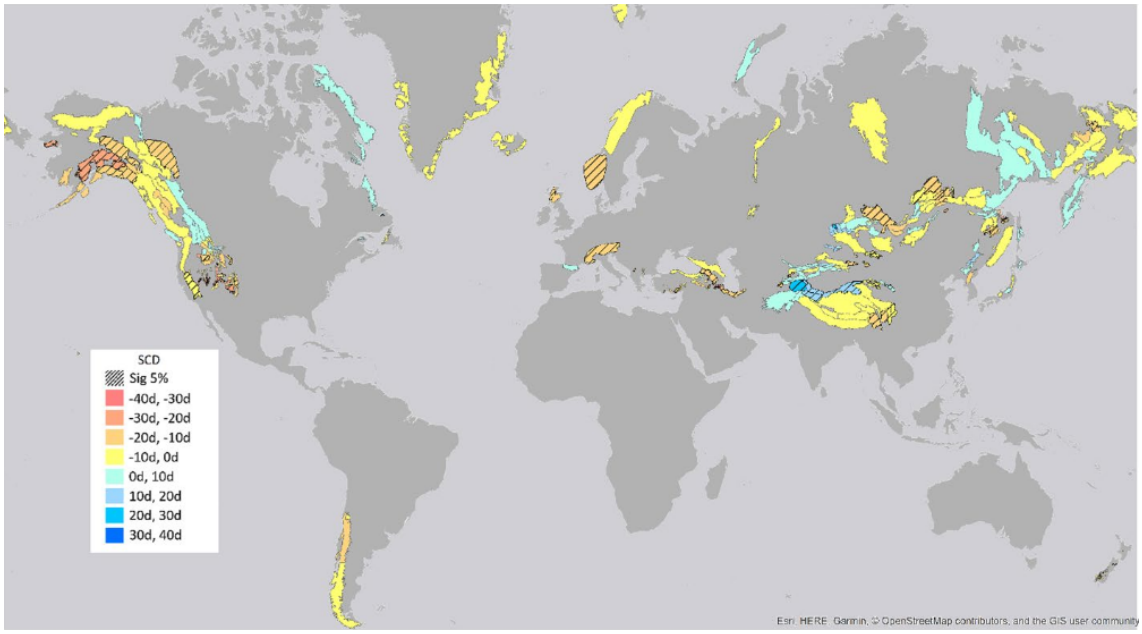

Figure 2.9: Trends in snow cover duration over mountainous terrain between 1982 and 2020. Underlined areas indicate trends that are significant at 95% confidence level. Image adapted from Notarnicola 2022.

the snow cover duration across the Pamir range has been increasing up to 27.4 days between 1982 and 2020 as a result of changes in the winter westerly disturbance, whereas the neighboring Himalayas experienced significant losses (Notarnicola 2022).

Within this heterogeneous picture the European Alps emerge as a particularly critical region. This vulnerability arises from the combined effects of their mid-latitude location and relatively modest elevations, which cause a substantial fraction of the snowpack to remain close to the melting point throughout the snow season. Under such conditions, even a slight increase in air temperature can lead to sudden

changes in snowfall and snow ablation patterns. This is documented by the Swiss meteorological network, whose records show a clearly decreasing trend in the ratio of snowfall days to precipitation days between 1961 and 2008. In particular, present day values for December, January, and February are similar to those observed in November and March during the 1960s, and the decline is more pronounced below 1400 m a.s.l., where winter temperatures are already close to 0 °C (Serquet et al. 2013). The same elevation dependence is found in snow depth across the entire European Alps: below 1000 m a.s.l. the average HS over November-May decreased by up to 10 % per decade between 1971 and 2019, whereas above 2000 m a.s.l. the reduction was only about 2.4 % every ten years. This asymmetry is reflected in the duration of snow cover, which over the same period shortened on average by 6.5 % per decade below 1000 m a.s.l. and just by 0.5 % per decade above 2000 m of elevation (Matiu et al. 2021). The shortening of the snow season is driven, as on the hemispherical scale, by an earlier melt-out of the snowpack, rather than by a delayed onset of snow accumulation. This phenomenon, combined with a declining amount of water stored within the snowpack, is causing a shift in the hydrological cycle potentially exposing millions of people to summer droughts, as the snowmelt contribution to streamflow is no longer sufficient to compensate water deficits later in the season.

Model projections suggest that these negative trends will persist throughout the century, as a further decrease in seasonal snowpack extent, depth and SWE is certain regardless of elevation and emission scenario. Mountainous areas below 1200 m a.s.l. will be the most affected, with simulations showing almost snow free conditions by the end of the century. In a non-intervention scenario, significant losses are projected to affect the highest peaks as well, with a decrease in HS by about 50 % above 3000 m of elevation. Under these circumstances, the accompanying changes in snow cover patterns would be roughly equivalent to an upward elevation shift of 750 m. By contrast, a substantial cut in greenhouse gas emissions would eventually lead to a stabilization in snow characteristics around the turn of the twenty-second century, with losses settling at about 30 % relative to present day rather than the 70 % projected if no action were taken (Marty et al. 2017).

Chapter 3

Snow products

The previous chapter introduced the main physical processes governing the snowpack, and showed that snow cover is a highly dynamic system which can be investigated by means of several measurements techniques. None of them, however, is able on its own to describe in detail every property of the snowpack over a wide and morphologically complex region. Manual surveys and automatic weather stations provide accurate local observations, but their spatial representativeness is limited, since monitoring networks are often too sparse to capture the variability of the snowpack and are typically biased toward low to intermediate elevations. Remote sensing, in contrast, provides the spatial coverage that ground observations lack, but at the price of significant trade-offs in resolution, revisit time and the type of variable that can be retrieved. MODIS, for instance, yields daily snow cover maps at 500 m resolution, but provides no hydrological insight, as it only discriminates between snow-covered and snow-free pixels. Remote sensing techniques targeting snow water equivalent (SWE), on the other hand, typically operate at resolutions of tens to hundreds of kilometers, too coarse to resolve local topography and therefore suitable only for hydrological studies at the continental scale.

To overcome these limitations and obtain continuous information on snow variables across an entire region, numerical models have been developed to simulate the evolution of the snowpack under meteorological forcing. The simplest approaches rely solely on air temperature and total precipitation, regarded as the main drivers of snow accumulation and snowmelt. More advanced models solve the mass and energy balance equations (see Chapter 2) at every grid point, accounting for the effects of complex terrain and for microphysical processes such as snow metamorphism, with the goal of reproducing the state of the snowpack as accurately as possible. Their input fields, which must be spatialized beforehand, include incoming solar radiation, air temperature, relative humidity and wind speed, among others.

Some of these models, such as ALPINE3D, are already employed as tools for avalanche forecasting and danger assessment, since the simulations can accurately identify weak snow layers that are prone to failure (Lehning et al. 2006). Snow products are equally relevant in hydrological applications: an accurate estimate of

the amount of water stored within the snowpack and of the timing of its release can substantially improve discharge predictions, which in turn are essential for the management of hydropower production. Beyond these practical applications, snow models play a fundamental role in climate change research, as they allow spatially distributed trend analysis and even make it possible to predict the response of the snowpack to different emission scenarios.

This flexibility, however, comes at a cost, since the snow variables resulting from numerical simulations inherit the uncertainties of the meteorological forcing. Moreover, the modelling philosophy behind each product may neglect physical processes whose omission is critical in specific applications: most models are therefore task-oriented, and should be used only for the class of problems they were designed to address. Since different models may yield substantially different results for the same snowpack, assessing the extent of such disagreement is a prerequisite for producing reliable snow predictions in a given domain. This is exactly the aim of this chapter: the following sections introduce and compare the main snow datasets available for the study area of this thesis, highlighting their different approaches, the discrepancies that emerge over the investigated region and the inadequacy for trend estimations, thereby motivating the need for a new product.

3.1 Study area

This thesis focuses on the Italian Hydrological portion of the Greater Alpine Region (IH-GAR), comprising all Italian territory located north of 43 °N together with the Swiss portion of the Po basin. The main geological features of this territory are respectively:

- *the Italian Alps*, which are the southernmost of the Alpine Arc. They extend for approximately 1200 km, covering an area of about 50 000 km^2. The highest peak, Monte Bianco, reaches 4810 m a.s.l., and summits above 3000 m a.s.l. are fairly common throughout the range. Such elevations allow snow to persist throughout summer at the highest altitudes, sustaining 901 glaciers, whose total surface area amounted to roughly 368 km^2 in 2016. (Smiraglia and Diolaiuti 2016).

- *the Po Plain*, which is a predominantly flat region shaped by an initial phase of marine sedimentation followed by later fluvial deposition. It is drained by the Po river, Italy's longest, which flows for approximately 652 km. Together with the Venetian and Friulian plains, it forms the most productive agricultural and industrial area of the country, home to roughly 20 million people.

- *the Northern Apennines*, formed by the collision between the western margin of the African Promontory and the Sardinia-Corsica block. Their highest peak reaches only 2165 m a.s.l., insufficient to prevent the complete melt of the seasonal snowpack (Soldati and Marchetti 2018).

The high elevations of the Italian Alps, combined with low winter temperatures, allow substantial volumes of water to be stored in solid form. Snow also accumulates over the Northern Apennines, although in far smaller amounts and for shorter periods, since the peaks are substantially lower. These differences in snow cover give rise to distinct hydrological regimes: up to 50% of the annual streamflow of rivers rising in the Alpine Arc consist of meltwater (Colombo et al. 2023), whereas rivers draining the Apennines rely predominantly on rainfall, and often experience low flow, or even run completely dry, during summer. The complex orography of the study area, examined in greater detail in Figure 3.1, also plays a significant role in governing the climate of Northern Italy. Indeed, the Italian Alps act as a barrier against cold air

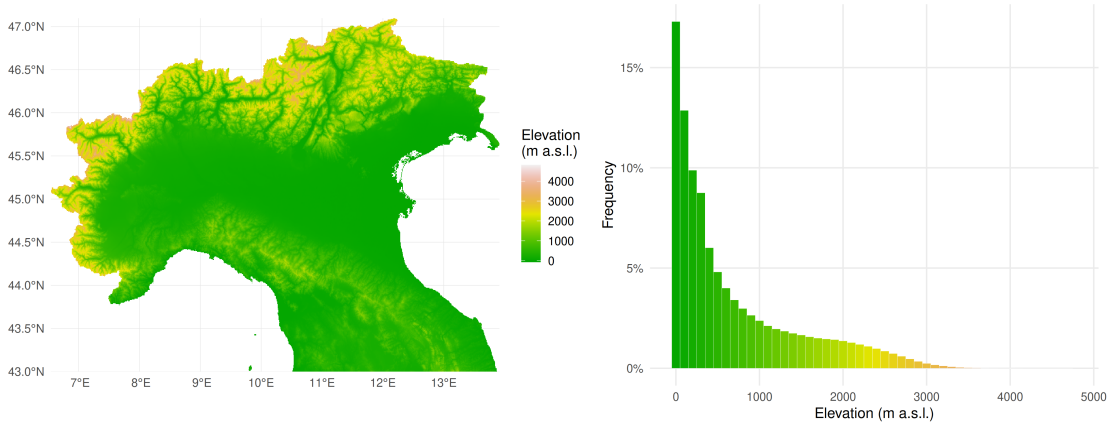

Figure 3.1: Topography and elevation distribution of the study area. The frequency histogram is based on 100 meters wide elevation bins. Elevation data were obtained from ALOS AW3D30 global digital elevation model (DEM).

masses from the north, while the Apennines shield the Po Valley from the milder influence of the Tyrrhenian Sea. Two main climate types can thus be identified within the study area:

- *the Alpine climate*, which prevails over the Alps and the Northern Apennines and is characterized by low nighttime and winter temperatures, with precipitation peaking in summer;
- *the Po Valley climate*, with cold winters, hot summers, and precipitation

concentrated during spring and autumn (Manara, Brunetti, et al. 2026; Ministry for the Environment, Land and Sea 2007).

This complex climatological and topographical setting gives rise to a substantial variability in the snowpack, which makes the IH-GAR region an ideal test bed for comparing the performance of snow models and remote sensing products in reproducing the actual state of the snow cover. The strong dependence on orography can be investigated by means of satellite observations alone, whereas the accurate streamflow estimates required by agriculture and hydropower production call for more complex and physically based modelling. To date, three spatially distributed snow datasets are available over the area: MODIS, IT-SNOW and a product specifically developed for the mountainous portion of the Po River District. None of them matches the IH-GAR boundaries exactly, but all overlap with the investigated domain to a large extent. Each dataset is discussed in a dedicated section below. The main results presented hereafter were computed as a part of the preliminary work of this thesis, which was made possible by the public online availability of the three datasets. In each case, the methodological framework outlined in the corresponding literature was adopted.

3.2 MODIS

Owing to its multispectral configuration, MODIS has already proven to be a key tool for mapping snow cover extent and duration across a variety of regions, such as the Karakoram (Almagioni et al. 2025) or the Greater Alpine Region (Fugazza et al. 2021). One of the most recent studies based on this sensor is devoted to the Italian territory (Manara, Almagioni, et al. 2025) and thus covers almost the entirety of the IH-GAR. The resulting dataset, curated by the Department of Enviromental Science and Policy of the University of Milan, provides daily snow cover maps at the national scale and, in its latest version, spans the period from 24 February 2000 to 30 September 2025, thereby including 25 hydrological years. It consists of binary rasters with a spatial resolution of approximately 500 m, in which a pixel is assigned a value of 1 when it's at least half-covered in snow, and a value of 0 when the ground is bare.

3.2.1 Dataset generation and gap-filling strategy

To obtain binary rasters describing the presence or absence of snow cover, Fugazza et al. 2021 proposed an operational processing chain designed to maximize prediction

reliability. The procedure, was also used by Manara, Almagioni, et al. 2025 to produce snow cover maps across Italy, and takes as input MODIS Terra Snow Cover Daily L3 Global 500 m SIN Grid V061 (MOD10A1 V61) and MODIS Aqua Snow Cover Daily L3 Global 500 m SIN Grid V061 (MYD10A1 V61) products, both distributed by the US National Snow and Ice Data Center (NSIDC). In these datasets, an integer value is assigned to each grid cell according to the criteria summarized in Table 3.1.

Table 3.1: Integer values in NSIDC snow product.

Value	Condition
0	Snow free
1-100	Snow covered (NDSI value x 100)
200	Missing data
201	No decision
211	Recorded during the nigth
237	Fresh water
239	Sea water
250	Cloud covered
254	Detector saturated
255	Fill value

To address the observation gaps caused by cloud cover or by sensor saturation, Fugazza et al. 2021 first reduce the complexity of the dataset to three surface classes: snow covered, snow free and missing data. Integer values between 1 and 100 are reclassified as snow covered whenever the corresponding NDSI exceeded the critical threshold of 40, and as snow free otherwise. Pixels acquired at night, affected by cloud cover or flagged with an error condition are all labelled as missing data, whereas freshwater pixels are reclassified as missing only when freshwater presence has been detected in less than 10 % of the available observations. Once the number of classes has been reduced to three, Fugazza et al. 2021 propose a gap filling procedure structured into six sequential phases, each exploiting different spatial and temporal information to progressively reclassify the missing pixels until the record is complete. These steps, adapted from the previous work of Gafurov and Bárdossy 2009 consist of the following:

- *Terra-Aqua combination*, which exploits the few hours offset between Terra and Aqua overpasses. Since Terra is considered more accurate than its counterpart, the snow map is built primarily from Terra values whenever snow presence or

absence is known. If a Terra data is missing, the corresponding Aqua value is examined instead and retained if it is not itself missing. This step assumes that no snowmelt or snowfall occurs between the two overpasses.

- *Short range temporal filling*, which relies on combining snow cover information from adjacent dates to try filling the remaining gaps. For each grid point, the values recorded on the previous and following day are considered: if neither is missing and both agree on the pixel classification, the missing value is reclassified accordingly.

- *Elevation based filling*, which relies on the vertical extent of snow cover on a given day. For each day, the maximum snow line elevation H_s^{max} (above which every pixel is snow covered), and the minimum snow line elevation H_s^{min} (below which every pixel is snow free) are computed. A missing pixel located above H_s^{max} is thus reclassified as snow covered, while one located below H_s^{min} is labelled as snow free.

- *Direct neighbour spatial filling*, during which each missing pixel is compared with its four adjacent pixels. If at least three share the same class, that class is assigned to the missing pixel.

- *Elevation weighted spatial filling*, in which the 3×3 neighbourhood surrounding each missing grid point is examined. If any neighbouring pixel is snow covered and lies at a lower elevation than the central one, the missing pixel is reclassified as snow covered.

- *Pixel-wise temporal filling*, which consists of examining the subsequent values of a missing grid point until a snow covered or snow free classification is found. This value is then assigned to the missing pixel.

After the sixth step, the dataset is entirely free of missing values, providing snow cover information on a daily basis across the whole Italian peninsula.

3.2.2 Results

Following the approach adopted by the authors, particular attention was devoted to those metrics that describe the timing and the duration of the snow season. In particular, the snow cover duration (SCD) for each pixel was computed as

$$\mathrm{SCD}_{i,j} = \sum_{k=1}^{n} \mathrm{S}_{i,j,k}, \tag{3.1}$$

thereby summing, over a hydrological year spanning from the 1^{st} of October to the 30^{th} of September, the binary values recorded at each specific location. In Equation (3.1), k denotes the day of the hydrological year, whereas (i, j) represent the spatial coordinates of the pixel. Two further snow metrics considered in the analysis of this dataset are the snow onset date (SOD) and the snow end date (SED). The former is defined as

$$\mathrm{SOD}_{i,j} \;=\; \mathrm{D} \;-\; \sum_{k=1}^{D} \mathrm{S}_{i,j,k}, \tag{3.2}$$

where D corresponds to 116, namely the number of days between the beginning of the hydrological year and the 25^{th} of January. Conversely, the latter one can be computed pixel-wise as

$$\mathrm{SED}_{i,j} \;=\; \mathrm{D} \;+\; \sum_{k=D}^{n} \mathrm{S}_{i,j,k}, \tag{3.3}$$

with D retaining the same meaning as in Equation (3.2). In what follows, only the results for SCD are presented: for the analogous analysis of SOD and SED, the reader is referred to Appendix A.

The mean snow cover duration across Italy, computed as an average over every pixel and over the 2001-2025 period, is 22 days, with an interannual standard deviation of 5 days. This rather low value is significantly biased by the abundance of plains and low-elevation terrains, which in the climatic setting of the study region remain essentially snow free all year round. MODIS observations indeed reveal that the snowpack is characterized by pronounced spatial variability: the longest snow seasons occur in high-altitude mountainous catchments, whereas the lowlands receive only sporadic and ephemeral snowfall. In the Po Plain, for instance, elevations below 500 m a.s.l. experience less than 5 days of snow cover per year on average, while the surrounding peaks are often snow covered for over 250 days a year. The mean SCD map, shown in Figure 3.2, closely follows the orography of the main mountain ranges, with valleys exhibiting markedly lower snow cover durations than the adjacent slopes. It also underlines that the snow season in the Apennines is substantially shorter than in the Alps, mainly due to their lower average elevations. Only the highest massifs of the former mountain range, such as Gran Sasso and Maiella, exhibit snow patterns somewhat comparable to those observed in the Alpine area. SOD and SED show similar spatial variability, as mountainous terrain strongly increases the likelihood of an earlier onset and a later end of the snow season.

The predominant role of elevation in determining the duration of snow cover can be investigated by dividing the Italian territory into 5 meter wide elevation bands. The quantiles of the average SCD distribution, shown in Figure 3.3, exhibit a clear monotonic relationship with elevation, as higher regions are typically associated with

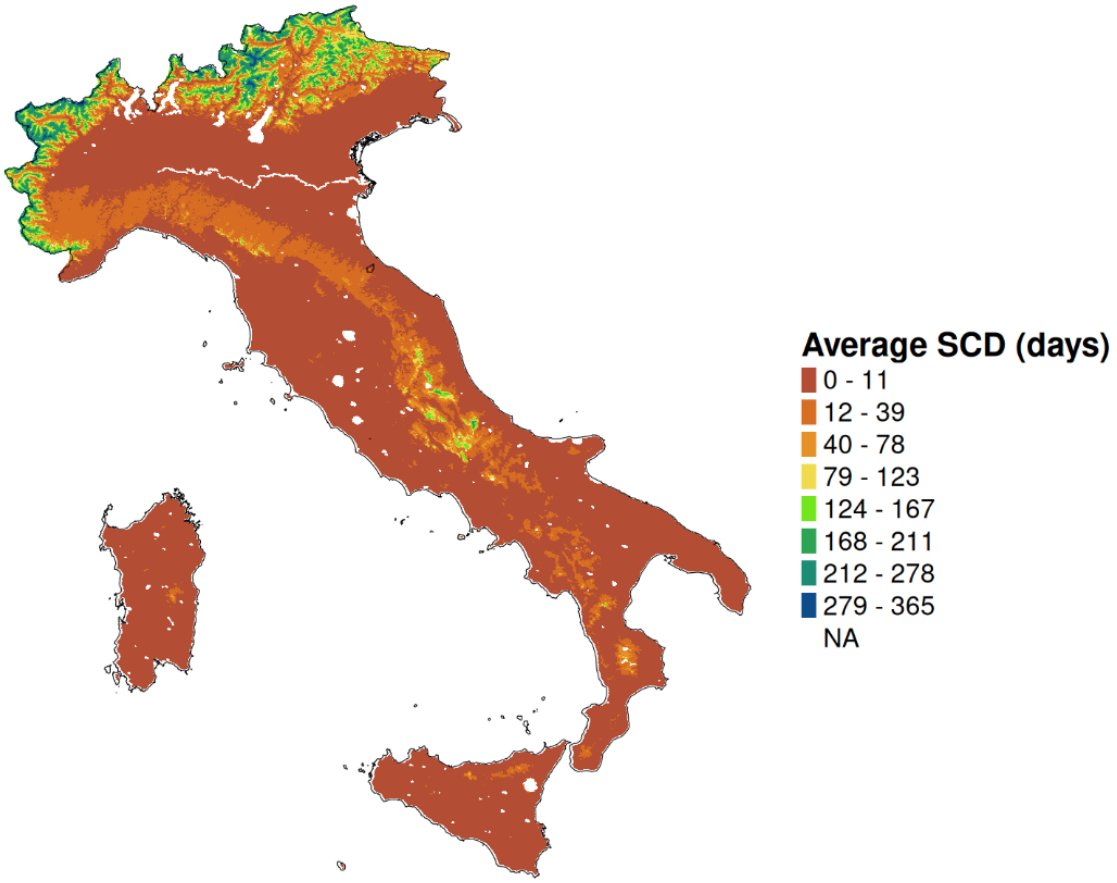

Figure 3.2: Average SCD map across the Italian peninsula for the 2001-2025 period. SCD values accurately follow the orography of the Italian Alps, as altitude strongly influence the persistence of snow on the ground. The lowest SCD values are associated with lowland plains, which remain snow free for most of the year.

longer-lasting snowpacks. However, the considerable spread observed within each elevation band suggest that additional factors, such as climatic regimes or additional topographic variables, may influence snow cover duration across the Italian peninsula. For instance, repeating the same analysis for individual sub-regions of the investigated domain shows that the steepness of the quantile curve depends on the specific region considered. The snow regimes in the Alps and the Eastern Apennines are similar, with a rapid increase of SCD with elevation, whereas the Western Apennines, Sicily and Sardinia show a more gradual trend. Moreover, a multilinear regression with respect to several topographic and geographic predictors shows that aspect and latitude also provide a non-negligible contribution to snowpack properties (Manara, Almagioni, et al. 2025). The combined effect of these climatic and topographic factors ultimately determines whether a certain portion of the study area retains more or less snow than its surroundings. At the same elevation, for instance, south

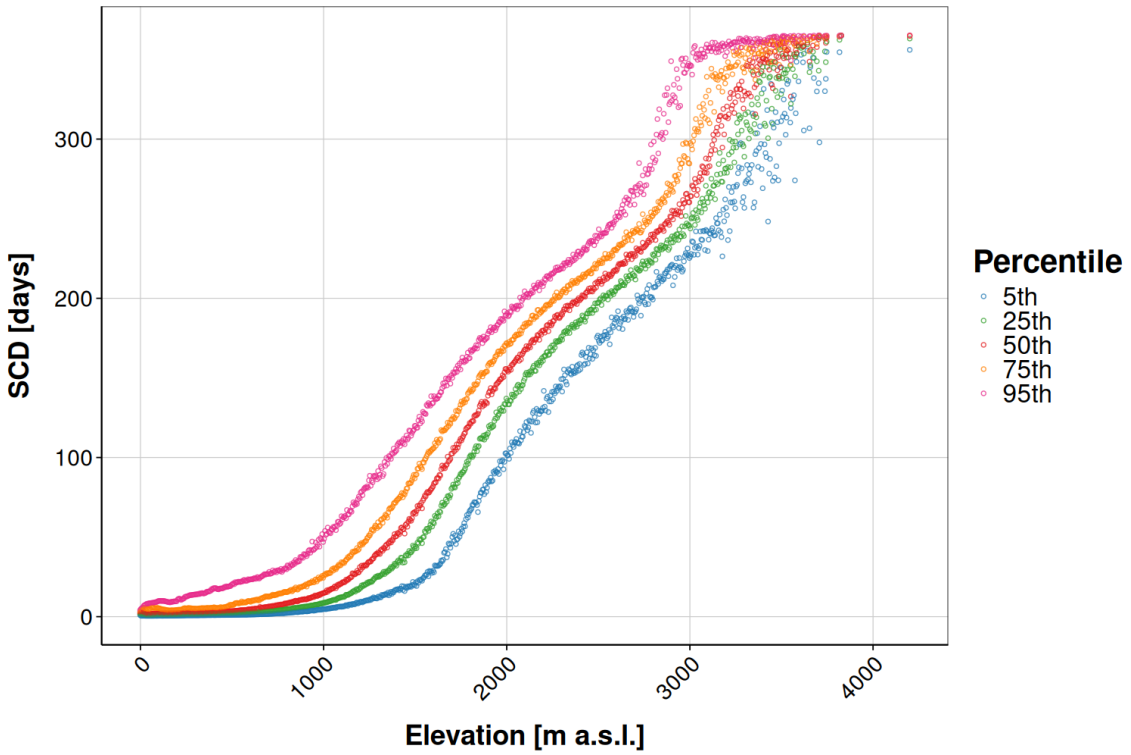

Figure 3.3: Quantiles of the average SCD distribution in each 5 m wide elevation band for the 2001-2025 period. The area taken into account is the whole Italian peninsula. A clear monotonic trend with elevation is visible.

facing slopes experience on average a shorter snow season than north facing ones, since aspect controls the amount of incoming shortwave radiation contributing to the snowpack energy balance. Similarly, enhanced snowfall resulting from the orographic lifting of air masses may lead to higher SCD values, as greater snowpacks require more time to melt completely. These processes are captured by MODIS observations, as shown in Figure 3.4, which reports the difference between the average SCD value of each pixel and the mean SCD of the 5 m elevation band it belongs to. A striking feature of this map is that the Eastern Apennines are significantly snowier than the Italian average at comparable elevations. This particular behaviour is driven by cold continental air masses originating from the Balkan region, which become enriched with moisture as they cross the Adriatic Sea, resulting in consistent snowfall through the whole mountain arc (Capozzi et al. 2025). This east to west contrast progressively weakens toward the Southern Apennines, which typically exhibit a shorter snow season than the other Italian mountain ranges.

Since the snowpack is highly sensitive to meteorological conditions, such as thermal regimes and precipitation patterns, the extent and duration of snow cover observed in any given hydrological year cannot be expected to exactly follow the

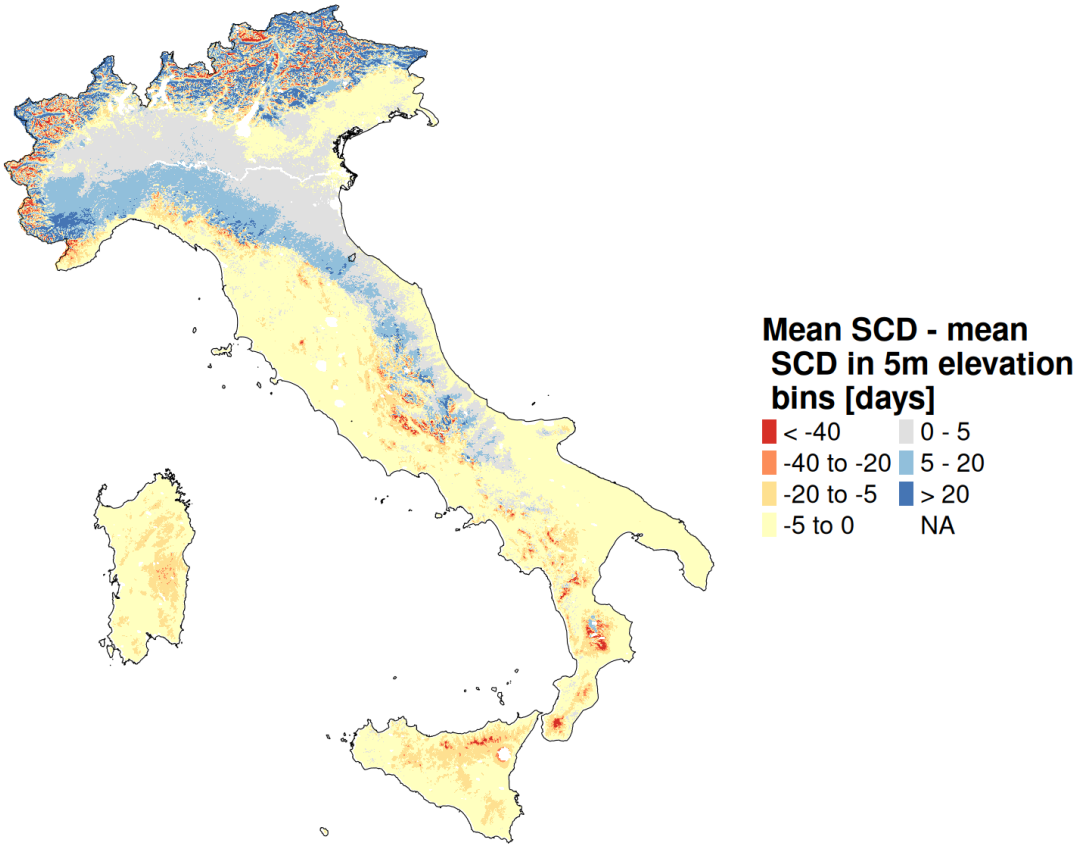

Figure 3.4: Spatial distribution of the difference between the average SCD of each pixel and the average SCD per 5 m wide elevation band it belongs to.

2001-2025 average. The substantial interannual variability of the snowpack can be quantified through the standard deviation of SCD from its mean value across the investigated period. This quantity, as shown in Figure 3.5, increases steadily with elevation, peaking at approximately 1800 m a.s.l. and then gradually decreasing up to 3000 m a.s.l., beyond which a sharp drop is observed. This complex relationship with elevation may be linked to both thermal regimes and glacier presence. Low-elevation mountainous terrain is indeed more sensitive to year-to-year variability, as temperatures are typically close to freezing point, and even a slight increase in temperature may lead to liquid precipitation instead of snowfall or to a substantially earlier end of the snow season due to warmer spring months, whose impact at higher elevations is less pronounced. Moreover, above 3000 m a.s.l. a large portion of the study area is ice-covered all year round, thereby reducing the variability of SCD values since MODIS does not distinguish between snow and glacier ice.

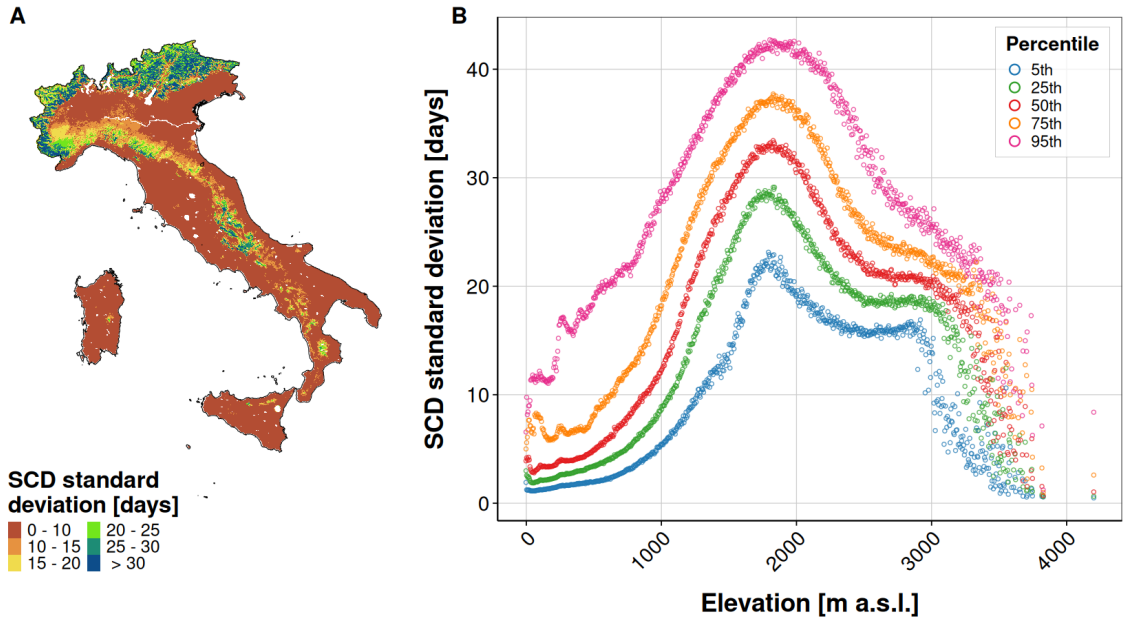

Figure 3.5: (a) Spatial distribution of the standard deviation of SCD from the mean, computed over the 2001-2025 period. (b) Quantiles of the same quantity as a function of elevation. In order to produce this plot, standard deviation values across the whole Italian territory have been organized into 5 meters wide elevation bands.

Ultimately, this snow dataset can be used to assess the response of the cryosphere to climate change over the Italian territory. Although the actual temporal coverage of 25 hydrological years is still relatively limited and falls five years short of the thirty-year period conventionally required for trend analyses, it is nevertheless sufficient to provide an initial perspective of persistent changes in snow cover dynamics. SCD is experiencing widespread reductions across the whole investigated domain, as shown in Figure 3.6 where the results of Mann-Kendall (trend significance) and Sen-Theil (trend magnitude) non-parametric tests are presented. Decreasing trends exceeding -15 days/decade are observed in the Western Alps and in the Central Apennines, whereas over the lowlands the reduction does not exceed -2.5 days/decade, consistent with the much shorter snow seasons of these areas. Most of the statistically significant trends are found in mountainous area, highlighting the sensitivity of those regions to the ongoing warming. This analysis also reveals that trend magnitude has a non-trivial dependence with elevation: the most pronounced changes occur in the 1500–2000 m a.s.l. elevation band where on average the snow season is shrinking by 10 days per decade, whereas areas above 3000 m are experiencing substantially weaker changes, maxing out at 5 days per decade.

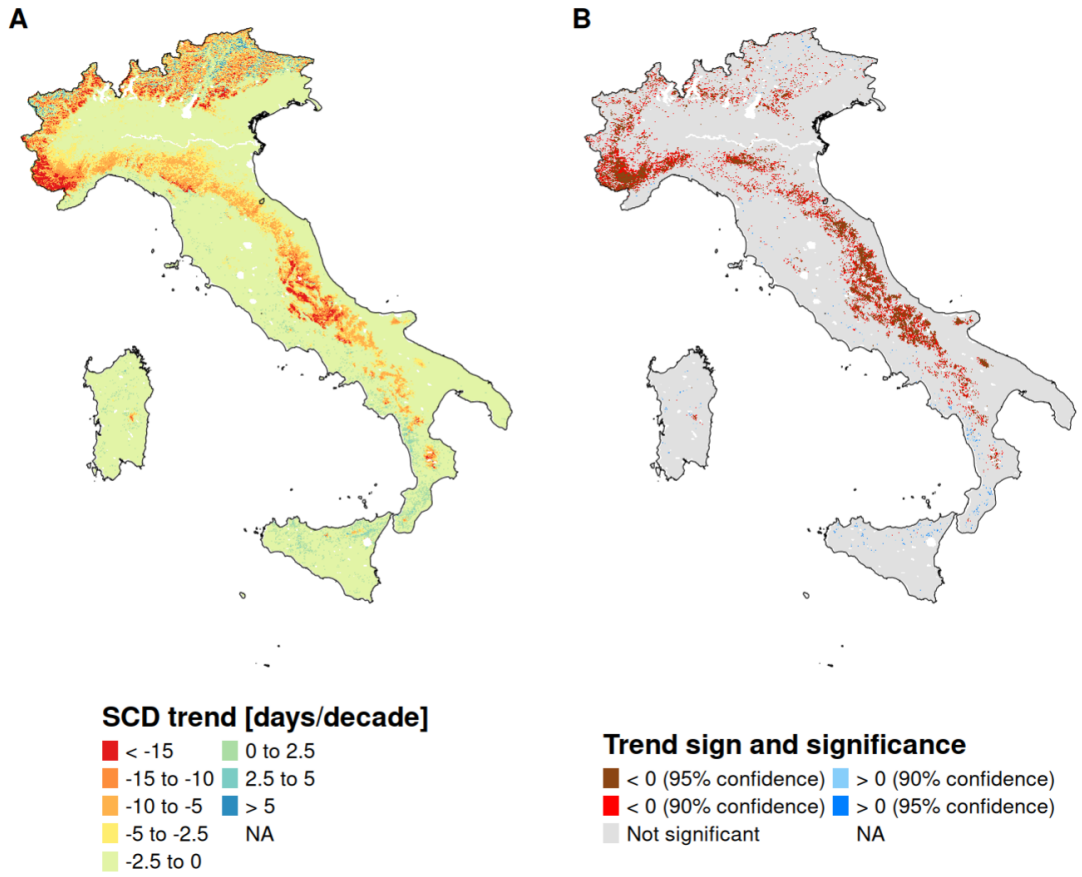

Figure 3.6: (a) SCD trend slope computed by means of Sen-Theil non-parametric test over the 2001–2025 period. (b) Trend sign and significance over the 2001-2025 period computed by means of Mann-Kendall non-parametric test.

3.3 IT-SNOW

IT-SNOW is a snow reanalysis covering the entire Italian peninsula at a spatial resolution of roughly 500 m. This dataset consist of daily maps of snow water equivalent, snow depth, bulk snow density and liquid water content of the snowpack, spanning the period from 1 September 2010 to 31 August 2025 in its latest release. IT-SNOW is the output of S3M Italy, a modelling chain developed by CIMA Research Foundation and specifically designed for large scale applications, with the aim of serving as a valuable tool within flood forecasting frameworks. This model achieves a high level of performance by focusing on the key drivers of snow evolution, namely total precipitation, air temperature, relative humidity, and total radiation, thereby providing a spatially consistent reconstruction of snowpack dynamics. Assimilation of in situ HS measurements and MODIS products is employed to ensure data reliability (Avanzi, Gabellani, Delogu, Silvestro, Pignone, et al. 2023).

3.3.1 S3M

S3M, also known as *Snow Multidata Mapping and Modelling*, is a hydrology oriented cryospheric model developed to simulate the spatial and temporal evolution of seasonal snowpacks. It adopts a raster based approach, in which neighbouring pixels are treated as independent units, so that horizontal mass and energy transports are neglected. All input and output variables are therefore spatially distributed, and the simulated snowpack represents the average conditions within each grid cell. The model provides daily (or even hourly if needed) rasters of snow water equivalent (SWE), snow depth (HS), bulk snow density and liquid water content. These outputs can then be used to compute, during post-processing, snow cover maps spanning across the investigated domain, which could be used for direct comparison with remote sensing products. The snow cover fraction of each pixel can indeed be evaluated as

$$\mathrm{SCF} \;=\; \min\left(1,\, \frac{\mathrm{SWE}\ [\mathrm{kg\,m^{-2}}]}{0.1 \cdot \rho_s}\right), \tag{3.4}$$

where ρ_s is the bulk snow density (Kouki et al. 2023). Since a pixel is conventionally regarded as snow covered when at least half of its area is, binary snow cover maps can be obtained by assigning a value of one to all grid points with SCF > 0.5, and zero elsewhere.

Within S3M, seasonal snow is partitioned into a dry component, corresponding to the ice matrix, and a wet component, given by the interstitial liquid water. These two constituents are modelled separately, and their mutual interactions govern the overall snowpack dynamics. The total snow water equivalent is thus obtained as

$$\mathrm{SWE} \;=\; \mathrm{SWE}_D \;+\; \mathrm{SWE}_W, \tag{3.5}$$

so as the sum of its dry (SWE_D) and wet (SWE_W) contributions. The evolution of these two quantities is described by the following mass conservation equations:

$$\frac{d}{dt}\left(\mathrm{SWE}_D\right) \;=\; S_f \;-\; M \;+\; R, \tag{3.6}$$

$$\frac{d}{dt}\left(\mathrm{SWE}_W\right) \;=\; R_f \;+\; M \;-\; R \;-\; O, \tag{3.7}$$

where S_f and R_f denote the snowfall and rainfall mass fluxes, M and R the snowmelt and refreezing rates and O the snowpack runoff mass flux. All the terms in Equations (3.6) and (3.7) are expressed in mm w.e. $\mathrm{h^{-1}}$ (Avanzi, Gabellani, Delogu, Silvestro, Cremonese, et al. 2022). Solving this system of ordinary differential equations (ODEs) is a key step in simulating snowpack dynamics: in order to do so it is essential to

make specific parametrizations for each of the mass fluxes involved.

Snowfall and rainfall mass fluxes are estimated from the total precipitation P through a phase partitioning scheme based on air temperature and relative humidity. These two quantities are respectively computed as

$$\begin{cases} S_f \;=\; (1 \,-\, p_r)\, P \\ R_f \;=\; p_r P \end{cases}, \tag{3.8}$$

where p_r is the rainfall probability, defined as

$$p_r \;=\; \frac{1}{1 \,+\, e^{a + bT_{air} + cRH}}. \tag{3.9}$$

In the previous equation, T_{air} is the air temperature in °C and RH is the relative humidity in percent, while the fixed parameter $a = 22$, $b = -2.7$ and $c = -0.2$ are taken from Froidurot et al. 2014.

Snowmelt occurs only when both the current air temperature and its 10-day running mean exceed a threshold T_τ, which in S3M is set by default to 1 °C. This dual condition provides a simple way of accounting for the snowpack's cold content, since by requiring sustained mild conditions, rather than a single warm hour, the model avoids triggering unphysical melt events that the snowpack's thermal inertia would not allow. S3M estimates the melt rate through a hybrid temperature index and radiation driven melt parametrization. This formulation improves the model's ability to reproduce the strong spatial variability induced by topography on snow cover extent and duration, since slope and aspect play a significant role in mountainous terrain that cannot be fully captured by air temperature alone. The hybrid approach adopted in S3M thus reads

$$M \;=\; m_{rad} \left[\frac{(1 \,-\, \alpha)\, \mathrm{SW}\downarrow}{\rho_W \lambda_f} \right] \Delta t \;+\; c_M m_r \left(T_{air} \,-\, T_\tau \right), \tag{3.10}$$

where m_{rad} is a dimensionless coefficient that converts the potential radiative melt into actual melt, c_M is a timestep-adjusting parameter, α is the snow albedo, λ_f the specific latent heat of fusion and m_r the degree day parameter [mm °C^{-1} h^{-1}]. Both m_{rad} and m_r are calibration based.

In S3M, refreezing occurs whenever the air temperature drops below T_τ and is computed through a degree day approach as

$$R = -c_M m_r \left(T_{air} \,-\, T_\tau \right). \tag{3.11}$$

Since this expression depends solely on air temperature, refreezing is assumed to respond instantaneously to atmospheric cooling, without accounting for the internal structure or thermal gradient of the snowpack.

Ultimately, the runoff O is estimated by simulating liquid water movement through the snowpack using a matrix-flow approximation. This liquid flux is governed by physical properties of the snow cover, as

$$O = \beta \frac{\rho_W g K_W}{\mu_W}, \tag{3.12}$$

where β is a conversion parameter, g the acceleration due to gravity, K_W the intrinsic permeability of water in snow and μ_W is water dynamic viscosity. This parametrization completes the system of ODEs that, together with those governing the evolution of snow density, are solved at each grid point using the forward Euler method. By default, S3M adopts an hourly integration time step, although this can be adjusted by the user to meet specific application needs.

3.3.2 Results

According to the IT-SNOW reanalysis over the 2011-2025 period, the yearly maximum volume of water stored within the snowpack on the national scale is on average $(11.9 \pm 5.2)\,\mathrm{Gm}^3$, with a maximum of $23.1\,\mathrm{Gm}^3$ in 2014 and a minimum of $5.4\,\mathrm{Gm}^3$ in 2022. This peak is typically reached on 4 March $\pm$ 17 days, with the earliest occurrence being on 26 January 2023 and the latest on 26 March 2013. This means that over the investigated period the peak SWE condition is always achieved before 1 April, which is the date commonly adopted in literature as a reference for maximum SWE. The temporal evolution of the total SWE, which is shown in Figure 3.7, exhibits a great year-to-year variability. In particular, the last years of the record stand out for SWE values markedly below the long term mean, with the partial exception of 2024 since an unusually wet spring helped to balance the deficits accumulated during the early snow season. This pattern provides further evidence of the persistent snow drought conditions which have been detected in the Italian Alps during the early 2020s, driven by the combined effects of reduced precipitation and higher temperatures (Colombo et al. 2023). Besides a strong temporal variability, the modelled snow cover also displays a substantial spatial heterogeneity across Italy, as shown in Figure 3.8. The simulated snowpack results indeed deeper and longer-lasting in high altitude catchments, whereas the lowlands only experience an ephemeral snow cover, owing to their milder temperatures. Water storage is also unevenly distributed between mountain ranges, since according to IT-SNOW estimates almost 87 % of

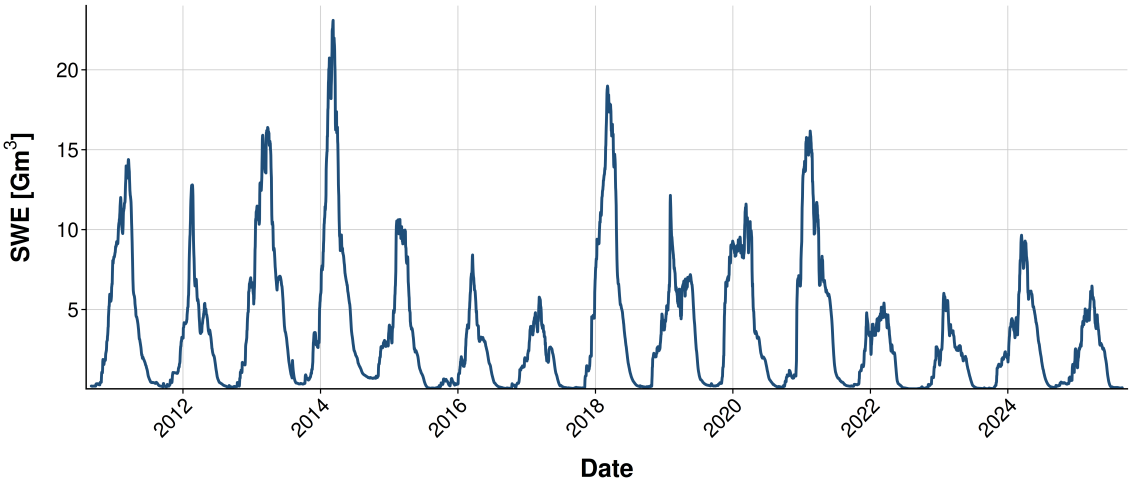

Figure 3.7: Daily evolution of the total water volume stored in the Italian snowpack, computed as snow water equivalent (SWE) integrated over the whole national domain, from 1 September 2010 to 31 August 2025. Data are from the IT-SNOW reanalysis (Avanzi, Gabellani, Delogu, Silvestro, Pignone, et al. 2023).

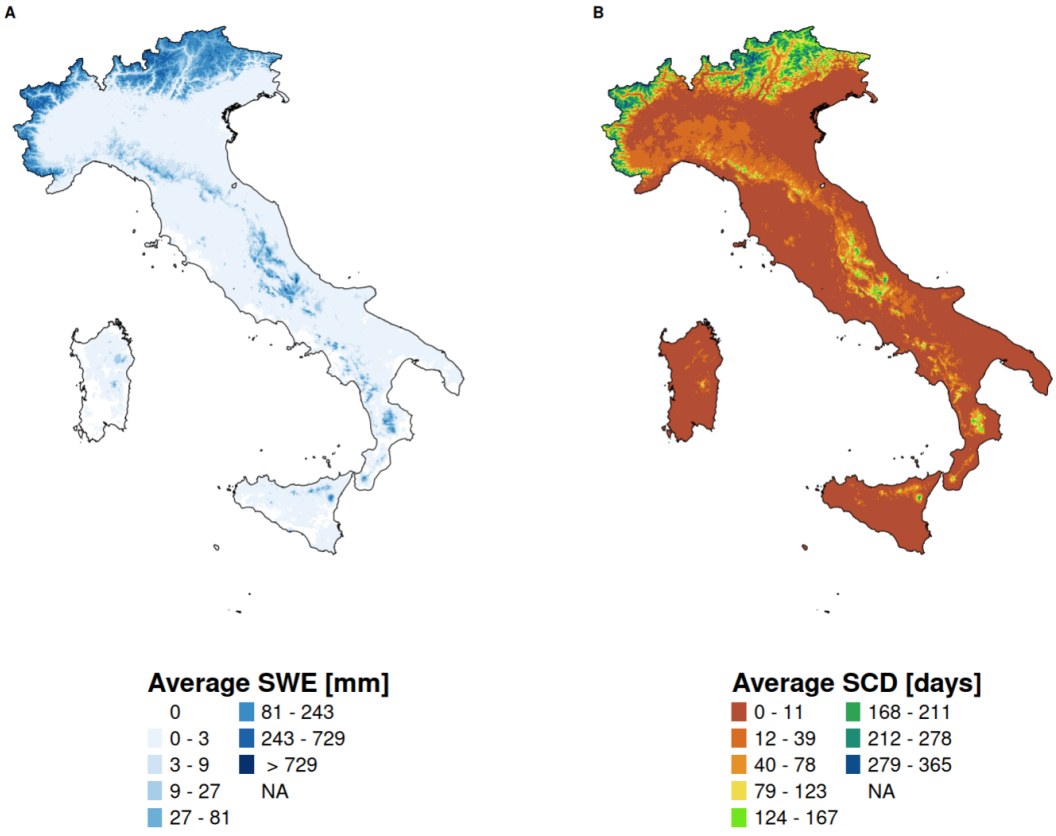

Figure 3.8: (a) Average seasonal snow water equivalent (SWE), computed as the mean of the November–June averages of each hydrological year considered in this reanalysis. (b) Average snow cover duration (SCD), computed as the mean of the annual SCD maps across the same period.

the national SWE is found within the five main Alpine river basins, namely the Po, Adige, Piave, Tagliamento and Brenta. The Central Apennines are Italy's second most important water tower, accounting for about 7% of total SWE, whereas the remaining fraction is distributed across the rest of the country.

3.4 Po River District dataset

This snow dataset focuses on the mountainous portion of the Po River District, an over-regional entity that cover almost 87 000 km^2 and comprises the Po River Basin together with some smaller adjacent catchments, such as Conca and Reno. This snow product is generated by an operational chain known as "J-snow", which is a hybrid approach that combines a physically based snow model named GEOtop 2.0 with in situ snow depth measurements and remote sensing snow cover extent estimates. SWE maps with a spatial resolution of approximately 500 m are available on a daily basis for all the hydrological years between 1992 and 2021. Within each hydrological year, maps are available only between October 3^{rd} and July 1^{st}, a period that roughly corresponds to the longest snow covers in non glacierized areas (Dall'Amico et al. 2025).

3.4.1 GEOtop 2.0

GEOtop 2.0 is a physically based model that simulates the heat and water exchanges at and below the ground surface. Its primary focus is thus the accurate reproduction of soil dynamics, including freezing and thawing processes as well as snow accumulation and melting, whenever thermal and meteorological conditions allow.

The simulated soil is discretized into several cells, which are typically thinner near the ground surface in order to model as accurately as possible the strong interactions with the atmosphere taking place in the upper layers. The depth of the investigated domain is a user-defined parameter, which could range from a few meters to several hundred meters. The soil is treated with a rigid-soil scheme, meaning that expansion of the ground matrix is not allowed because GEOtop 2.0 lacks a mechanical model of porous media. This numerical compromise implies that no volume change is allowed during freezing and thawing processes and, consequently, the densities of liquid and solid water are both set at 1000 $kg\,m^{-3}$. The simulated snowpack is directly coupled to the modelled soil, since the snow cover buffers the exchanges of energy and mass between the atmosphere and the ground. The heat and water content equations of the two media are solved as a single system by means of a time-lagged approach, whereby the coupling terms at time t are replaced by their values at the previous

time step $t-1$. This numerical method reduces the complexity of the problem, as each equation can now be solved separately, while preserving stability and first order accuracy. At each timestep, the snowpack is simulated by computing in sequence:

- *heat exchanges*, whose treatment follows the general framework presented in Chapter 2, since in GEOtop 2.0 internal energy varies under the forcing of shortwave and longwave radiative budgets, turbulent heat fluxes, precipitation heat fluxes and ground heat fluxes. A distinctive feature of this model is that these terms are computed taking into account the complexity of the terrain, which is particularly relevant in mountainous areas. The direct component of the shortwave radiation is indeed corrected for shading, since the surrounding topography can hide the simulated pixel from the Sun, whereas the diffuse SW component is computed as the sum of the solar radiation scattered by the atmosphere and of the shortwave radiation reflected by the portion of terrain visible from the pixel. Analogous corrections are applied to the thermal radiation, which in mountainous regions typically originates both from the atmosphere and from the surrounding slopes. The topography also affects turbulent heat fluxes, since it induces localized wind patterns resulting in a marked variability of latent and sensible heat exchanges.

- *snow metamorphism*, restricted to its destructive form. Constructive metamorphism, which is described in detail in Chapter 2, is not taken into account, as it is mainly relevant to avalanche forecasting and danger assessment, a purpose that differs considerably from the hydrological one of this model. Compaction can therefore only result from destructive metamorphism acting on newly fallen snow and from the overburden exerted by the overlying snowpack, so that the thickness of a snow layer evolves according to

$$\frac{1}{D}\frac{\partial D}{\partial t} = C_1 + C_2, \tag{3.13}$$

  where D is the vertical extent of each grid cell, C_1 is the compaction rate resulting from destructive metamorphism and C_2 is the compaction rate caused by the overlying snow. Equation (3.13) is integrated with the same time step of the heat equation.

- *water percolation*, which can be evaluated by solving a mass conservation equation. The water balance of each snow layer is indeed given by

$$\frac{\partial \theta_{\mathrm{w}}^{\mathrm{fl}}}{\partial t} + \nabla \cdot \boldsymbol{J}_{\mathrm{w}} + S_{\mathrm{w}} = 0, \tag{3.14}$$

where $\theta_{\mathrm{w}}^{\mathrm{fl}}$ is the liquid water content, $\boldsymbol{J}_{\mathrm{w}}$ the flux of liquid water and S_{w} the liquid water sink term. No percolation occurs until the liquid water content exceeds the saturation threshold of 3% to 5% of the snow layer mass (see Chapter 2). Refreezing in sub-freezing layers is also accounted for: the internal energy carried by the water flux is tracked, and latent heat is released whenever refreezing occurs.

- *snow accumulation*, which is computed by means of a two air temperature threshold approach. Precipitation is assumed to fall entirely as rain above the upper threshold and entirely as snow below the lower one, while in between the liquid and solid phases coexist and the solid fraction is obtained by linear interpolation. The new snow density depends on both air temperature and wind speed, since it is evaluated as

$$\rho_{\mathrm{ns}} = \begin{cases} 500 - 475.5 \exp\left[-1.4(5.0 - T)^{-1.15} - 8 \cdot \dfrac{v_{\mathrm{w}}^{1.7}}{1000}\right] \\ \qquad \text{if } T > -13^{\circ}\mathrm{C} \\ 500 - 452.0 \exp\left[-8 \cdot \dfrac{v_{\mathrm{w}}^{1.7}}{1000}\right] \\ \qquad \text{elsewhere,} \end{cases} \tag{3.15}$$

where T is the mean air temperature over the time-step and v_w is the average wind speed.

The model predictions are subsequently corrected using in situ snow measurements to account for precipitation under-catch by rain gauges during the snow season. Remote sensing readings, such as those provided by MODIS sensors, are also employed to compare the simulated and observed extent of the snow cover with the aim of adjusting the melting rate in order to ensure a more accurate representation of the actual snow cover.

3.4.2 Results

The total volume of water stored within the snowpack exhibits substantial year-to-year variability, as shown in Figure 3.9. The average peak SWE value is $6.7 \pm 2.3\,\mathrm{Gm}^3$, typically reached on 20 March $\pm$ 25 days. This timing is consistent, within one standard deviation, with 1 April, the date conventionally used in snow literature as a proxy for maximum SWE. The largest amount of water stored within the study area, according to J-snow estimates, was $12.6\,\mathrm{Gm}^3$ on 6 March 2009, whereas the

minimum annual peak of SWE was recorded on 18 April 2005 and amounted to 3.5 Gm^3.

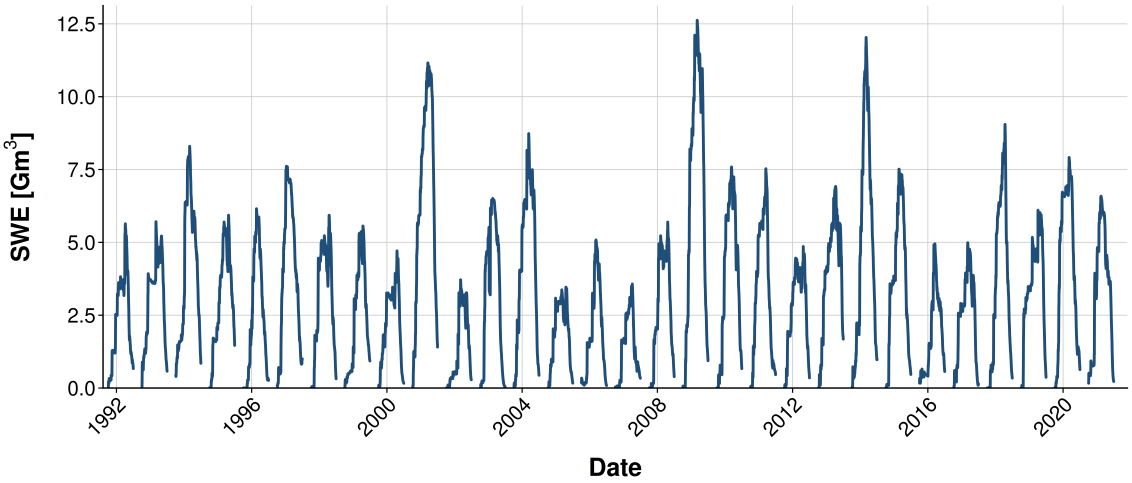

Figure 3.9: Daily SWE evolution in the mountainous portion of the Po River District between 3 October 1991 to 1 July 2021. Data are from the Po River District snow dataset (Wani et al. 2025)

Snow accumulation can continue beyond 20 March in some portions of the Po River District, as the timing of runoff varies substantially across a complex terrain. This variability can be captured by dividing the study area into 500 meter wide elevation bands. As reported in Table 3.2 the maximum amount of water stored within the snowpack below 500 m a.s.l. is reached at the beginning on January, during the coldest period of winter. In contrast, above 2500 m a.s.l. the snow cover begins to

Table 3.2: Elevation wise peak SWE, timing of peak SWE and SWE trends.

Elevation band	**Peak SWE**	**Peak date**	**SWE trend**
0 - 500 m	$0.08 \pm 0.06\,Gm^3$	10 Jan ± 33 days	$-0.01\,Gm^3$/decade (p = 0.11)
500 - 1000 m	$0.22 \pm 0.11\,Gm^3$	13 Jan ± 32 days	$-0.02\,Gm^3$/decade (p = 0.41)
1000 - 1500 m	$0.63 \pm 0.39\,Gm^3$	04 Feb ± 29 days	$-0.06\,Gm^3$/decade (p = 0.30)
1500 - 2000 m	$1.6 \pm 0.8\,Gm^3$	06 Mar ± 23 days	$+0.06\,Gm^3$/decade (p = 0.50)
2000 - 2500 m	$2.7 \pm 0.9\,Gm^3$	05 Apr ± 19 days	$+0.21\,Gm^3$/decade (p = 0.27)
> 2500 m	$2.3 \pm 0.6\,Gm^3$	30 Apr ± 17 days	$+0.19\,Gm^3$/decade (p = 0.20)

melt significantly only towards the end of April. Moreover, temporal fluctuations in the date of peak SWE tend to decrease with elevation, as the highest areas experience more persistent sub-freezing conditions and are therefore less affected by interannual temperature variability. Strikingly, according to this snow product, the maximum SWE values recorded above 2000 m show an increasing tendency between 1992 and

2021, although no statistically significant trend has been identified. This behaviour may provide further evidence of the recent SWE and HS trends observed across the European Alps, as several monitoring networks recorded a sharp (almost step-like) decrease in snow cover at the beginning of the 1990s, followed by a partial recovery in the following decade (Matiu et al. 2021).

3.5 Intercomparison of snow products

The snow datasets presented above differ in several respects. IT-SNOW and the Po River District products are the outputs of two distinct modelling chains, both primarily focused on snow water equivalent. IT-SNOW, however, provides more comprehensive information, as it also includes maps of snow density, snow depth and liquid water content, allowing for a more detailed characterization of the snowpack. In contrast, MODIS describes snow cover alone through binary rasters: this discrepancy does not preclude a comparison, since SWE maps can be converted into binary snow cover maps by means of Equation (3.4) or similar approaches. These datasets also differ in spatial coverage, resolution, coordinate reference system, and temporal extent, as summarized in Table 3.3. Hereafter, the comparisons are carried out over the largest common spatial and temporal overlap.

Table 3.3: Main characteristics of each snow product.

Dataset	Data type	Start date	End date	Study area	Spatial resolution	Reference system
MODIS	Snow cover	24-02-2000	30-09-2025	Italy	0.00417 °	WGS 84 (EPSG:4326)
IT-SNOW	SWE	01-09-2010	31-08-2025	Italy	0.00506 °	WGS 84 (EPSG:4326)
Po Basin	SWE	03-10-1991	01-07-2021	Po Basin	500 m	ETRS89 (EPSG:3035)

3.5.1 IT-SNOW and Po River District dataset

It is particularly interesting to compare the two hydrology oriented snow reanalyses, namely IT-SNOW and the Po River District product. Both are generated by a modelling chain that couples a physically based snow model with a data assimilation scheme, which uses in situ HS measurements and satellite observations to improve prediction accuracy and reliability. Despite this overall similar structure, the two

snow models, S3M for IT-SNOW and GEOtop 2.0 for the Po River District, represent the main snowpack dynamics in markedly different ways. S3M is designed for computational efficiency, so that its predictions could be valuable in real-time flood forecasting, whereas GEOtop 2.0 models the entire energy balance of the snowpack, providing a much more detailed picture of the complex physical phenomena governing snow cover dynamics. Before the comparison of the two products, it is necessary to reproject the Po River District maps onto the coordinate reference system (CRS) and grid of IT-SNOW. The analysis is then carried out over the Italian mountainous portion of the Po River District (i.e. the overlap area), and spans the hydrological years between 2011 and 2021, which is the longest common period. Within this spatial and temporal overlap, the average peak volume of water stored in the snowpack is rather comparable between the two products: IT-SNOW suggest a maximum SWE of 6.8 ± 2.3 Gm^3 occurring on average on 4 March $\pm$ 15 days, whereas the Po River District estimate is more conservative, predicting a peak SWE of 5.9 ± 1.7 Gm^3 happening later in the season, namely on 19 March $\pm$ 22 days. Moreover, the two daily SWE time series also show good agreement over the investigated period, as $R^2 = 0.88$, RMSE $= 0.76\,Gm^3$ and MAE $= 0.52\,Gm^3$. However, this overall consistency may conceal substantial differences in the spatial distribution of the snowpack, since the total volume of water stored within the snowpack is an aggregated metric over the entire study area. Consequently, over and underestimates in different areas may compensate for each other, resulting in similar total values despite significant local differences. To investigate this aspect in greater detail, the common domain was partitioned into three elevation bands: low elevations (0 - 1000 m a.s.l.), medium elevations (1000 - 2000 m a.s.l.) and high elevations (above 2000 m a.s.l.). The elevation-based comparison highlights a substantial difference below 1000 m, as the coefficient of determination (i.e. R^2) indicates only a moderate correlation between the two datasets. Above 1000 m a.s.l., in contrast, the two snow products are in better agreement, as shown by the statistical indicators reported in Table 3.4.

Table 3.4: Elevation-wise comparison between IT-SNOW and Po River District SWE series.

Elevation band	**R^2**	**RMSE**	**MAE**
0 - 1000 m	0.56	$0.24\,Gm^3$	$0.10\,Gm^3$
1000 - 2000 m	0.92	$0.37\,Gm^3$	$0.20\,Gm^3$
$>$ 2000 m	0.86	$0.57\,Gm^3$	$0.44\,Gm^3$

The strong agreement observed for the SWE series at medium and high elevations

can be also appreciated in Figure 3.10, which shows the elevation-dependent series computed over the period of temporal overlap between the two datasets. In particular,

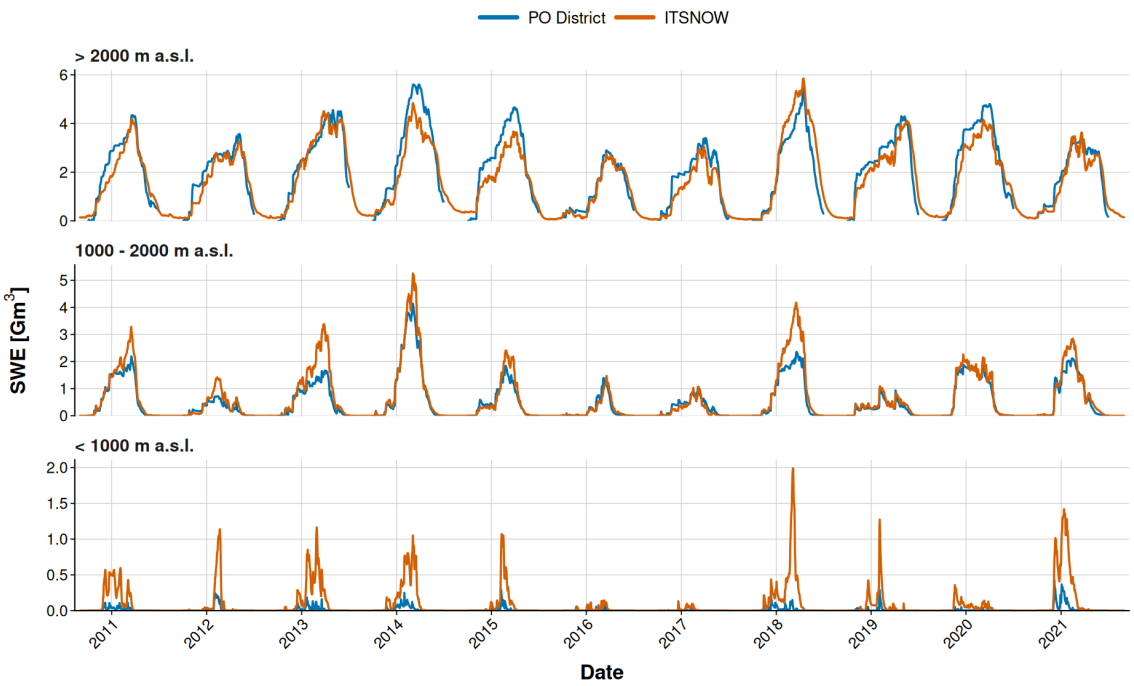

Figure 3.10: Elevation-wise SWE series comparison between IT-SNOW and the Po River District snow product.

IT-SNOW shows a higher SWE volume during peak accumulation at lower elevations, whereas the Po River District product slightly overestimates the water volume stored above 2000 m a.s.l. with respect to its counterpart. The best agreement between the two snow product is at medium elevations. Figure 3.10 also highlights a significant elevation dependent bias in the "J-snow" modelling chain used to generate the Po River District product: the discrepancy with respect to IT-SNOW decreases steadily with elevation, until the underestimation eventually turns into an overestimation. This behaviour could be partially related to the meteorological forcing used by GEOtop 2.0, which is based on precipitation records spatialized and, where the monitoring network is too sparse, corrected using ERA5 reanalysis estimates, which systematically overestimates total precipitations in high elevation Alpine catchments (Wani et al. 2025). A more detailed assessment of ERA5 precipitation biases, which is beyond the scope of this thesis, can be found in Cavalleri et al. 2024. In this context, it is also worth noting that the fraction of SWE stored above 2000 m is larger in the PO River District product than it is in IT-SNOW, which may also help explain the delayed timing of peak SWE in the former dataset.

Even when considering the seasonal average SWE over the overlap region, clear differences between the two snow products can be observed. For instance, as shown

in Figure 3.11, IT-SNOW generally overestimates the amount of water stored within the highlands of the Aosta Valley at the seasonal scale, whereas it substantially underestimates SWE over the Adamello and Rhaetian Alps with respect to the Po River District dataset. Regardless of elevation gradients, a slight overall overestimation by IT-SNOW can also be identified across the entire study area, as the number of pixel with a positive bias is higher than that with a negative bias, as shown in Panel (b) of Figure 3.11. This result is consistent with the higher peak water volume estimated by IT-SNOW over the investigated domain.

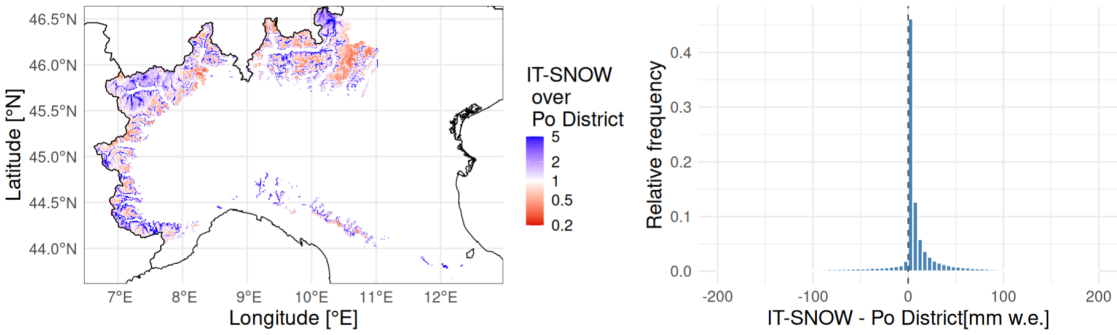

Figure 3.11: (a) Spatial distribution of the ratio between IT-SNOW and Po River District SWE estimates for each pixel. (b) Frequency histogram of IT-SNOW - Po River district SWE values.

Another aspect to consider when comparing the two snow datasets is their reliability to make trend analyses. IT-SNOW provides snow water equivalent estimates from 1 September 2010 to 31 August 2025, which is insufficient for assessing long term changes in snow cover metrics, as the results would still be strongly affected by interannual variability. The Po River District product, on the other hand, meets the temporal requirements for reliable trend analyses, since it spans over 30 hydrological years. However, some modelling choices made during the generation of the dataset limit its suitability for this purpose. In particular, daily maps are not available between 2 July and 2 October of each year preventing the identification of ephemeral snowfall that may still occur in high alpine catchments during summer. Moreover, the final stages of snowmelt at high elevations are not taken into account during those years when favorable conditions allow the snowpack to persist beyond 1 July. This is evident in panel (a) of Figure 3.10, as the SWE record is interrupted before reaching zero during several years, such as 2014. Finally, the spatial coverage of the Po River District product is relatively limited, meaning that any trend estimate derived from this dataset could not be generalized to the entire Alps or to the lowlands.

3.5.2 Modelled and observed SCD

The comparison between the two hydrological oriented products with MODIS is somewhat redundant, since the modelling chains used to generate both IT-SNOW and the Po River District dataset already incorporate MODIS observations through the assimilation schemes discussed above. However, the gap filling strategy adopted by Fugazza et al. 2021 and Manara, Almagioni, et al. 2025 to reconstruct missing values gaps plays a role in determining the extent of snow cover on a given day and, to date, has not been extensively validated. A brief comparison can therefore still provide useful information. By comparing satellite observations with IT-SNOW snow cover duration maps, computed using Equation (3.4), it is evident that satellite readings substantially underestimate SCD across the whole peninsula, as shown in Figure 3.12. This systematic difference becomes more pronounced with increasing elevation, as differences of up to 50 snow cover days can be detected in mountainous terrain. The tendency to underestimation is also clearly visible in the difference histogram. The origin of this behaviour, which can be observed also while comparing the Po River District product with MODIS estimates, is still unclear and may result from a combination of factors related to the satellite gap-filling procedure and to the modelling philosophies of the two hydrologically oriented modelling chains. For example, the NDSI threshold of 0.4 may be overly stringent, potentially leading to substantial underestimations. Moreover, the main differences visible in Figure 3.12 appear to occur in densely forested areas, suggesting that significant tree cover

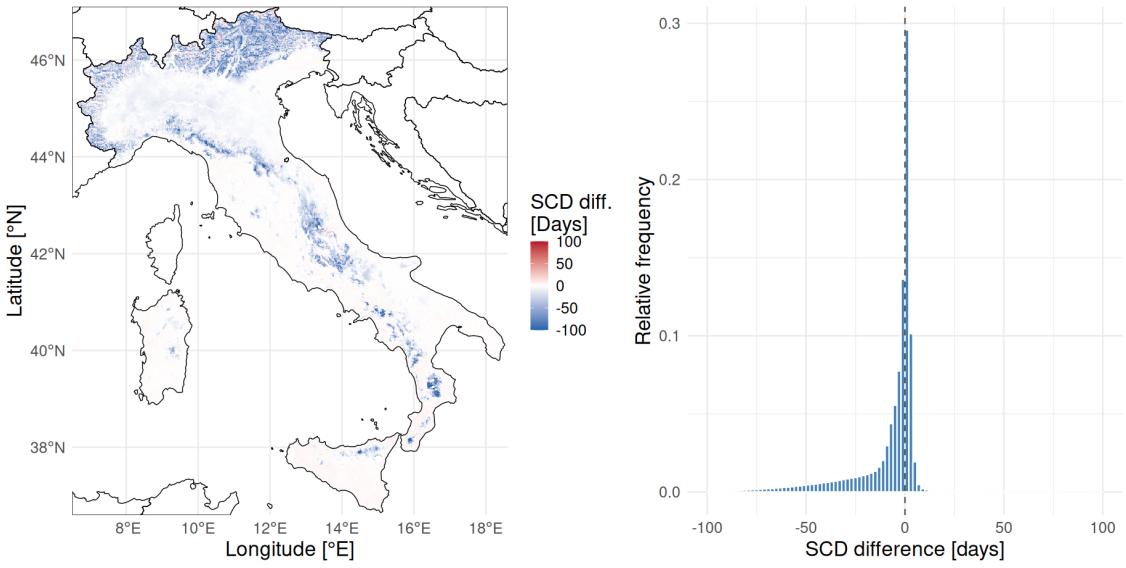

Figure 3.12: (a) Spatial distribution of the SCD difference between MODIS and IT-SNOW estimates. (b) Relative frequency histogram of SCD difference values between MODIS and IT-SNOW estimates.

could introduce additional biases. However, a more detailed discussion of this underestimation is beyond the scope of this thesis.

It is instead more important to focus on the reliability of MODIS snow cover maps for estimating trends. At the time of publication of Manara, Almagioni, et al. 2025, MODIS snow cover data were available over the Italian territory only up to 2022, and trend analyses were thus limited to that year. At the time of this thesis, however, MODIS observations were available up to 30 September 2025. This additional three years, which were mainly characterized by snow droughts across Italy, made it possible to extend trend analysis and assess the effect of this additional period on the results. The inclusion of only a few extra years led to a noticeable increase in the number of pixels exhibiting statistically significant decreasing trends, as shown in Figure 3.13. This suggests that the relatively limited MODIS temporal coverage may still be strongly affected by interannual variability and may therefore be insufficient for obtaining robust long term trend estimates. Despite this limitation, MODIS remains highly relevant for snow cover studies, as this dataset gets continuously updated and

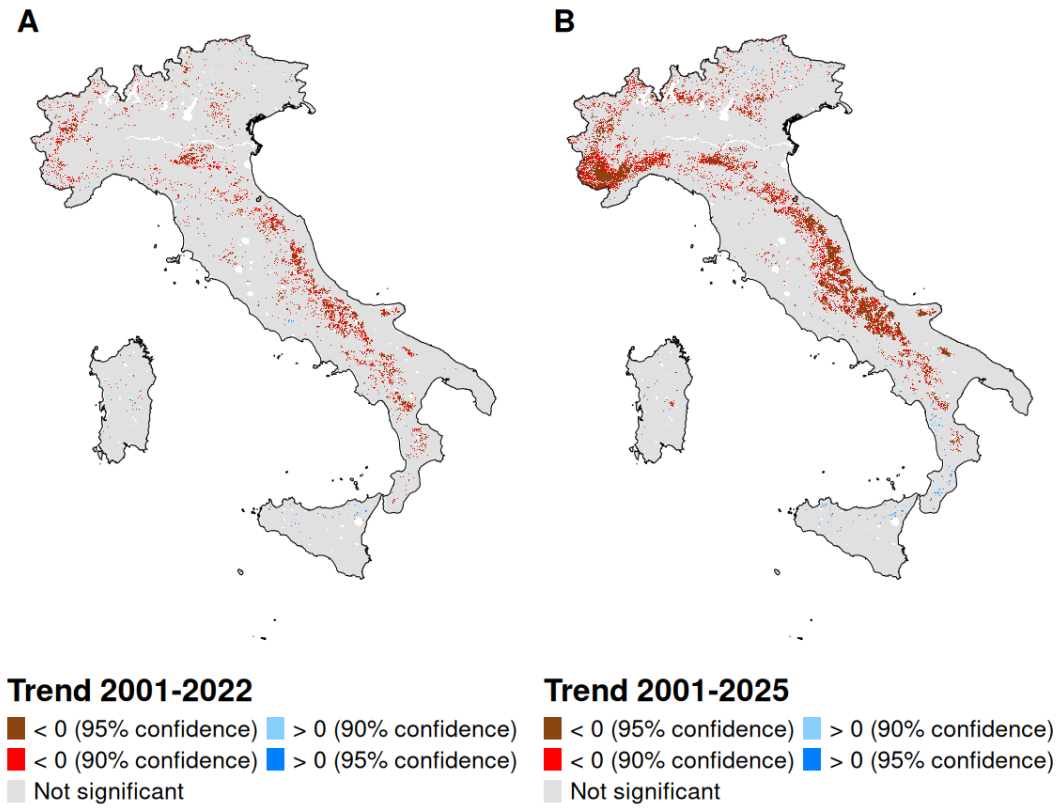

Figure 3.13: (a) Sign and significance of SCD trends derived from MODIS observations over the 2001–2022 period. (b) Sign and significance of SCD trends derived from MODIS observations over the 2001–2025 period. The comparison highlights the strong impact of three snow scarce years on trends across the Italian peninsula.

represent one of the few sources of spatially distributed observations that do not rely on numerical simulations of the snowpack.

3.5.3 Conclusions

The three spatially distributed snow datasets currently available over the IH-GAR provide different insights on snow cover properties and variability. MODIS observations provide direct estimates of snow cover extent in the form of binary rasters, but are not suitable for robust trend estimations, since the dataset spans only 25 hydrological years and interannual variability may still substantially affect the resulting trends. IT-SNOW and the Po River District product show a relatively good agreement over the investigated area, although elevation plays a significant role: below 1000 m a.s.l. the two products provide substantially different estimates of the amount of water stored within the snowpack, while at medium and high elevations the agreement improves considerably. Moreover, neither dataset can be considered a reliable source for trend analysis. The temporal coverage of IT-SNOW, which spans only 15 years, is too short to limit the influence of interannual variability, whereas the Po River District dataset does not provide complete coverage for the 1992–2021 period. In particular, SWE rasters are not available between 2 July and 2 October of each year, which results in neglecting ephemeral snowfall on the highest peaks during summer, as well as the latest stages of snowmelt in particularly snowy years, such as 2014.

These limitations do not diminish the value of the existing snow datasets, but rather highlight the need for a new snow product specifically designed for climatological applications over the IH-GAR. Such a product should cover a sufficiently long period to allow reliable trend estimations while also providing spatially and temporally continuous information across the study area. Furthermore, the model should be driven by input data that do not need to be corrected using reanalysis products, in order to avoid introducing additional biases into the simulated snow cover. The aim of this thesis is therefore to address these limitations by developing a modelling procedure capable of reproducing the evolution of SWE over the 1951–2023 period, thereby providing improved estimates and new insights in snow cover changes across the last 70 years.

Chapter 4

Snow model

The limitations of existing snow products, discussed in the previous Chapter, highlight the need for a long term snow reconstruction over IH-GAR based exclusively on observational data. To model the snowpack as accurately as possible, it would be necessary to take into account all the physical processes that characterize its evolution throughout the snow season, starting from an estimate of all terms contributing to the snowpack energy balance to then focus on compaction and water percolation, which ultimately precedes runoff. The main issue that makes such an approach effectively inapplicable within the framework of this thesis is that it requires a wide variety of input variables, such as the spatial distribution of incoming shortwave radiation, wind speed maps and soil temperature measurements, which have only been recorded through a spatially dense station network since the 2000s. For this reason, the model presented in this chapter adopts a substantially minimalist approach, given the limited number of meteorological inputs available over a period as long as 1951–2023. Indeed, to simulate snowpack dynamics, only temperature and precipitation data are used: these two variables are therefore the only drivers of snow accumulation and snowmelt.

In the following sections, the temperature and precipitation datasets, as well as the procedure adopted for their spatial reconstruction, are presented first. The second half of this chapter is instead dedicated to the actual description of the snow model. The model calibration process, carried out by comparing model outputs to observational snow depth time series, is addressed separately in Chapter 5.

4.1 Model inputs

The gridded temperature and precipitation datasets, used as model inputs, have both a spatial resolution of 30 arc-seconds. They were developed by the Department of Environmental Science and Policy of the University of Milan, building on earlier work aimed at rescuing new station records and improving the quality of existing ones. The numerical procedure chain used to generate these rasters is based on the anomaly method (Manara, Brunetti, et al. 2026), which assumes that the spatio-temporal

structure of a meteorological signal can be described as the superposition of two fields, namely the climatological normals over a reference period of thirty years and the departures from them. In the following, only the precipitation version is presented. For temperature, the only difference is that are used additive anomalies rather than multiplicative ones.

4.1.1 Gridding procedure

The gridding procedure used to produce daily precipitation maps requires a spatially dense and temporally continuous dataset of station observations, which has undergone rigorous quality checks to identify spikes and missing values. Such control is essential for preventing systematic errors from propagating into the gridded fields. This method is composed of three main steps, namely:

- *calculation of high-resolution monthly climatologies*, which are required as a reference for the reconstruction of daily series. The first step in deriving the gridded climatologies consisted of computing the 1991–2020 precipitation monthly normals for each station of the dataset. The method used to generate the climatology rasters is the Local Weighted Linear Regression (LWLR) of precipitation against elevation, which had already proven to be a robust approach for reconstructing gridded fields (Crespi et al. 2018). At a specific location $(x,\, y)$, the monthly normals were computed as

$$P_{clima}\,(x,\, y) \;=\; q\,(x,\, y) \;+\; m\,(x,\, y) \cdot h\,(x,\, y)\,, \tag{4.1}$$

  where P_{clima} is the reconstructed normal, q and m are the location-specific regression coefficients, and h the elevation of the considered grid point. The weights used to evaluate the regression coefficients account for both the distances and the level of orographic similarity between each station and the target grid point. In particular, the weight assigned to the i-th station for a grid point $(x,\, y)$ was given by

$$w_i\,(x,\, y) \;=\; w_i^{rad}\,(x,\, y) \cdot w_i^{h}\,(x,\, y) \cdot w_i^{st}\,(x,\, y) \cdot w_i^{facet}\,(x,\, y) \cdot w_i^{dsea}\,(x,\, y)\,, \tag{4.2}$$

  where the superscripts refer, respectively, to radial distance, elevation, slope steepness, slope orientation, and distance from the sea. To attain these orographic features, the 7.5 arc-second resolution GTMED2010 DEM was used. (Manara, Brunetti, et al. 2026). Each factor on the right side of Equation (4.2)

was defined through a Gaussian function, so that

$$w_i^{par}(x, y) = \exp\left\{-\frac{[\Delta_i^{par}(x, y)]^2}{c^{par}}\right\}, \tag{4.3}$$

where Δ_i^{par} represents the difference in features between the selected station and the specific grid point, while c^{par} controls the width of the Gaussian. The c^{par} parameters were locally optimized to maximize the agreement between estimated and observed normals. For each grid point, the regression coefficients were estimated taking into account the 30 stations with the highest weights. Furthermore, to evaluate the robustness of the estimates, the LWLR was performed 30 times removing a single station at each iteration. If the difference between the lowest and highest estimates exceeded 30 % of the weighted mean obtained using all 30 stations, the c^{par} were relaxed by 10 % to prevent individual stations from dictating the normal values (Manara, Brunetti, et al. 2026).

- *computation of monthly precipitation maps*, which relies on monthly anomaly maps to reconstruct the actual monthly totals over the entire investigated period as

$$P_{month}(x, y) = P_{clima}(x, y) \cdot A_{month}(x, y), \tag{4.4}$$

where P_{month} and A_{month} are, respectively, the total precipitation and the multiplicative anomaly for a given month at the grid point (x, y). To generate the anomaly maps, all station series were converted into anomaly series relative to the 1991–2020 monthly normals. The departures from the climatologies were then computed at each grid point as a weighted mean of the values from the 10 nearest stations (Manara, Brunetti, et al. 2026). The weights accounted for distance, elevation difference, and the angular distribution of the stations around the grid point being reconstructed. The distance weight, which had a Gaussian form as in Equation (4.3), was halved at the distance of the third closest station. The elevation one, on the other hand, got halved at an elevation difference of 800 meters for grid points below 1600 m a.s.l. or at half the elevation of the grid point otherwise (Manara, Brunetti, et al. 2026). This adaptive scaling scheme allowed the weights to decrease more gradually in areas with a lower station density, while limiting the influence of less compatible stations whenever there was high data availability.

- *estimation of daily precipitation values*, which, at a given location (x, y), were

computed as

$$P\,(x,\,y)\;=\;P_{month}\,(x,\,y)\cdot F_{day}\,(x,\,y)\,, \tag{4.5}$$

where P denotes the daily precipitation value and $F_{day}\,(x,\,y)$ is the ratio between the daily precipitation and the corresponding monthly total. To obtain $F_{day}\,(x,\,y)$ at each grid point, the station time series were first converted into daily precipitation fractions by dividing each daily value by the corresponding monthly one. The resulting fractions were then interpolated over the spatial grid using the same method described in the previous bullet point (Manara, Brunetti, et al. 2026).

4.1.2 Precipitation data

The gridded precipitation dataset used as model input in this thesis is publicly available through the UNIMI Dataverse. Over the IH-GAR, the dataset indicates an average annual precipitation of 1090 ± 190 mm for the 1991–2020 period. This spatially aggregated value, however, masks substantial heterogeneity in the annual precipitation climatology, as shown in Figure 4.1. The highest totals are indeed

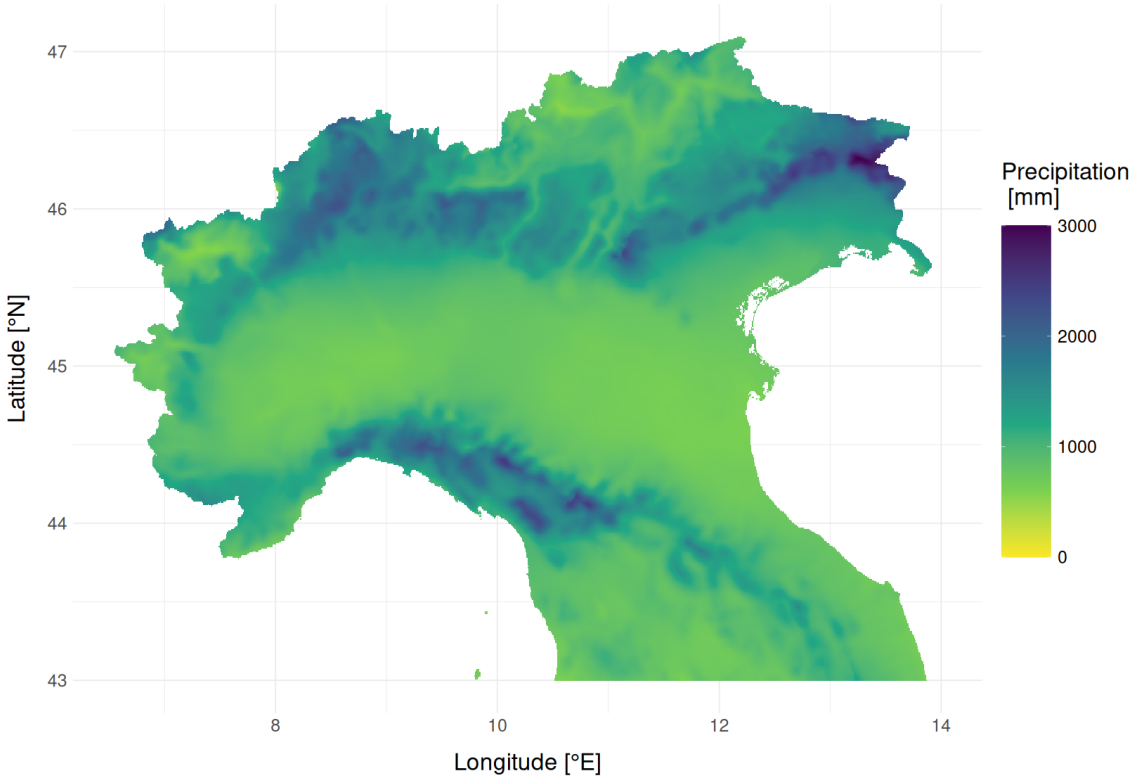

Figure 4.1: Annual precipitation climatology for the Italian Hydrological part of the Greater Alpine Arc (IH-GAR), computed over the 1991–2020 period at a spatial resolution of 30 arc-seconds. The highest totals occur along the Alpine foothills, where air masses undergo orographic lifting.

observed at the foothills of the Alps and in the Northern Apennines, mainly as a result of the stronger orographic uplift of air masses. Conversely, the lowest values occur in the Po River Delta and in the lowlands of the innermost Alpine valleys, such as the Aosta and Venosta ones. The timing of precipitation maxima is site-specific, as different climatic regimes are present in the study region. In the lowlands, precipitation is typically higher during spring and autumn, whereas the inner Alpine regions experience a single peak during summer. The transition zone, located along the foothills of the main mountain ranges, is instead characterized by a markedly wetter autumn season. In particular, close to the border between Italy and Slovenia, cumulative precipitation can exceed 1000 mm over the course of these three months alone.

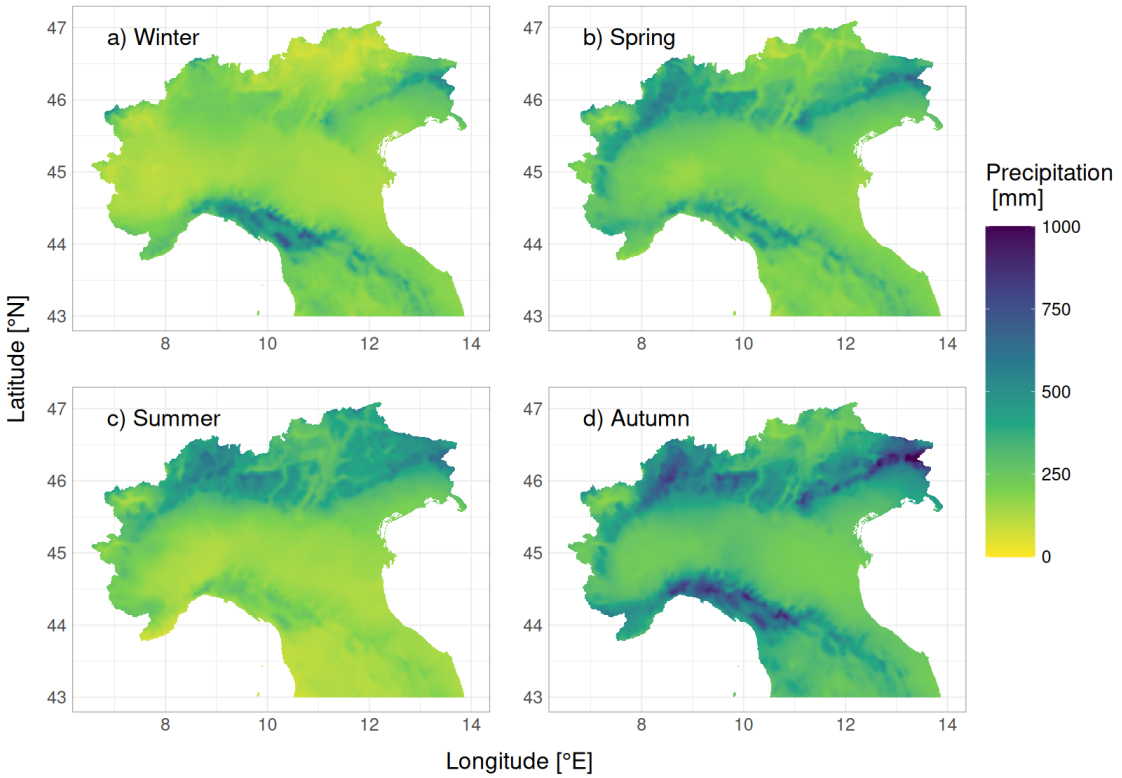

Figure 4.2: Seasonal precipitation climatology for the Italian Hydrological part of the Greater Alpine Arc (IH-GAR), computed over the 1991–2020 period at a spatial resolution of 30 arc-seconds. Autumn is the wettest season on average over the lowlands and the Alpine foothills, whereas the internal Alps experience enhanced precipitation during summer.

Besides this marked spatial and seasonal variability, the precipitation regime also exhibits substantial interannual variability. In Figure 4.3 the additive precipitation anomalies are shown relative to the spatially aggregated annual normal over the 1991–2020 period. An irregular alternation between dry and wet years is evident throughout the record, while no clear long term trend is visible. Indeed, the 21-

year moving average of the anomalies never deviates substantially from zero. This behaviour is consistent with that detected over the Mediterranean region, where precipitation patterns are typically dominated by interannual variability rather than by pronounced long-term trends (Vicente-Serrano et al. 2025).

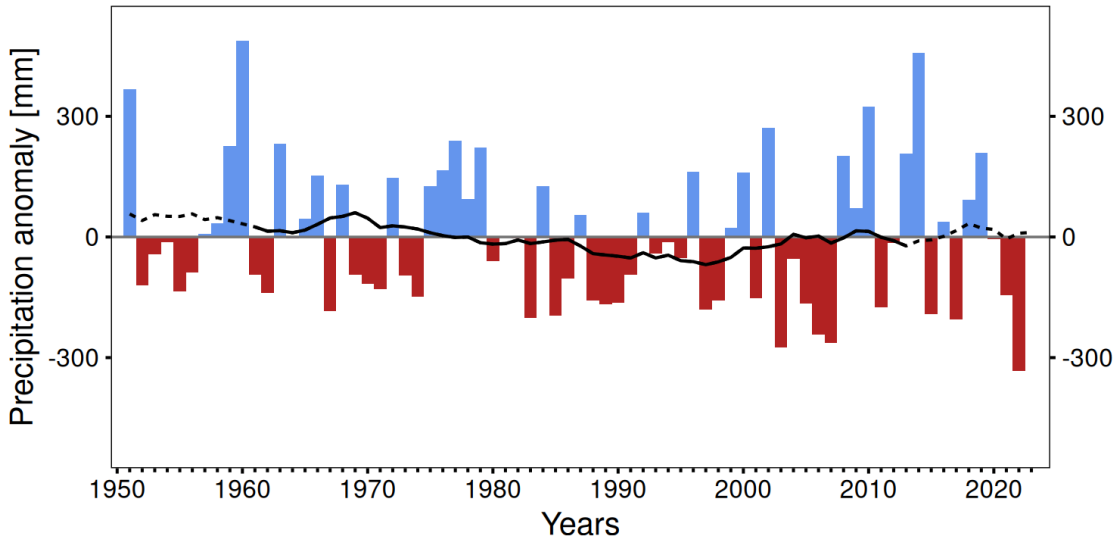

Figure 4.3: Annual precipitation additive anomalies [mm] averaged over the Italian Hydrological part of the Greater Alpine Arc (IH-GAR) for the period 1951–2023, relative to the 1991–2020 climatology. Blue (red) bars indicate positive (negative) anomalies. The black line is a 21-year running mean: its dashed segments, at the beginning and at the end of the record, are computed over a truncated time window.

4.1.3 Temperature data

The paper describing the generation of these temperature rasters is currently under review: the data will be made publicly available through the Unimi Dataverse once the study has been accepted for publication. Three temperature fields were reconstructed on a daily basis, namely mean, minimum and maximum temperature.

As expected, the temperature dataset indicates that the IH-GAR has experienced marked changes in response to global warming. However, this trend is not characterized by a steady increase over the 1951–2023 period, but rather by a pronounced shift in behaviour in the late 1970s, as shown in Figure 4.4. This is consistent with the broader European climate evolution: until the mid-70s, most temperature related indices exhibited either statistically non-significant cooling trends or approximately stationary long-term conditions, with departures from climatology mainly associated with interannual variability (Fioravanti et al. 2016). From the 1980s onwards, however, warming in the IH-GAR has peaked up significantly, with mean

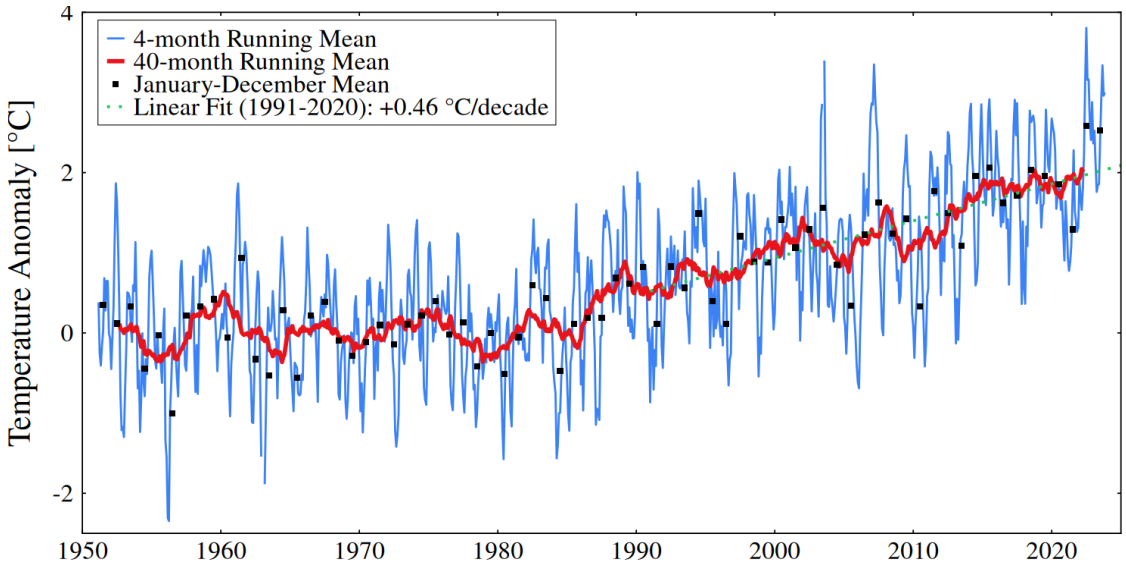

Figure 4.4: Mean daily temperature anomaly series with respect to the 1951-1980 climatology over the IH-GAR. The 4 month and 40 month running means are shown, as well as a linear fit of the 1991–2020 period. The spatially averaged mean daily temperatures are increasing by $+0.46\,°C$/decade.

daily temperatures increasing at a rate of $+0.46\,°C$/decade over the last thirty years. Maximum daily temperatures exhibit a slightly stronger trend of $+0.48\,°C$/decade over the same period, whereas minimum daily temperatures show a somewhat lower increase of $+0.44\,°C$/decade. Such rapid warming represents one of the main drivers of glacier melt, which has intensified dramatically over recent decades (Smiraglia and Diolaiuti 2016). This phenomenon is also having significant consequences for snow resources within the IH-GAR as, for instance, the snowline has been estimated to rise by approximately 166 m for each additional degree of warming during the winter months (Hantel et al. 2012).

In particular, air temperature strongly influences the persistence of snow cover, especially during the middle and final stages of the ablation season. In this context, it is important to consider not only the increase in mean temperature, but also how the thermal regime is evolving with elevation and throughout the year. For example, between 2000 and 2500 m a.s.l., the input dataset clearly shows that both the maximum and minimum temperature distributions during April, while shifting towards higher values, have also significantly widened as a consequence of prolonged mild conditions during the ablation months. This is illustrated in Figure 4.5, where the distributions for the periods 1951–1980 and 1991–2020 are compared.

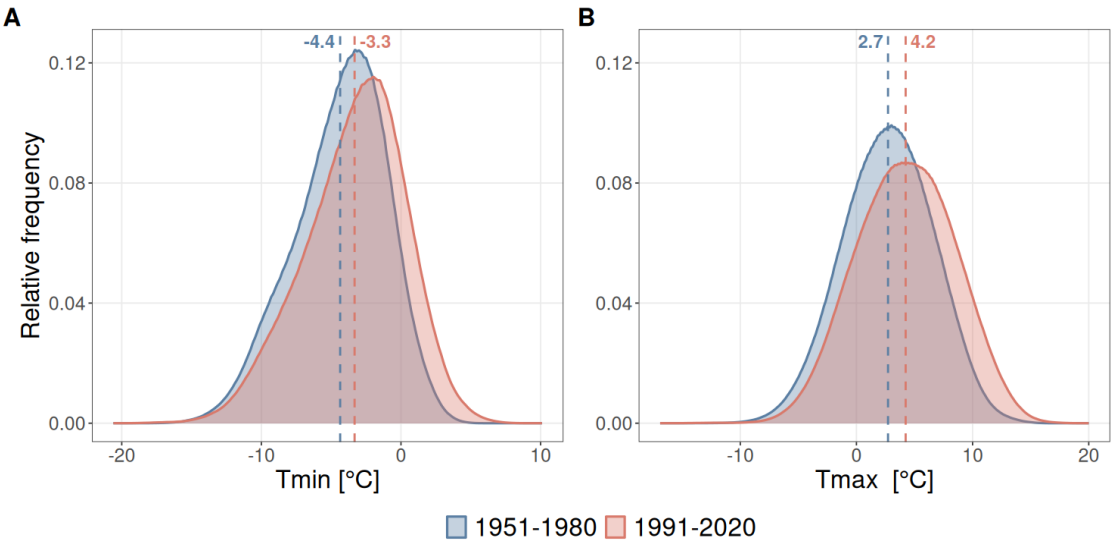

Figure 4.5: Minimum (a) and maximum (b) temperature distributions in April during the 2000–2500 m elevation band. In both panels, the distributions for the periods 1951–1980 and 1991–2020 are compared.

4.2 Model structure

Since both temperature and precipitation inputs are gridded fields, the snow model developed in this thesis adopts a raster based approach, in which adjacent pixels are treated independently, therefore neglecting every kind of horizontal transport. Model outputs retain the same temporal and spatial resolution of the inputs, respectively being 1 day and 30 arc-seconds. The following subsections describe the operational workflow developed to reconstruct the snow water equivalent time series. The snow model was implemented in R language, and the corresponding code is provided in Appendix B.

4.2.1 Snow accumulation

The simplest way to estimate daily snowfall when only air temperature and total precipitation are available is to compare the mean daily temperature with a threshold temperature T_τ in order to determine the precipitation phase. This approach, however, completely neglects the variability of air temperature within the day, and therefore yields only rough estimates. When daily maximum and minimum temperatures are also available, such as in this thesis, the fraction of precipitation that reaches the ground as snow can be evaluated more realistically, since the full daily temperature

range is known. In particular, the snowfall S_f at each grid point is evaluated as

$$S_f = \begin{cases} P & T_{\max} < T_\tau \\ P \cdot \dfrac{T_\tau - T_{\min}}{T_{\max} - T_{\min}} & T_{\min} < T_\tau < T_{\max} \\ 0 & T_{\min} > T_\tau, \end{cases} \tag{4.6}$$

where $T_{\max}$ and $T_{\min}$ are the daily maximum and minimum temperatures respectively, and P the total daily precipitation.

4.2.2 Snowmelt

The snow models discussed in Chapter 3 adopted different approaches to snowmelt estimation. "J-snow", the modelling chain that led to the generation of the Po River District dataset, explicitly simulated the energy balance of the snowpack, whereas "S3M" employed a hybrid degree-day method, accounting for both air temperature and shortwave radiation. Neither method, however, is suitable for investigating snowpack dynamics over the 1951–2023 period, as both require input variables that are not continuously available throughout the entire study period.

The modelling approach adopted is therefore a canonical degree-day method, in which mean air temperature is used as a proxy for the various energy fluxes affecting the energy state of the snowpack. In particular, the model follows the formulation proposed by Magnusson et al. 2014, whereby the daily snowmelt at each grid point is computed as

$$\mathrm{M} = \mathrm{DDF}\left\{(T_{mean} - T_{melt}) + m_m \cdot \ln\left[1 + \exp\left(-\frac{T_{mean} - T_{melt}}{m_m}\right)\right]\right\}, \tag{4.7}$$

where T_{mean} is the mean daily temperature, computed as the average between the maximum and minimum ones, T_{melt} is the threshold snowmelt temperature, and m_m is a smoothing parameter measured in °C. The term enclosed in curly brackets can be interpreted as an effective melt temperature: the logarithmic contribution ensures a smooth transition between melting and freezing conditions around T_{melt}.

In Equation (4.7), DDF denotes the degree-day factor, which is a parameter measured in $\mathrm{mm\,{}^\circ C^{-1}\,d^{-1}}$: its multiplication by the temperature dependent term yields the amount of snow melted away on a given day. To account, at least partially, for the seasonal variability of shortwave radiation, which plays a significant role in the snowpack energy balance, the degree-day factor is prescribed with an annual

cycle according to

$$\mathrm{DDF}_t \;=\; \mathrm{DDF}_{ave} \;+\; \mathrm{DDF}_{amp} \cdot \sin\left[\frac{2\pi\,(t \;-\; 81)}{366}\right], \tag{4.8}$$

where t is the day of the solar year, DDF_{ave} is the mean degree day factor, whereas DDF_{amp} is the amplitude of its oscillation. With this formulation, the DDF peaks on summer solstice, in qualitative agreement with the seasonal evolution of incoming solar radiation. Moreover, during non-leap years, the DDF value corresponding to 29 February is neglected.

4.2.3 SWE evaluation

Finally, the temporal evolution of snow water equivalent over the investigated period is simulated by accumulating, on a daily basis, the difference between solid precipitation and snowmelt. The SWE on day t is therefore computed at each pixel as

$$\mathrm{SWE}(t) \;=\; \mathrm{SWE}(t \;-\; 1) \;+\; S_f(t) \;-\; M(t), \tag{4.9}$$

where t or $t \;-\; 1$ denote the day on which each quantity is evaluated. When the computed snowmelt exceeds the available snow water equivalent at a given location, the corresponding pixel value is set to zero, as this condition indicates the complete meltout of the snowpack. Application of the snow model over the IH-GAR territory requires the calibration of the parameters (T_τ, T_{melt}, DDF_{ave}, DDF_{amp}, m_m). The procedure employed to determine their optimal values is described in the following chapter.

Chapter 5

Model calibration

5.1 HS data

The snow depth dataset used to calibrate the degree-day model described in the previous chapter was created through a data rescue activity. It combines HS time series from Matiu et al. 2021 with data provided by national and regional meteorological and hydrological services, including ARPA Piemonte, Centro Funzionale della Valle d'Aosta, Meteoswiss, WSL Institute for Snow and Avalanche Research SLF, ARPA Lombardia, Meteotrentino, ARPA Veneto, and the Austrian Hydrological Service (Maugeri et al. 2026). The final dataset includes 1052 stations, mostly automatic, distributed across the entire Alpine Arc, as shown in Figure 5.1. Depending on the station, these time series can extend back to the 1960s, although most observations date from 1990 onwards.

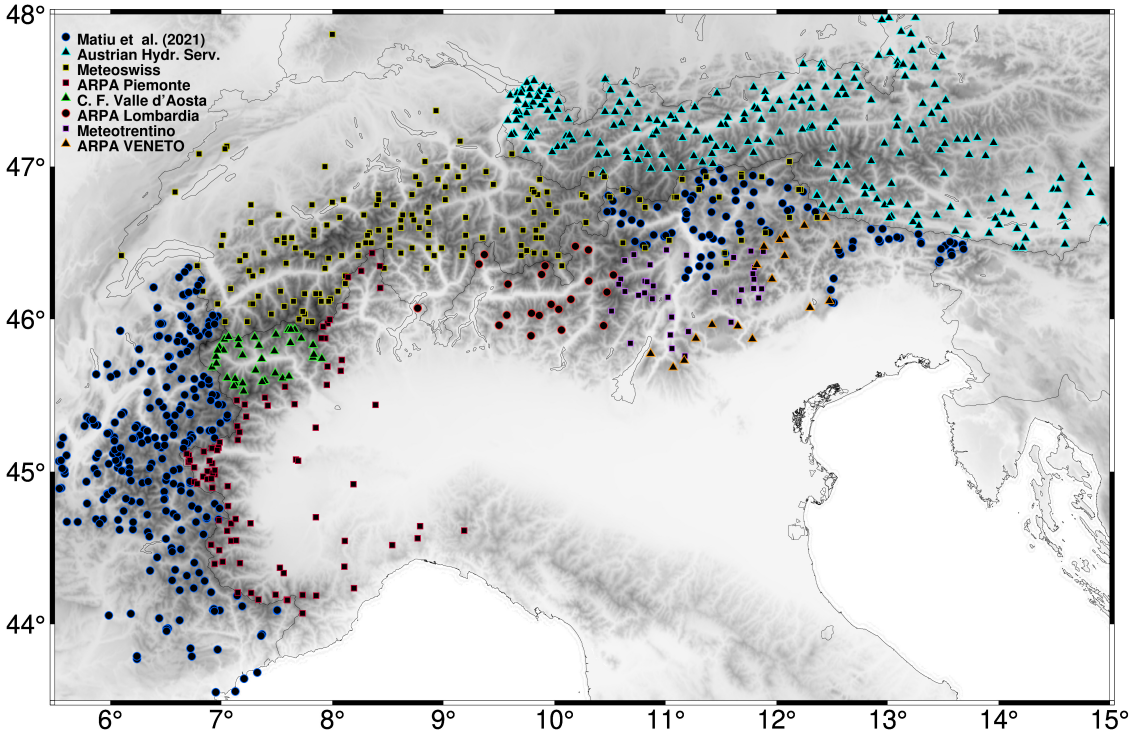

Figure 5.1: Spatial distribution of the stations that made up the original dataset. Each station is marked with the symbol associated with the data source from which the data were obtained. Image from Maugeri et al. 2026.

The original HS dataset, however, presented several issues that needed to be addressed before it could be considered suitable for further analysis. The first reliability check that was done focused on the accuracy of the station metadata, specifically longitude, latitude and elevation. To this end, the ALOS AW3D30 digital elevation model (DEM) was used to compare the reported elevation of each station with the elevation of the DEM pixel nearest to its coordinates. A compatibility

threshold of 12 m was adopted. Although this criterion is relatively strict given the complex topography of the Alpine Arc, it was considered appropriate due to the high spatial resolution of the DEM, namely 30 m. Based on this threshold, more than half of the stations were classified as unreliable, as show in Figure 5.2.

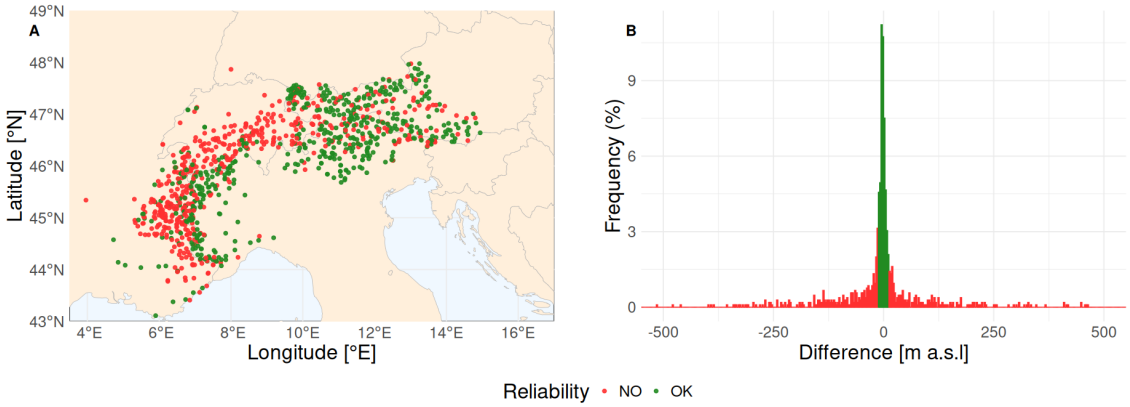

Figure 5.2: (a) Spatial distribution of stations with reliable (unreliable) metadata. (b) Frequency histogram of the differences between the reported station elevations and those derived from the ALOS AW3D30 DEM. In both panels, green represents reliable stations, for which the absolute elevation difference didn't exceed 12 m, while red indicates stations that didn't meet this criterion.

Moreover, a second quality check revealed that not only the station metadata, but also the snow depth time series required further processing. In fact, to investigate potential issues within the snow depth dataset, HS series were compared with MODIS observations for stations falling within the study area and the period covered by Manara, Almagioni, et al. 2025. For each station in this subset, the MODIS time series was extracted from the nearest grid point, while the HS data were converted into binary snow cover series using a threshold of 0 cm: positive snow depth values were assigned a value of one, whereas zeros remained unchanged. Both time series were then organized in pseudo-hydrological years, extending from 1 September of a given year to 31 August of the following one, to allow for a direct comparison. A perfect match was not expected, as station measurements are collected in-situ, whereas MODIS provides spatially distributed observations with pixels classified as snow free if the snow cover fraction does not exceed 0.5. Nevertheless, the comparison shown in Figure 5.3 revealed substantial discrepancies between the two datasets. In particular, the comparison between MODIS and HS time series showed a considerable scatter in SCD values, as visible in panel (a) of Figure 5.3. Two distinct groups of data point deviating markedly from the identity line can indeed be identified: the first is concentrated in the upper-left corner of the plot, while the second one lies much

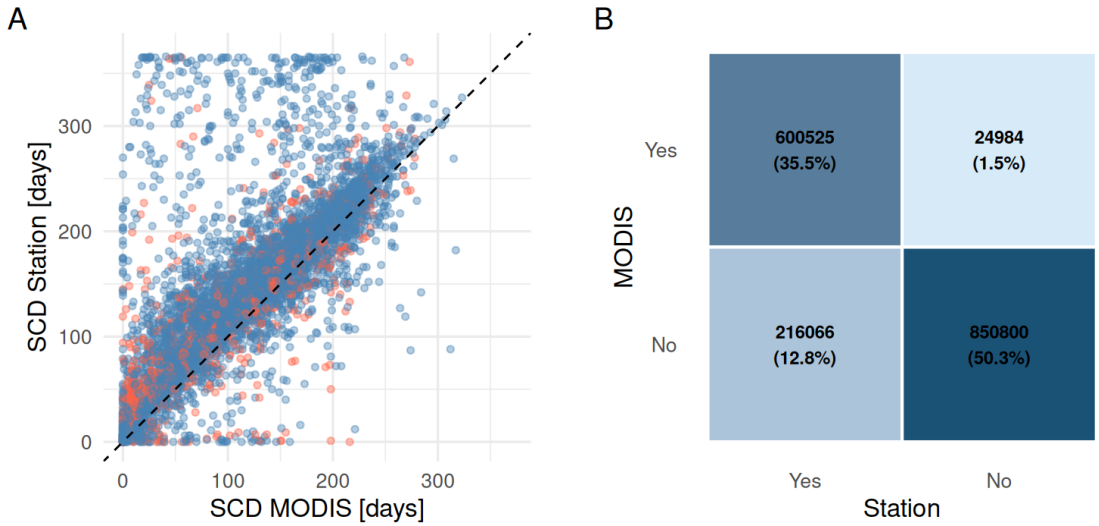

Figure 5.3: (a) Comparison between SCD values computed from HS series and MODIS data. Each dot represents a pseudo-hydrological year, which is the period spanning from 1 September of a given year to 31 August of the following one. Blue dots are data retrieved from reliable stations, whereas orange ones are not. (b) Confusion matrix comparing snow cover observations according to MODIS and HS data.

closer to the x-axis than expected. The latter group mainly consists of hydrological years characterized by numerous and/or prolonged gaps in the snow depth records, with observations missing for much of the snow season. A representative example is shown in Figure 5.4.

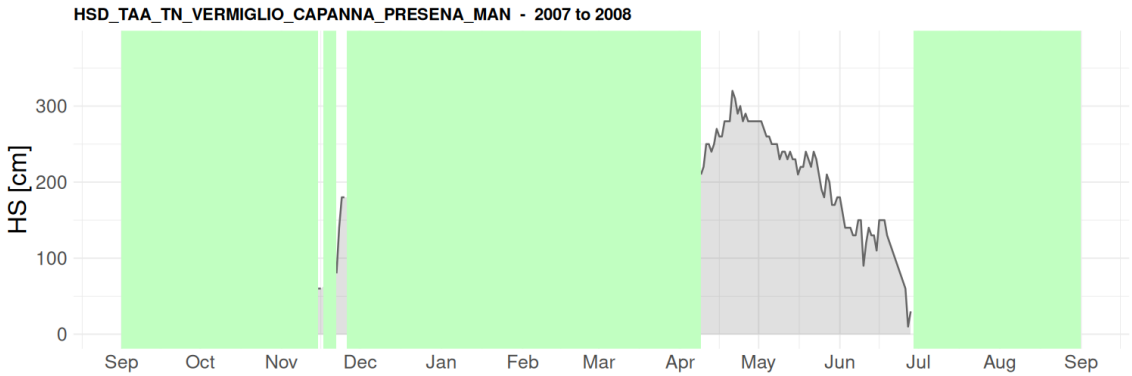

Figure 5.4: Example of an HS time series affected by extensive data gaps (highlighted in green). This series cannot be reliably reconstructed using standard gap-filling methods because the SOD (snow onset date) is unknown.

The former group, instead, is characterized by HS measurements that, when converted into snow cover series, indicate the presence of snow on the ground for most

of the hydrological years. This behaviour results from snow depth values that rarely reach zero. Such anomalies may be caused by a bias in the measuring instrument, which incorrectly identifies the ground level as being a few centimetres above its actual position, or by systematic errors in recording snow free conditions during summer. An example of each of these cases is shown in Figure 5.5.

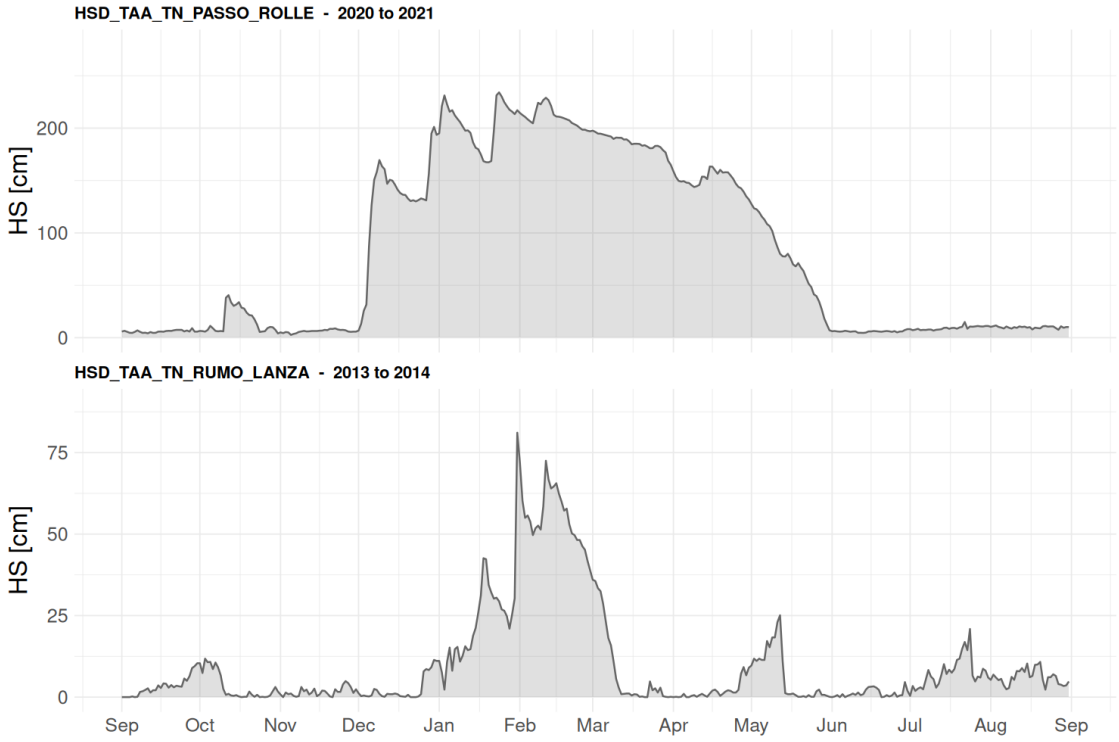

Figure 5.5: (a) HS time series showing an incorrect identification of the ground level, with snow depth values remaining approximately constant at $\sim 10\,\text{cm}$ throughout the summer. (b) HS time series showing anomalous summer spikes, clearly caused by erroneous readings. Snow cover during the hottest months is indeed not possible at Rumo Lanza, which is located at 1100 m a.s.l.

Clearly, missing values and erroneous summer readings are not mutually exclusive, and more complex numerical anomalies may therefore occur within the same pseudo-hydrological year. Moreover, several HS series are affected by other numerical artifacts, such as spikes or sudden drops in snow depth. Given these issues, specific procedures were developed to improve the quality of the snow depth dataset before proceeding with the calibration. The remainder of this section is devoted to describing separately the methods adopted to improve the accuracy of the station metadata and to correct, whenever possible and necessary, the numerical artifacts that initially compromised the reliability of the data.

5.1.1 Position correction

The main issue with the unreliable coordinates was that most of them had been rounded to the nearest arc-minute. In the complex topography of the Alps, this spatial resolution, corresponding to roughly 2 km, can result in elevation differences of several hundred meters, potentially affecting the representativeness of the reported station location. For this reason, the metadata of each station with unreliable coordinates were manually reviewed to identify its precise position, an important step for ensuring the accuracy of the subsequent model calibration. The review involved consulting the catalogues of the agencies responsible for the different stations. While this process did not lead to any improvement in some cases, such as for ARPA Lombardia, it allowed the retrieval of the exact coordinates of dozen of stations managed by WSL Institute for Snow and Avalanche Research SLF, Meteoswiss, Meteo France and the Austrian Hydrological Service. The updated coordinates were then inspected using Google Earth to visually verify whether the identified sites corresponded to meteorological stations, as illustrated in panel (a) of Figure 5.6. Panel (b), on the other hand, displays the original locations relative to their true positions, highlighting that most corrections involved shifts of less than 1 km.

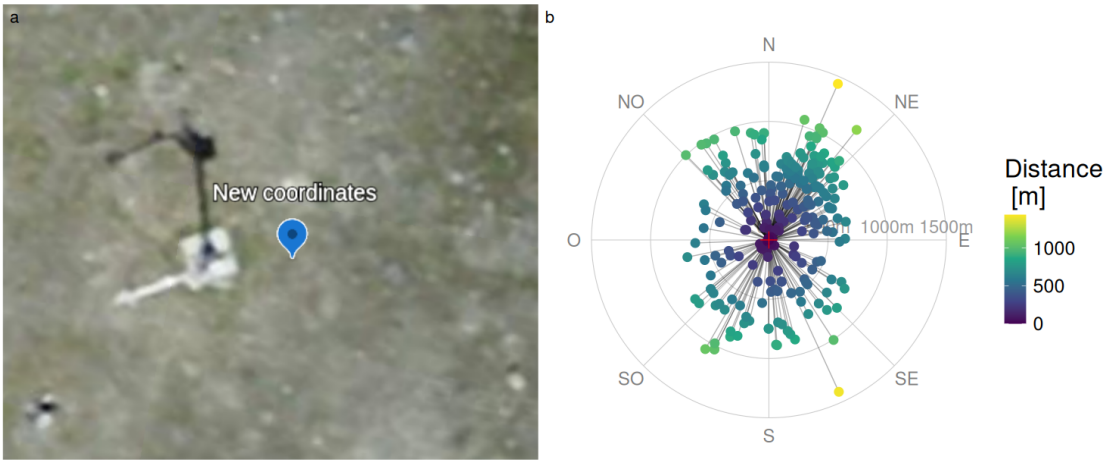

Figure 5.6: (a) Example of a Swiss AWS viewed from satellite imagery. The original coordinates placed the station approximately 1 km east of its actual location. (b) Original station locations relative to their corrected positions, indicated by the red cross at the center of the plot.

Overall, this procedure allowed the precise coordinates of approximately 300 stations to be recovered, increasing the number of stations whose elevations were consistent with the DEM from 532 to 829. For the remaining stations, the original coordinates were retained because no sufficiently reliable sources were available to support a correction.

5.1.2 HS series correction

Before applying any corrections to the snow depth time series, the data were visually inspected to identify the main issues and develop suitable correction procedures. For each station, comparison plots were generated for all hydrological years containing at least one non-missing HS value, as illustrated in Figure 5.7. Each plot consisted of three panels: the first displayed the original HS time series, the second one showed the corresponding snow water equivalent (SWE) obtained using the ΔSnow model, while the third presented the SWE simulated using the melt parameters optimized by Magnusson et al. 2014 and a temperature threshold of 2.5 °C for precipitation phase partitioning. Although using model outputs to identify errors in the observations may

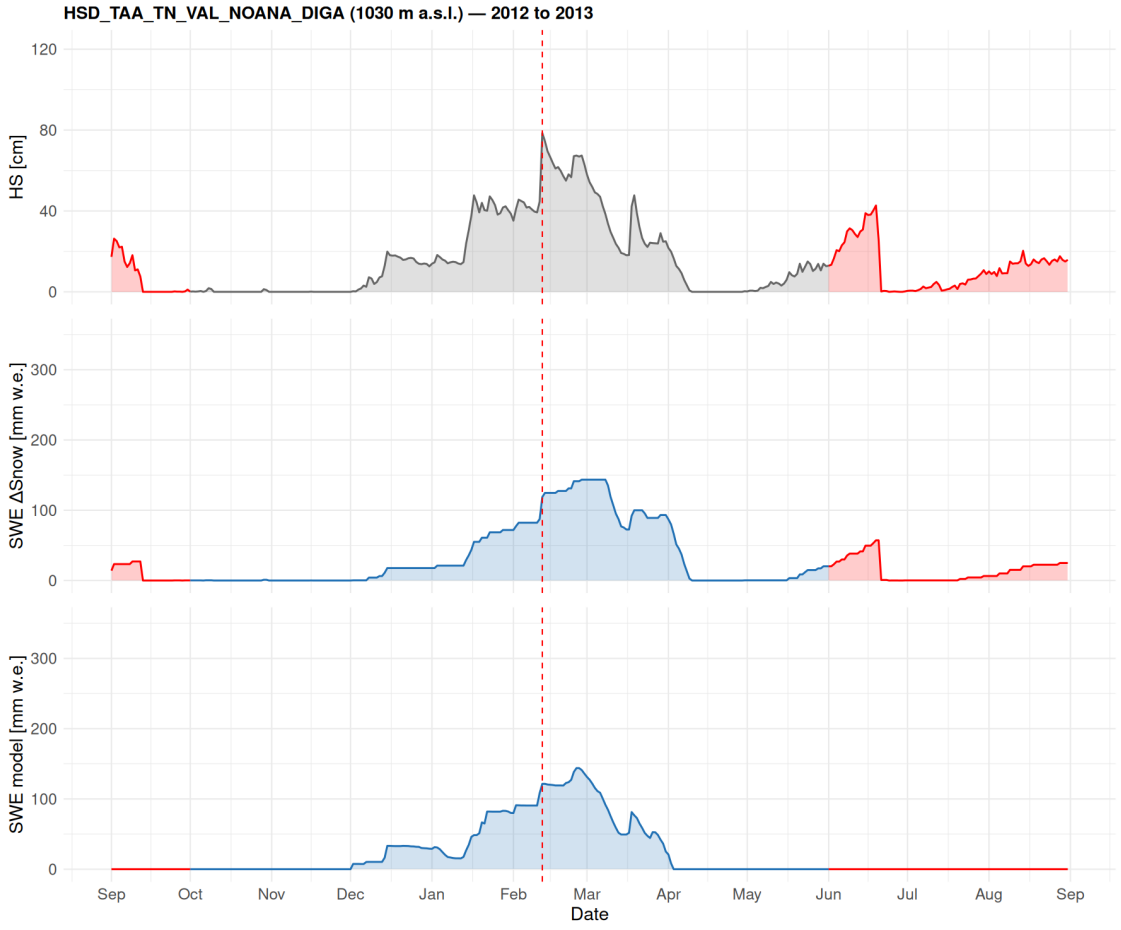

Figure 5.7: Example of a comparison plot used to visually identify numerical artifacts in the HS series. The three panels show (a) the original snow depth measurements, (b) the corresponding SWE values obtained using the ΔSnow model, and (c) the simulated SWE. The latter was obtained using the melt parameters optimized by Magnusson et al. 2014 and a temperature threshold of 2.5 °C for phase partitioning. The red shaded areas indicate the summer window to be automatically set to zero.

appear circular, the relatively high temperature threshold adopted for precipitation phase partitioning provides additional confidence in distinguishing numerical issues from actual snowfall events. In particular, when a spike in snow depth, such as the one observed in early September in Figure 5.7, is not accompanied by a corresponding increase in the simulated SWE, it can reasonably be considered a numerical artifact since with a high temperature threshold the number of snowfall events is overestimated, since even rainfall sometimes meets the requirements for solid precipitation computation. Each comparison plot also included a red shaded interval corresponding to a period typically associated with bare ground. HS values within this window were proposed to be automatically set to zero. The duration of this interval varies depending on the elevation of the station as shown in Table 5.1: During the preliminary visual inspection, hydrological years unsuitable for

Table 5.1: Proposed time windows to be automatically set to zero.

Ele. band	**Start date**	**End date**
0 – 500 m	1 May	31 October
500 – 1000 m	15 May	15 October
1000 – 1500 m	1 June	30 September
1500 – 2000 m	15 June	15 September
2000 – 2500 m	1 July	31 August
Above 2500 m	15 July	15 August

reconstruction were identified and flagged for exclusion. This was necessary when missing values were too numerous to extract meaningful information from the HS series, or when measurement noise could not reliably be distinguished from the actual snow signal. In particular, all series ending before the snow depth had returned to zero, or beginning after snow accumulation had already started, were discarded. In these cases the onset or end of the snow season could not be determined accurately, making a proper reconstruction impossible. This truncation of the HS series led to the exclusion of the majority of stations located in ski areas, where measurements typically did not cover the entire snow cycle. Moreover, the main numerical artifacts for each station were recorded, together with any necessary adjustments to the proposed summer zeroing window. Based on these observations, a set of correction steps was developed and subsequently applied to each station separately. Although this semi-automated approach required substantial manual effort, it was necessary to account for the different types of numerical artifacts affecting individual stations. The denoising and gap-filling procedure consisted of the following steps:

- *series filtering*, which consisted of discarding all pseudo-hydrological year previously flagged as unsuitable for signal reconstruction. In addition, negative HS values were reclassified as missing data, since snow depth must always be bigger or equal to zero.

- *summer zeroing*, in which the selected summer window was automatically set to zero. For pseudo-hydrological years characterized by a ground-level bias over the entire period, as in panel (a) of Figure 5.5, the average HS value was first computed and subtracted from every daily measurement, and only then was the summer window zeroed. Any values that became negative because of noise fluctuations were clipped to zero at this stage.

- *spike and drop detection*, aimed at identifying such numerical artifacts and reclassifying them as missing data. ?articular attention was paid to spiked higher than 80 cm and to drops larger than 50 cm.

- *gap filling*, which consisted of linearly interpolating between the endpoints of every gap bounded by two strictly positive values or by two zeros. This approach clearly cannot reconstruct the actual snow depth on each day within the gap, but nevertheless makes it possible to distinguish between bare and snow covered ground. Requiring both endpoints to be positive or zero limits the cases in which the interpolation assigns positive snow depth values to days when the presence of snow is uncertain.

- *first visual inspection*, carried out to verify that the gaps had been filled correctly and to decide how to proceed. Whenever the gap filling strategy produced faulty reconstructions, it was repeated selecting which gap to fill.

- *gap to zero*, which consisted of replacing all remaining missing values with zeros. Although this may seem somewhat crude, it does not introduce significant artifacts, because series whose gaps substantially affected the start or the end of the snow season had already been discarded during series filtering. In the worst case, only a couple of days per pseudo-hydrological year are forced to zero.

- *zero under threshold*, applied only when the first visual inspection had revealed minor residual noise. In such cases, all days with HS below a given threshold, typically a couple of centimetres, were set to zero.

- *final visual inspection*, performed to assess the plausibility of the corrected series. Any remaining issues were addressed case by case.

Figure 5.8 illustrates how a typical HS time series changed throughout the main stages of the correction procedure.

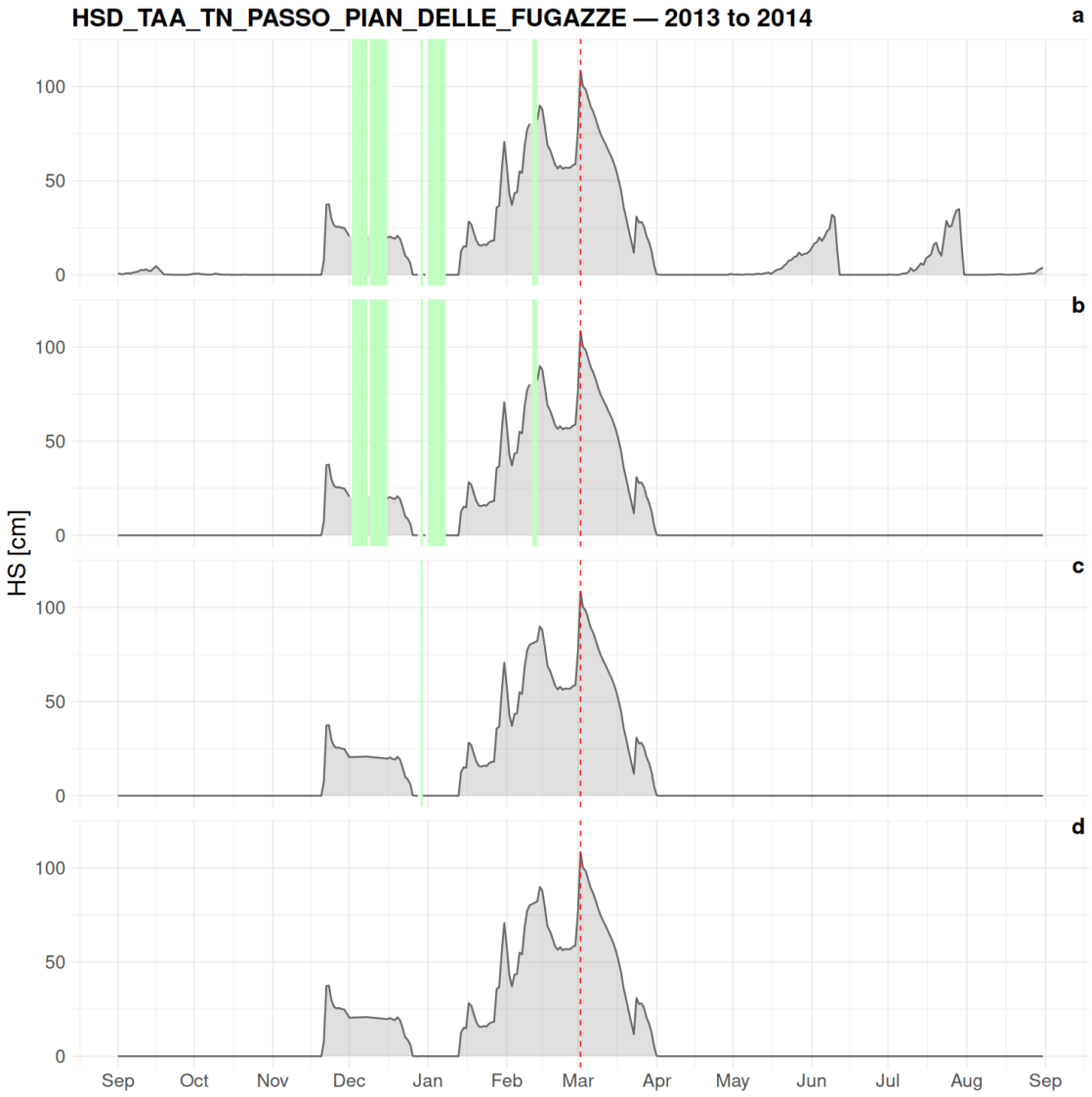

Figure 5.8: Example of the HS correction procedure. (a) Original HS series. (b) HS series after setting the summer window to zero. (c) HS series after gap filling. (d) Final corrected HS series.

5.1.3 Final dataset

The two quality-control procedures described in the previous subsections led to the removal of 220 stations, reducing the network from 1052 to 832. The total number of pseudo-hydrological records also decreased from 27 388 to 20 248, corresponding to a reduction of approximately 26%. Despite these changes, the spatial coverage and distribution of the network remained largely unchanged, as station positions

were adjusted by no more than 1.5 km. The elevation distribution also remained similar, with both the original and corrected networks mainly concentrated between 1000 and 2000 m above sea level. A substantial change was instead observed in the consistency between the reported station elevations and the ALOS AW3D30 DEM, as illustrated in Figure 5.9. In fact, following the corrections, 714 out of the 832 remaining stations passed the reliability check on elevation, and therefore appear green in panel (b).

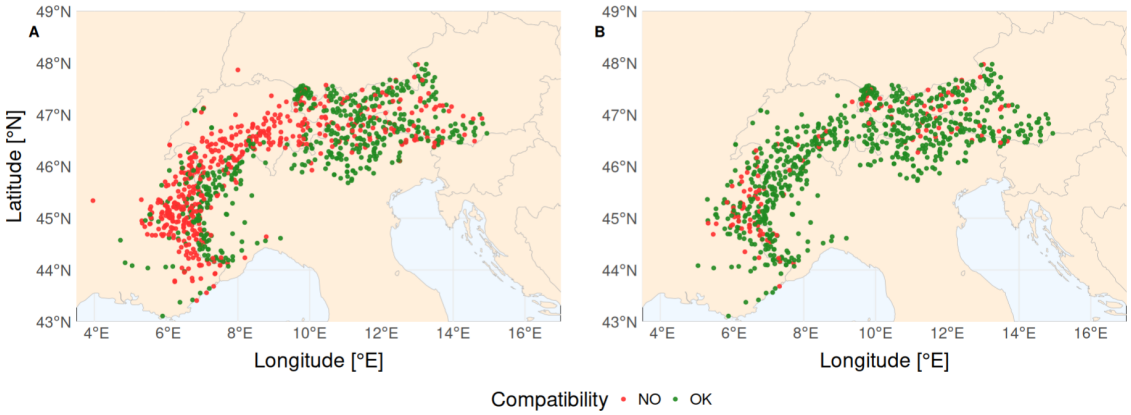

Figure 5.9: Compatibility between reported station elevations and the nearest DEM pixel before (a) and after (b) correcting station metadata. In both instances the ALOS AW3D30 global DEM was used. Stations are classified as reliable (OK) if the elevation difference is smaller than 12 m.

In addition to the visual inspection, which represented the final step of the denoising and gap filling procedure, the comparison with MODIS observations was repeated to assess whether the corrections had also reduced the discrepancies between the two datasets. The resulting scatter plot, shown in panel (a) of Figure 5.10, indicates a substantial improvement. In particular, all the data points now cluster around the identity line, suggesting that the issues responsible for the two previously anomalous groups of points have been successfully solved. The slight tendency to SCD overestimation by station measurements is consistent with expectations, considering that MODIS pixels are classified as snow free whenever at least half of their area consists of bare ground. However, a considerable spread in data point remains. To investigate the source of this scatter, a more detailed comparison was conducted revealing that MODIS appear to struggle over highly forested areas, consistently with the findings discussed in Chapter 3. Indeed, comparing individual HS series with the corresponding MODIS ones often revealed patterns similar to the one shown in Figure 5.11, where the snow covered periods identified by satellite observations (light blue) are noticeably shorter than those indicated by station measurements.

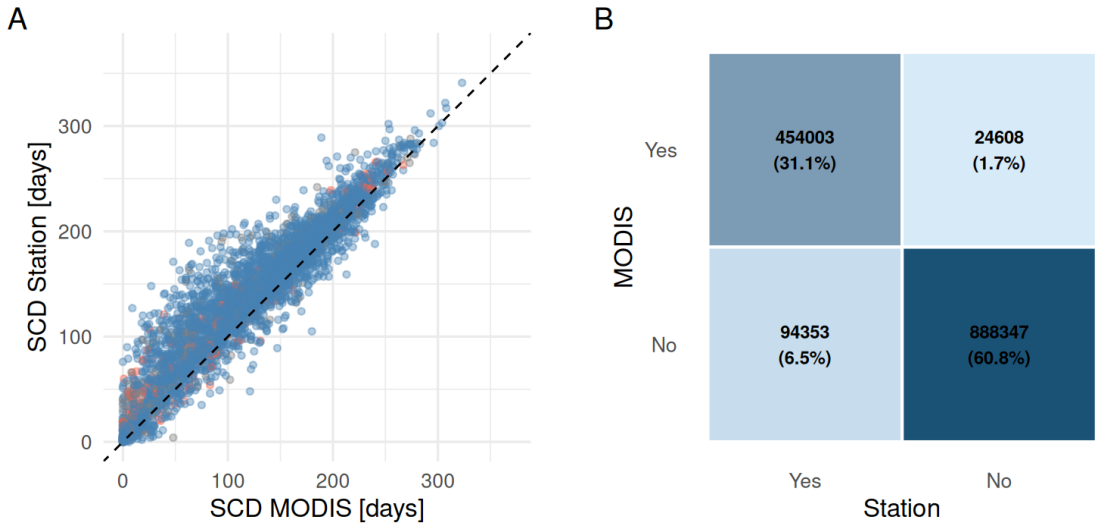

Figure 5.10: (a) Comparison between SCD values computed from corrected HS series and MODIS data. Each dot represents a pseudo-hydrological year. Blue dots are data retrieved from reliable stations, whereas orange ones are not. (b) Confusion matrix comparing snow cover observations according to MODIS and HS data.

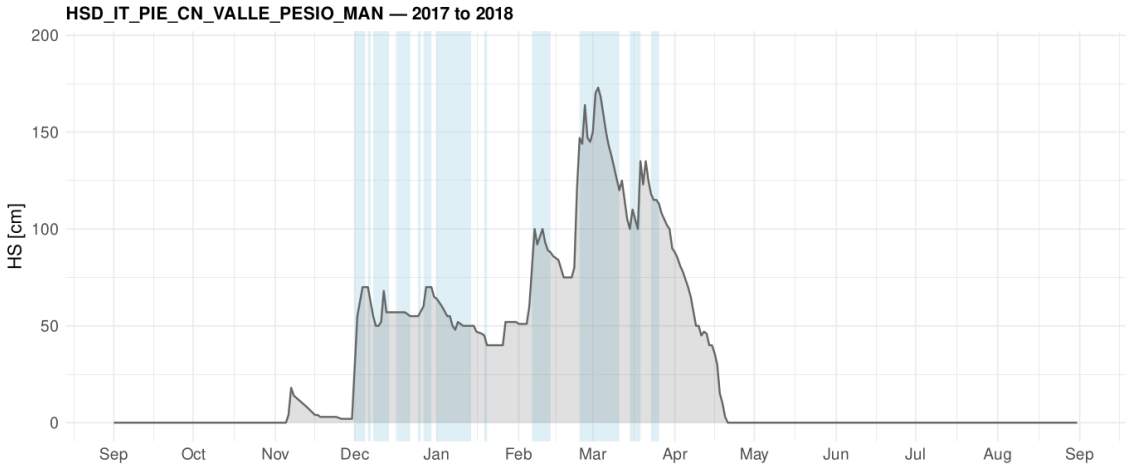

Figure 5.11: Comparison between HS and MODIS snow cover time series. Light blue shaded areas indicate periods classified as snow covere by MODIS.

5.2 Calibration procedure

The previously performed correction of the station metadata is particularly relevant at this stage, as accurate knowledge of the stations' geographical coordinates is essential to ensure proper calibration. From now on, only the 714 reliable stations of the 832 constituing the corrected dataset were taken into account. For each of their

coordinates, a reconstruction of daily temperature and precipitation fields over the 1951–2023 period was attempted. This procedure was successful for only 632 station out of the 714, since in some cases there were too few data to obtain reliable outputs. Having this temperature and precipitation data made it possible to reconstruct SWE series for each of the remaining stations, which could then be compared with the corresponding snow depth series, since both quantify the amount of snow within the snowpack, but through different unit of measurements. However, the conversion from one to another is not straightforward, since it requires the knowledge of the vertical profile of snow density, which in the framework of this thesis remains unknown. This issue can be solved by means of a snow model such as ΔSnow, which makes the conversion from snow depth to snow water equivalent under the assumption that every increase of more than a threshold τ in snow depth can be interpreted as a snowfall event. This multi layered simulation of the snowpack even predicts snow compaction by mean of a Newtonian approach. Within the framework of this thesis, however, it was decided to not make a direct comparison between model outputs and observational data, given that the usage of ΔSnow during the calibration would have made it susceptible to modelling errors and uncertainties within ΔSnow itself. The chosen approach consisted instead in converting both HS and SWE series into binary ones, labelling as zero the days with 0 cm of snow depth (or 0 mm of SWE), whereas positive values got labelled as ones. The main focus of the calibration was then the temporal allignement of these binary series, with the goal of making them overlap in snow presence prediction as much as possible. The loss function chosen to compare model outputs with snow depth time series is

$$\mathcal{L} = 1 - \mathrm{CSI}, \tag{5.1}$$

where CSI stands for the Critical Success Index, computed as

$$\mathrm{CSI} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP} + \mathrm{FN}}. \tag{5.2}$$

This means that the loss function effectively focuses on the snow event, disregarding true negative predictions (i.e. zero to zero correspondance in binary series).

5.2.1 Homogeneous calibration

The first step of the calibration procedure consisted on focusing on the temperature threshold for phase partitioning T_τ alone. The melt parameters are instead fixed, while the melt parameters were fixed to the values proposed by Magnusson et al.

2014. In particular, Considering that only T_τ was assumed as free, the optimization

Table 5.2: Model parameters proposed by Magnusson et al. 2014.

Parameter	Value
DDF_{ave}	2.2 mm w.e.
DDF_{amp}	1.7 mm w.e.
m_m	0.5 °C
T_{melt}	0.0 °C

approach consisted of uniformly sampling the interval between 0 and 3 degrees, experimentally known as the range where the transition from rain to snow occurs. The temperature step used in this phase was 1×10^{-3} °C, which is already too much given that temperatures are usually measured with a precision of one-tenth of a degree. The least loss value is

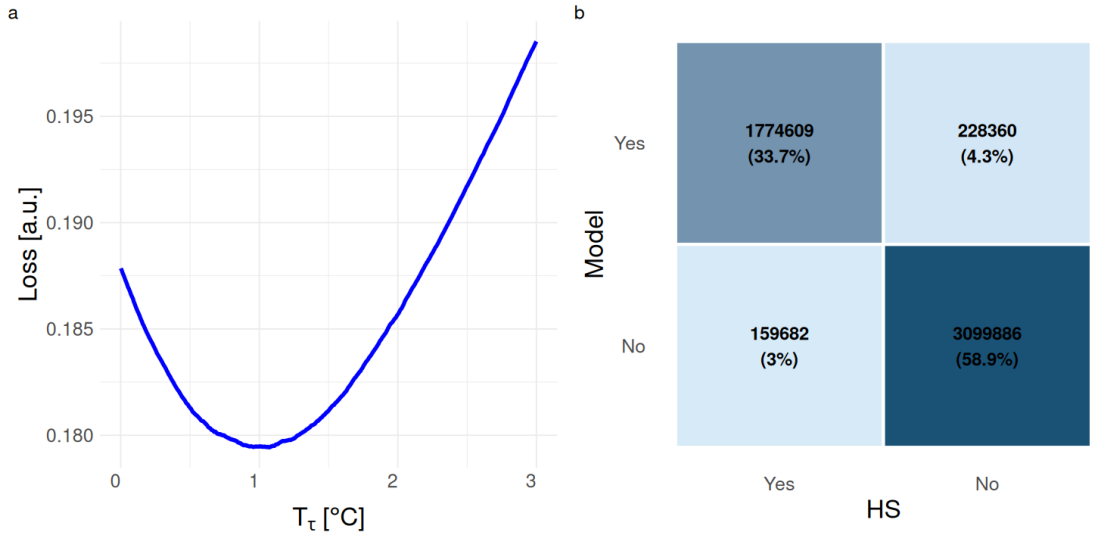

Figure 5.12

a
b
c
Model
Yes
No
286577
(19.4%)
85576
(5.8%)
24614
(1.7%)
1083985
(73.2%)
852636
(32.3%)
105118
(4%)
87890
(3.3%)
1598010
(60.4%)
635396
(55.8%)
37666
(3.3%)
47178
(4.1%)
417891
(36.7%)
Yes
No
HS

Figure 5.13

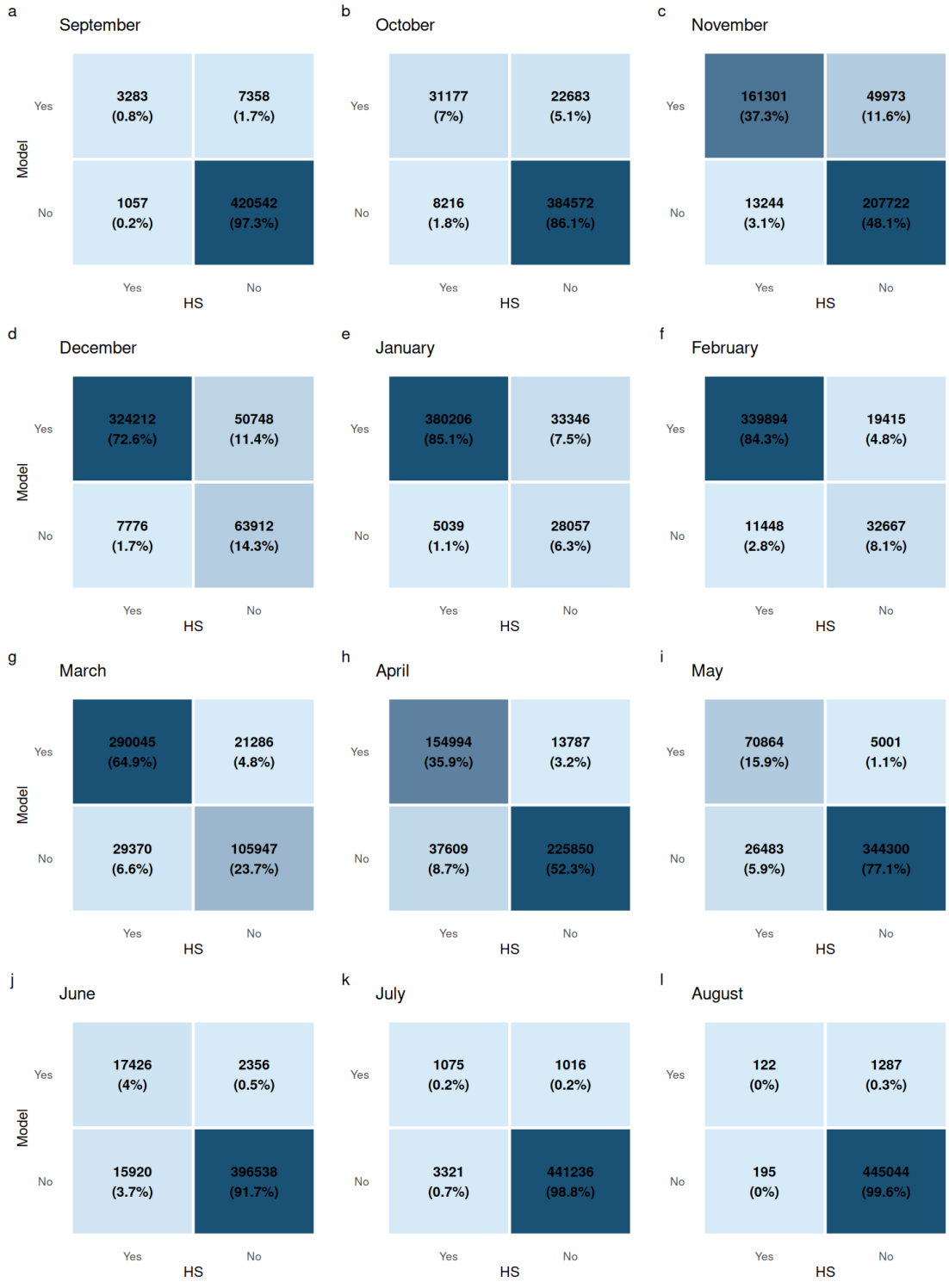

Figure 5.14

5.2.2 Simulated annealing

Chapter 6

Results

6.1 Glaciated areas

Appendix A

MODIS

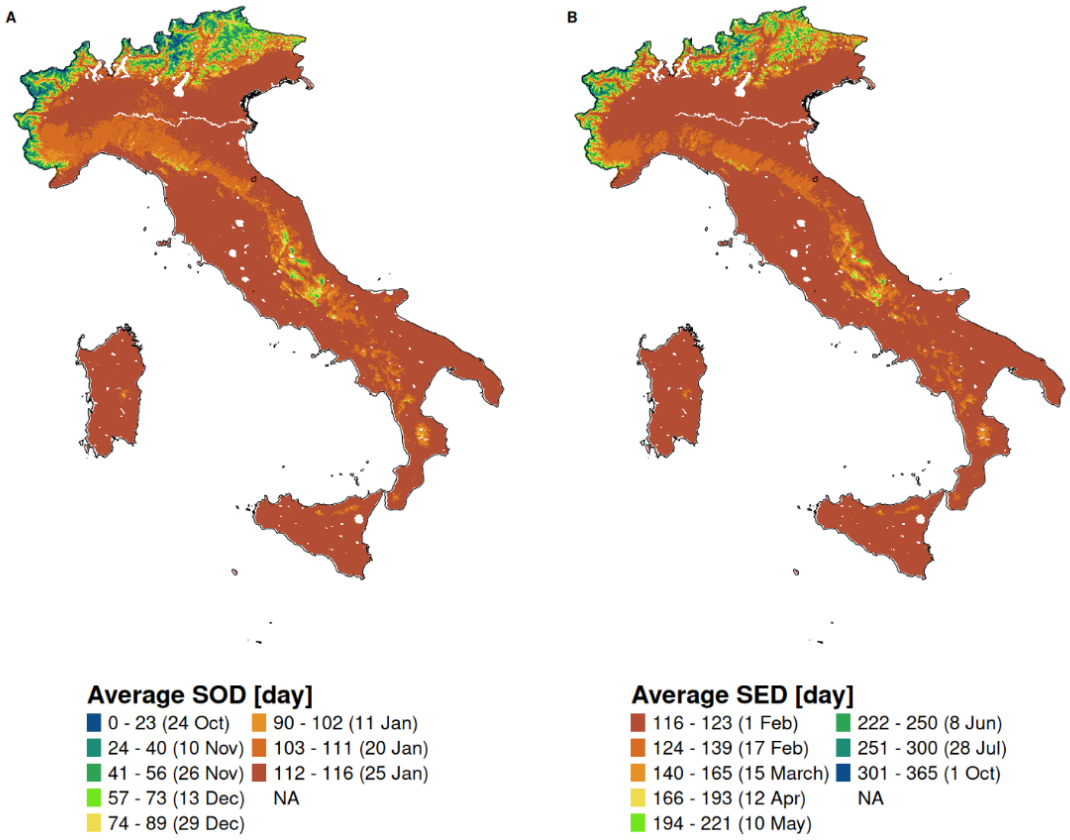

Figure A.1: (a) Spatial distribution of the average Snow Onset Date (SOD) over the 2001–2025 period. (b) Spatial distribution of the average Snow End Date (SED) over the 2001–2025 period. Both exhibit substantial heterogeneity across the investigated domain, since an earlier onset and a later end of the snow season can be detected in mountain ranges across the whole peninsula.

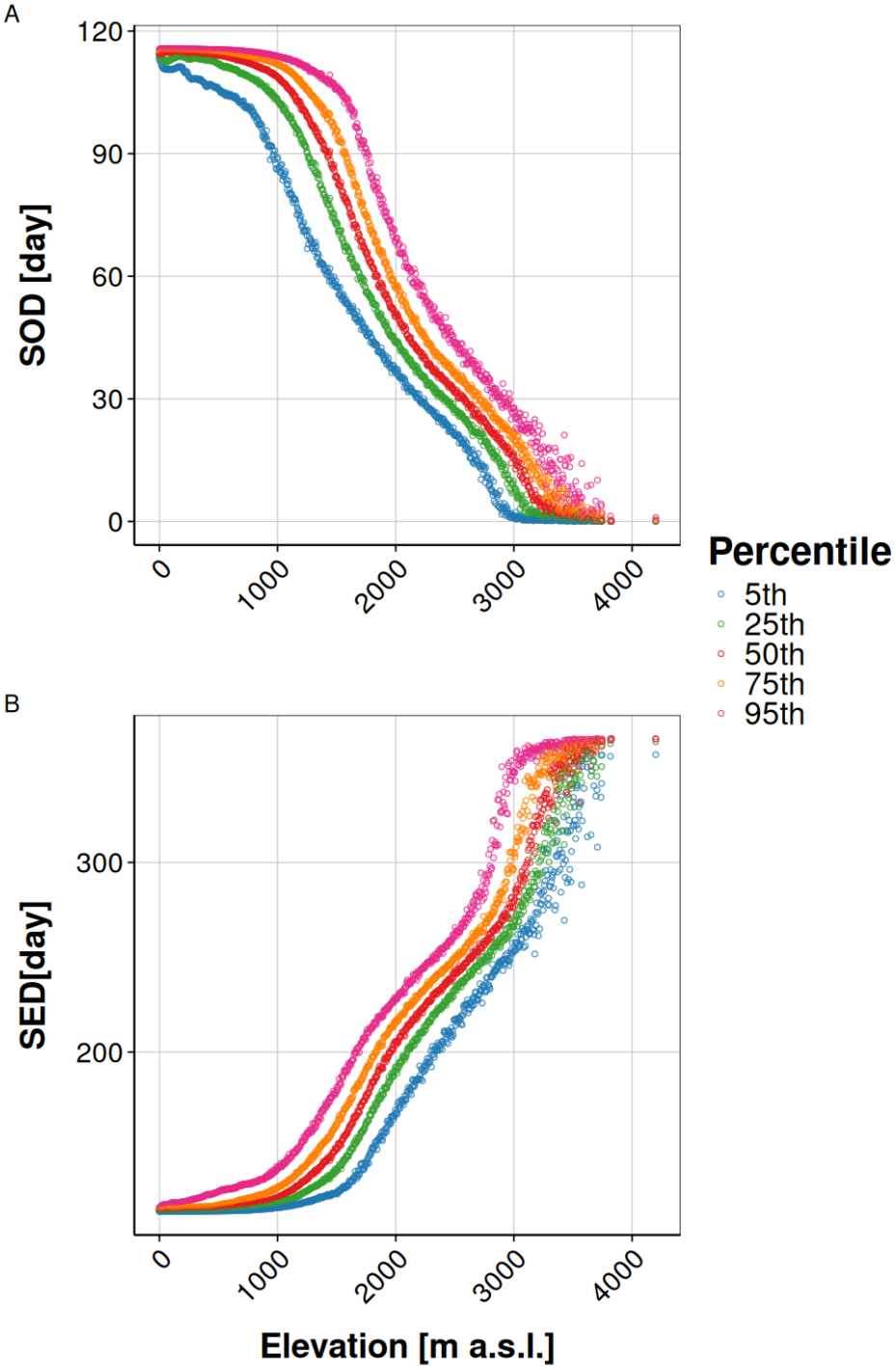

Figure A.2: Quantiles of the average Snow Onset Date (SOD, panel a) and Snow End Date (SED, panel b) distributions within each 5 m wide elevation band over the 2001-2025 period. Both panels show a clear monotonic relationship with elevation, since high altitude alpine catchments experience earlier onsets and later ends of the snow season. The relatively large spread within each band indicates that elevation alone cannot fully account for the observed variability in snow cover, highlighting the need to consider additional topographic variables.

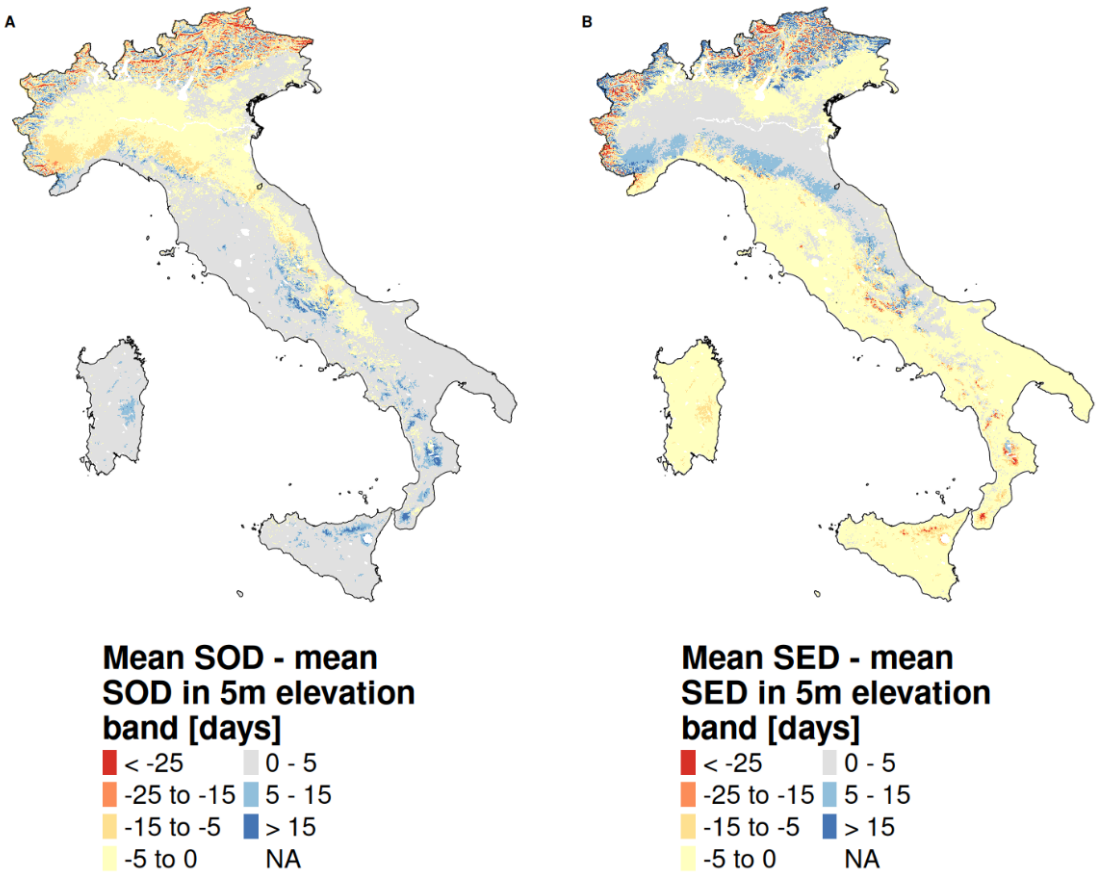

Figure A.3: (a) Spatial distribution of the difference between the SOD value of each individual pixel and the mean SOD of the corresponding 5 m wide elevation band. (b) Spatial distribution of the difference between the SED value of each individual pixel and the mean SED of the corresponding 5 m wide elevation band. The Eastern Apennines typically experience an earlier onset and a later end of the snow season relative to the elevation wise Italian average. This behaviour could be relater to cold air masses originating from the Balkans. In the Alps, pixel aspect plays a significant role, as south facing slopes often experience earlier melt.

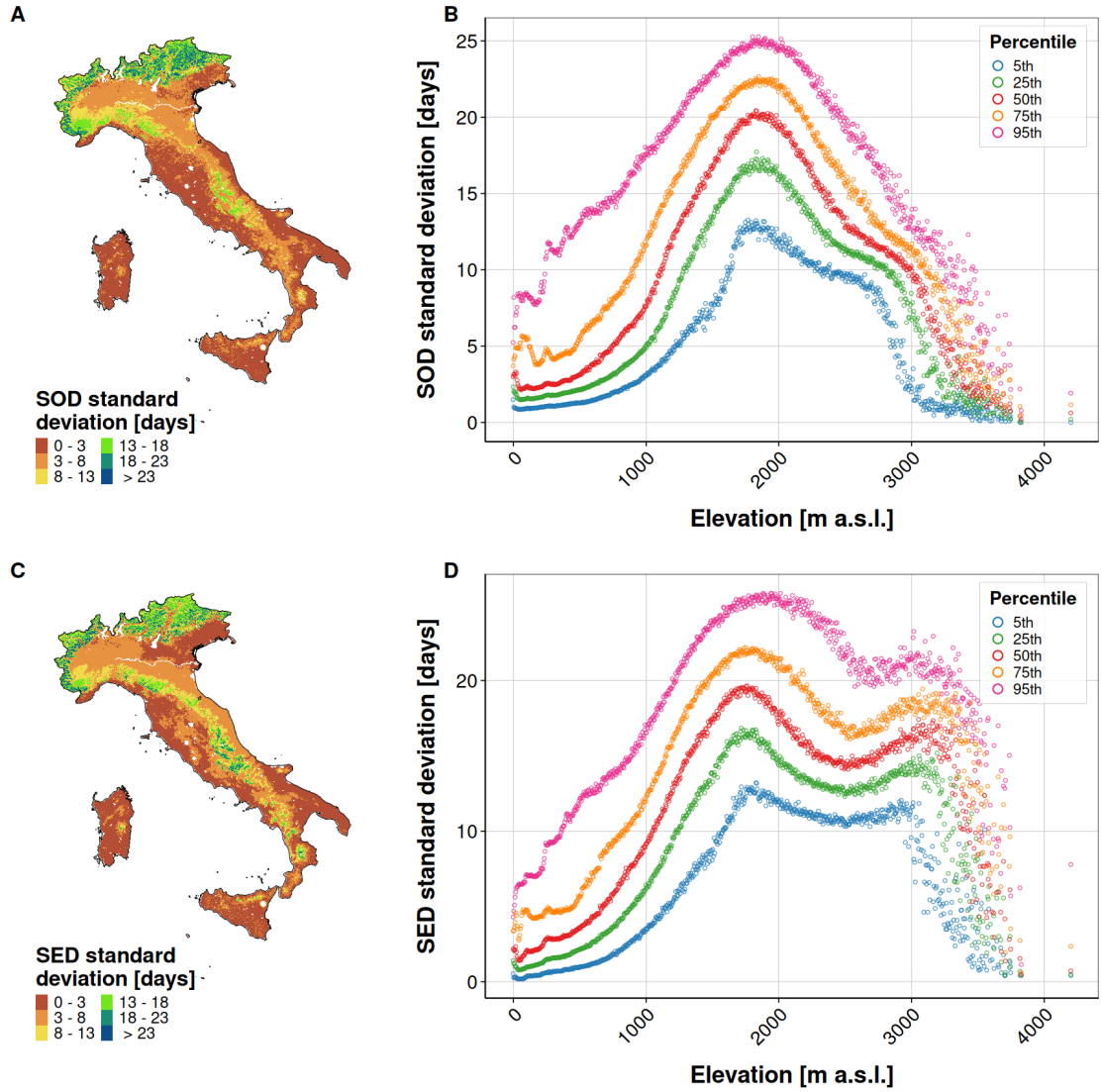

Figure A.4: (a - b) Spatial distribution and quantiles elevation wise distribution of the standard deviation of SOD from the mean, computed over the 2001-2025 period. (c-d) Spatial distribution and quantiles elevation wise distribution of the standard deviation of SED from the mean, computed over the 2001-2025 period. Both metrics exhibit an almost monotonic increase up to 1800 m a.s.l., beyond which a substantial decrease can be detected. It's worth noting that the SOD decrease is monotonic, whereas SED exhibits a local maximum at 3000 m. This secondary peak could be related to ice-covered pixels that are now ice-free as a consequence of retreating glaciers in a warming enviroment, thereby experiencing ordinary seasonal snow cover rather than a year-round one.

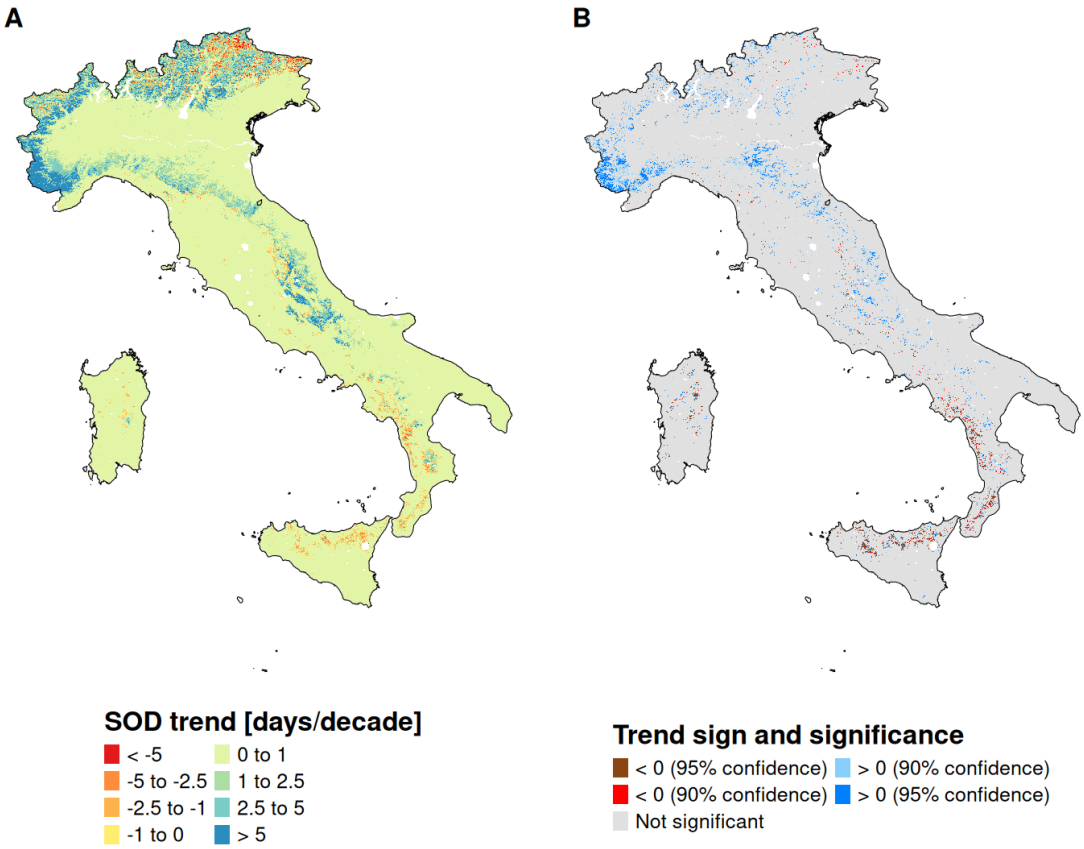

Figure A.5: (a) SOD trend slope computed by means of the non-parameter Sen-Theil estimator over the 2001–2025 period. (b) Trend sign and significance over the 2001-2025 period, computed by means of non-parametric Mann-Kendall test. SOD values are increasing over the Apennines and the Western Alps, meaning that fewer snow cover days occur between the 1 October and 25 January, whereas in the Eastern Alps there is no clear overall trend, as increasing SOD values are counterbalanced by decreasing ones.

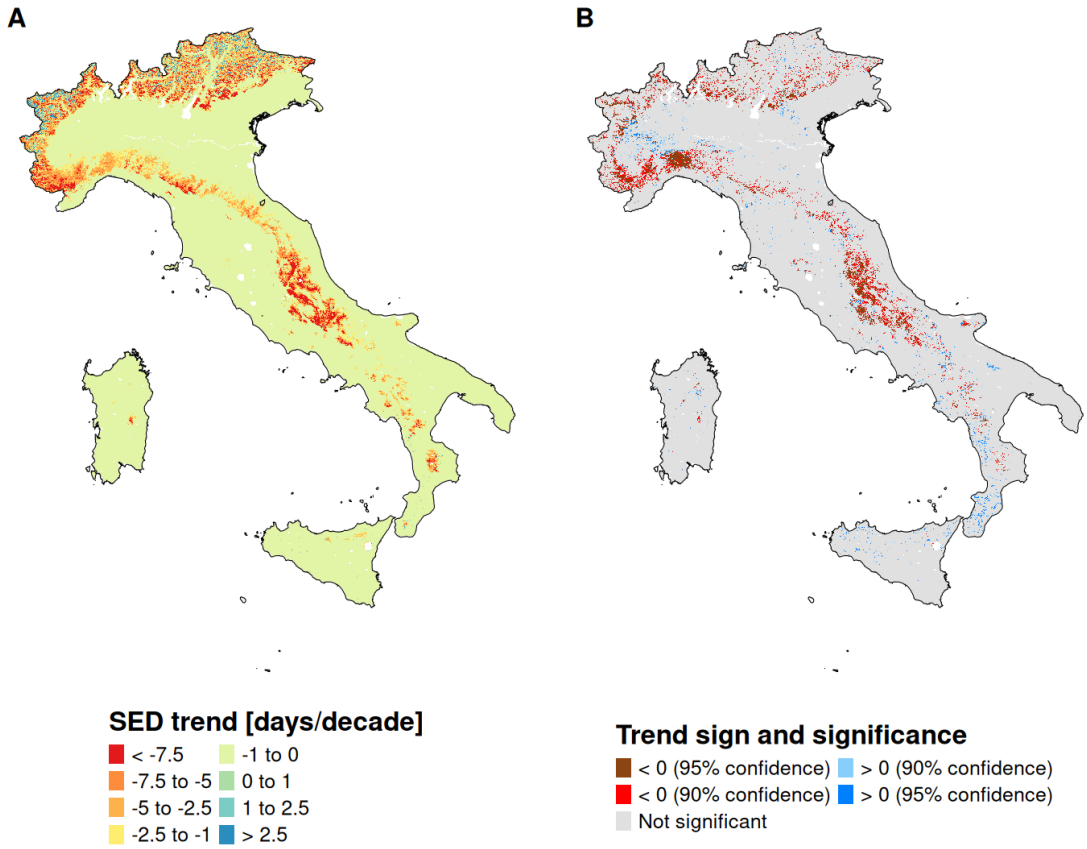

Figure A.6: (a) SED trend slope computed by means of the non-parameter Sen-Theil estimator over the 2001–2025 period. (b) Trend sign and significance over the 2001-2025 period, computed by means of non-parametric Mann-Kendall test. SED values are decreasing over the whole Italian Peninsula, with just a few exceptions in the north-east. By comparing these maps with those of SED and SCD values, it becomes evident that the main contributor to the reduction in snow season length is an earlier end of the snow season, rather than a delayed onset of snow cover.

Appendix B

Model implementation

```r
#######################################################################
#                       SWE model implementation                      #
#######################################################################
rm(list = ls()); invisible(gc())
library(ncdf4)

years <- 1951:2023
fname_in <- "input.dat"
setwd("/home/filippo/Desktop/Codicini/Master_Thesis/SWE_model/")


# Function to check whether two netCDF variables are defined on the
# same spatial grid, comparing directly the coordinate values
com_coord <- function(fname_1, var_1, fname_2, var_2, tol = 1e-6){

  fnames <- c(fname_1, fname_2)
  vars <- c(var_1, var_2)
  coords <- vector("list", 2)

  # Importing coordinates of the two variables
  for(i in 1:2){
    nc <- nc_open(fnames[i])
    v <- nc$var[[vars[i]]]
    if(is.null(v)){
      on.exit(nc_close(nc))
      stop(paste("Variable", vars[i], "not found in", fnames[i]))
    }
    stopifnot(length(v$dim) >= 2)

    coords[[i]] <- list(x = v$dim[[1]]$vals, y = v$dim[[2]]$vals)
    nc_close(nc); rm(nc, v); invisible(gc())
  }

  # Checking if the number of grid points matches along dimensions
  if(length(coords[[1]]$x) != length(coords[[2]]$x)) return(FALSE)
  if(length(coords[[1]]$y) != length(coords[[2]]$y)) return(FALSE)
```

```
37
38    # Checking if coordinates match one by one.
39    if(any(abs(coords[[1]]$x - coords[[2]]$x) > tol)) return(FALSE)
40    if(any(abs(coords[[1]]$y - coords[[2]]$y) > tol)) return(FALSE)
41    return(TRUE)
42  }
43
44
45  # Fun. to compute solid precipitations through a single temperature
46  # threshold approach: the threshold, together with the daily min
47  # and max temperatures, determines the fraction of precipitation
48  # that falls as snow or that falls as rain.
49  compute_solid_precipitations <- function(prec, tmnd, tmxd, t_th){
50
51    precs <- prec
52
53    # Threshold temperature is comprised within daily tmin and tmax
54    mask <- tmnd <= t_th & tmxd > t_th & !is.na(prec) & !is.na(tmnd)
        & !is.na(tmxd) & prec > 0
55    precs[mask] <- prec[mask]*(t_th - tmnd[mask])/(tmxd[mask] - tmnd[
        mask])
56    rm(mask); invisible(gc())
57
58
59    # Daily temperatures are too high for snowfall to occur
60    mask <- tmnd > t_th & !is.na(prec) & !is.na(tmnd) & !is.na(tmxd)
        & prec > 0
61    precs[mask] <- 0
62    rm(mask); invisible(gc())
63
64
65    # Alligning precipitation and temperature masks, in order to
66    # compute SWE only on the overlap area
67    mask <- !is.na(precs) & (is.na(tmnd) | is.na(tmxd))
68    precs[mask] <- NA
69    rm(mask); invisible(gc())
70
71    return(precs)
72  }
73
74
75  # Fun. to compute oscillating DDF. The period of the oscillation is
76  # 366 days, but for non-leap years the 29 February value is deleted
77  compute_ddf <- function(year, ddf_ave, ddf_ampl){
78    # DDF computation
```

```
  len <- 366; offset <- 81;idx <- 1:len
  ddf <- ddf_ave + ddf_ampl * sin(2 * pi * (idx - offset)/len)

  # Leap on non-leap year?
  mask <- year %% 4 == 0 && (year %% 100 != 0 || year %% 400 == 0)
  if(!mask) ddf <- ddf[-60]
  return(ddf)
}


# Fun. to compute snowmelt based on Magnusson approach. The degree
# day is corrected to ensure smoothness at threshold temperature.
compute_melt <- function(year, tmean, ddf_ave, ddf_ampl, expfact){

  # Computing oscillating DDF and degree day
  ddf <- compute_ddf(year, ddf_ave, ddf_ampl)
  deg_day <- tmean + expfact * log(1 + exp(-tmean/expfact))

  # Snowmelt computation (dimension match?)
  stopifnot(dim(tmean)[3] == length(ddf))
  melt <- sweep(deg_day, 3, ddf, "*")

  # Looking for eventual overflows
  stopifnot(all(is.finite(melt) | is.na(melt)))
  return(melt)
}


# Function to create the container that stores the SWE map, updated
# on a daily basis across the 1951--2023 period.
create_swe_container <- function(fname){
  nc <- nc_open(fname)
  prec <- ncvar_get(nc, "total_precipitation")
  nc_close(nc)

  appo <- array(0, dim = c(dim(prec)[1], dim(prec)[2]))
  rm(prec); invisible(gc())
  return(appo)
}


# Function to save swe maps during a given solar year to netCDF.
# The structure of the input files is used in order to keep the
# same architecture for input and output files
save_annual_swe <- function(total, fname_template, fname_out){
```

```
124
125   nc <- nc_open(fname_template)
126   v <- nc$var[["total_precipitation"]]
127   if(is.null(v)){
128     on.exit(nc_close(nc))
129     stop("Variable 'total_precipitation' not found. You can
130       choose between: ", paste(names(nc$var), collapse = ", "))
131   }
132   stopifnot(length(v$dim) == 3)
133
134   dims <- vector("list", 3)
135   for(i in seq_along(v$dim)){
136     d <- v$dim[[i]]
137     dims[[i]] <- ncdim_def(
138       name = d$name, units = d$units, vals = d$vals,
139       unlim = d$unlim, create_dimvar = isTRUE(d$create_dimvar),
140       calendar = if(is.null(d$calendar)) NA else d$calendar)
141   }
142   nc_close(nc); rm(nc, v); invisible(gc())
143
144
145   # Checking if netCDF container and SWE container dimensions match
146   lens <- vapply(dims, function(d) as.integer(d$len), integer(1))
147   stopifnot(identical(lens, as.integer(dim(total))))
148
149   swe <- ncvar_def(
150     name = "swe", units = "mm", dim = dims, missval = -9999,
151     longname = "Snow water equivalent (SWE)", prec = "float",
152     compression = 5)
153
154   nc_out <- nc_create(fname_out, swe, force_v4 = TRUE)
155   ncvar_put(nc_out, "swe", total)
156   nc_close(nc_out)
157
158   rm(nc_out, swe, dims); invisible(gc())
159 }
160
161
162
163
164
165
166
167
168 # Importing model parameters
```

```r
df_in <- read.table(fname_in, header = TRUE)
ddf_ave <- as.numeric(df_in$ddf_ave)
ddf_ampl <- as.numeric(df_in$ddf_ampl)
expfact <- as.numeric(df_in$expfact)
t_th <- as.numeric(df_in$tlim)

if((ddf_ave - ddf_ampl) < 0){ stop("DDF not correct: stopping! ") }


# Cycle over years
appo <- create_swe_container("Dataset/PCPD/1951.nc")
for(y in years){

  # Checking that netCDF maps are aligned
  fn_p <- paste0("Dataset/PCPD/", y, ".nc")
  fn_t <- paste0("Dataset/TEMP/", y, ".nc")
  com_mask <- !com_coord(fn_p, "total_precipitation", fn_t, "tmnd")
  if(com_mask) stop(paste("Input dataset not aligned in", y))


  # Loading precipitation and temperature grids (1 layer = 1 day)
  nc <- nc_open(fn_p)
  prec <- ncvar_get(nc, "total_precipitation")
  nc_close(nc)

  nc <- nc_open(fn_t)
  tmnd <- ncvar_get(nc, "tmnd"); tmxd <- ncvar_get(nc, "tmxd")
  stopifnot(identical(dim(tmnd), dim(tmxd)))
  tmean <- (tmnd + tmxd)/2
  nc_close(nc)

  rm(nc); invisible(gc());
  stopifnot(identical(dim(tmean), dim(prec)))


  # Computing solid precipitations
  precs <- compute_solid_precipitations(prec, tmnd, tmxd, t_th)
  rm(prec, tmnd, tmxd); invisible(gc())
  # Computing annual melt
  melt <- compute_melt(y, tmean, ddf_ave, ddf_ampl, expfact)
  rm(tmean); invisible(gc())


  total <- array(NA_real_, dim = dim(melt))
  for(i in 1:dim(total)[3]){
```

```
214
215     # Adding solid precipitations and melt to the SWE map. If some
216     # negative values appear, snow has melted all the way.
217     appo <- appo + precs[, , i] - melt[, , i]
218     mask <- appo < 0 & !is.na(appo)
219     appo[mask] <- 0
220
221     total[, , i] <- round(appo, 1)
222   }
223   rm(mask); invisible(gc())
224
225
226   # Saving SWE data to file
227   total[is.na(total)] <- -9999
228   fname_template <- paste0("Dataset/PCPD/", y, ".nc")
229   fname_out <- paste0("Results/SWE_", y, ".nc")
230   save_annual_swe(total, fname_template, fname_out)
231
232   rm(precs, melt, total); invisible(gc())
233   cat("Computed SWE maps for: ", y, "\n")
234 }
```

Listing B.1: Model implementation. This code was used to generate the SWE maps.

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